FriendMate
Books

# DROPBOX

A Book on Computer Education with Artificial Intelligence

Name ..........................................................................................

Class ................................................ Section ......................................

Witty

# 7

Shashank Joshi
(M.Tech)

PREPARATORY CURRICULUM

Inculcating 21st-Century Skills

FriendMate Books

A Unit of Bhumika Prakashan
86, Mansarovar Industrial Estate. (Delhi-NCR)
Sales Office:
Mayur Vihar, Phase-1, Pocket-5, Delhi-91

e-mail: bhumikapub@gmail.com
Website: www.bhumikaprakashan.com

New Edition

Printed By :
Bhumika Industry, Mawana (Delhi/NCR)



# Preface

In the context of the National Education Policy 2020, Computer Science education must undergo a paradigm shift where our children transition from being 'users' to 'developer's at a young age through mastery in coding. Newer technologies such as AI, Natural Language Processing, Data Science, Computer Vision and Neural Networks must be Introduced.

As per CBSE directives, the series stresses upon sustainable development goals while also ensuring that the right attitudes and values are inculated through of cyber Ethios.

**Special Features :**

- Carefully selected and graded content
- Stepwise explanation of programs with clear screen shots.
- Inculcation of in depth problem solving and programming.
- Sufficient Lab Activities to ensure practical leaning.
- A section on computers and Health including tips to minimize health issues while working on computers.
- A section on Olympiad Bot containing practice questions for Natural Cyber olympiad.

**—Author and Publisher**

# INCORPORATES NEP 2020

Created with new dimension of joyful learning along with completely mapped parameters of National Education Policy, 2020

### Critical Thinking

Strengthening decision making and innovation

### Creative Learning

Thinking out of the box

### Art Integration

Promoting art based learning

### Computational Thinking

Enables to help solve problems and make thinking more engaging.

### Inter Disciplinary

Connects knowledge across the curriculum for holistic development

### Subject Enrichment

Aimed at understanding and skill developing in specific subject

### Communication

Emphasis on soft skills and power of language

### Experimental Learning

Encourages learning by doing through hands-on activities

### Life Skills and Values

Inclined towards empowering moral values in learner

# INSIDE THE BOOK

**Outline** includes the topics to be learnt in the chapter.

Outlines
- Computer : An Electronic Machine
- Features of a Computer
- Types of a Computer

**Info Zone** include some facts related to chapter to enhance the knowledge of the students.

Info Zone
The first version of MS Paint was introduced with the first version of windows in 1985.

**Smart Tip** provide keyboard shortcuts for menu commands, to help users save time while performing routine operations.

Smart Tip
You can also open Paint by typing Paint in the Search Box next to the Start button. Then, click on the Paint button.

**Computer Etiquette** presents computer ethics and manners to be followed while working on a computer.

COMPUTER ETIQUETTES
While working on a computer, keep your neck relaxed. Practise this exercise to avoid any strain on your neck.

**Let's Implement** contains carefully graded exercises to test the knowledge of concepts learnt in a chapter.

Let's Implement

A. Tick (✓) the correct option.

1. A monitor is also called:
   (a) LCU (b) VDU
2. Which part of a computer is called the brain of a computer?
   (a) CPU (b) VDU
3. It is used to give the output of our work on paper.
   (a) Speakers (b) Printer

**Refreshment Corner** includes interesting exercises for enhancing the skills with fun.

Refreshment Corner

A. Colour the alphabet keys that spell your name with yellow.

**Practical Session** for practical hands-on exercises to strong then the concepts learned by doing.

Practical Session Experiential Learning

- Visit your computer lab and open Paint. Draw and colour the following.

**Racking Your Brain** includes the life skills and values questions.

RACKING YOUR BRAIN Life Skills and Values
Smart devices are used to make our tasks easier. Should we always use them at small tasks? Justify your answer.

**Just Have a Discussion** stimulates a small discussion for the topics being covered in the chapter which in turn improves the communication and logical skills.

Just Have a Discussion Communication
Discuss the different components of ScratchJr in class in detail.

**Teacher's Note** provides important information and suggestions on creative approaches to a chapter or a topic.

TEACHER'S NOTE
- Give enough exercises to students so that they get practice of using the tools of Paint as well as using the mouse.

# CONTENTS

1

# NUMBER SYSTEM

**Outlines**

- Binary Number System
- Conversion of Decimal into Binary Number System
- Octal Number System
- Conversion of Octal into Decimal Number System
- Conversion of Decimal into Hexadecimal Number System
- Decimal Number System
- Conversion of Binary into Decimal Number System
- Conversion of Decimal into Octal Number System
- Hexadecimal Number System
- Conversion of Hexadecimal into Decimal Number System

The computer uses sets of values to represent different quantities. Such sets of values include numbers (0–9), alphabets (A–Z, a–z) and some special characters. So, we can say that, a set of values used to represent different quantities is known as Number System.

The Number System is divided into four categories. These are:

- Binary Number System
- Decimal Number System
- Octal Number System
- Hexadecimal Number System

**Info Zone**

Gottfried Wilhelm Leibniz, a German mathematician, is credited with the invention of the binary number system.

## Binary Number System

The binary number system consists of 0 and 1. The digits 0 and 1 are known as binary digits or bits. Since this system uses two digits, it has base 2. This number system is used by all digital computers to convert the data input by you into its binary equivalent.

Some of the examples of the binary number system are — $(1100)_2$, $(111101)_2$, etc.

**Info Zone** The base or radix of the number system is the number of digits used in it.

The base of the number is written as Subscript.

## Decimal Number System

The number system that we use in our day-to-day life is the decimal number system. The Decimal Number System consists of 10 digits, 0, 1, 2, 3, 4, ..., 8, 9, i.e., 0 to 9. Since, this system uses ten digits, it has the base 10.

Some of the examples of the decimal number system are — (903)10, (14)10, etc.

## Conversion of Decimal into Binary Number System

To convert a decimal number system into the binary number system, follow these steps:

**Step 1.** Divide the decimal number by 2.

**Step 2.** Write the remainder on the right side and divide the quotient again by 2.

**Step 3.** Repeat step 2 till the quotient is zero.

**Note:** Always write the remainder in reverse order from bottom to top or from the Most Significant Digit to the Least Significant Digit to form a binary equivalent of the decimal number.

a. $(83)_{10} = (?)_2$

| | | | |
|---|---|---|---|
| 2 | 83 | | |
| 2 | 41 | 1 | Least significant digit |
| 2 | 20 | 1 | |
| 2 | 10 | 0 | |
| 2 | 5 | 0 | |
| 2 | 2 | 1 | |
| 2 | 1 | 0 | |
| | 0 | 1 | Most significant digit |

Therefore, $(83)_{10} = (1010011)_2$

b. $(762)_{10} = (?)_2$

| | | |
|---|---|---|
| 2 | 762 | |
| 2 | 381 | 0 |
| 2 | 190 | 1 |
| 2 | 95 | 0 |
| 2 | 47 | 1 |
| 2 | 23 | 1 |
| 2 | 11 | 1 |
| 2 | 5 | 1 |
| 2 | 2 | 1 |
| 2 | 1 | 1 |
| | 0 | 1 |

Therefore, $(762)_{10} = (1111111010)_2$

## Conversion of Binary into Decimal Number System

To convert a binary number system into decimal number system, follow these steps:

**Step 1.** Multiply each binary number with its positional value, which is in terms of powers of 2 (starting with $2^0$), starting from the extreme right digit.

**Step 2.** Increase the power one by one of 2.

**Step 3.** Now, sum up all the products to get the decimal number.

a. $(100010)_2$

1 1 0 0 1

$1 \times 2^0 = 1 \times 1 = 1$

$0 \times 2^1 = 0 = 0$

$0 \times 2^2 = 0 = 0$

$1 \times 2^3 = 1 \times 8 = 8$

$1 \times 2^4 = 1 \times 16 = 16$

$= 1 + 0 + 0 + 8 + 16 = 25$

$(11001)_2 = (25)_{10}$

b. $(100010)_2$

1 0 0 0 1 0

$0 \times 2^0 = 0$

$1 \times 2^1 = 2$

$0 \times 2^2 = 0$

$0 \times 2^3 = 0$

$0 \times 2^4 = 0$

$1 \times 2^5 = 32$

$= 0 + 2 + 0 + 0 + 0 + 32 = 34$

$(100010)_2 = (34)_{10}$

## Octal Number System

The octal number system consists of 8 digits, 0, 1, 2, ... 6, 7, i.e., 0 to 7. Since, this system uses eight digits, it has the base 8. Some of the examples of the octal number system are – $(436)_8$, $(730)_8$, $(14)_8$, etc.

## Conversion of Decimal into Octal Number System

To convert a decimal number system into the octal number system, follow these steps:

**Step 1.** Divide the decimal number by 8.

**Step 2.** Write the remainder on the right side and divide the quotient again by 8.

**Step 3.** Repeat step 2 till the quotient is zero.

**Note:** Always write the remainder in reverse order from bottom to top or from the Most Significant Digit to the Least Significant Digit to form an octal equivalent of a decimal number.

a. $(7488)_{10} = (?)_8$

| | | |
|---|---|---|
| 8 | 7488 | |
| 8 | 936 | 0 |
| 8 | 117 | 0 |
| 8 | 14 | 5 |
| 8 | 1 | 6 |
| | 0 | 1 |

$(7488)_{10} = (16500)_8$

b. $(642)_{10} = (?)_8$

| | | |
|---|---|---|
| 8 | 642 | |
| 8 | 80 | 2 |
| 8 | 10 | 0 |
| 8 | 1 | 2 |
| | 0 | 1 |

$(642)_{10} = (1202)_8$

## Conversion of Octal into Decimal Number System

To convert an octal number system into the decimal number system, follow these steps:

**Step 1.** Multiply each Octal number with its positional value which, in terms of powers of 8, is starting from the extreme right digit.

**Step 2.** Increase the power of 8 one by one.

**Step 3.** Now, sum up all the products to get the decimal number.

a. $(228)_8 = (?)_{10}$

2 2 8

$8 \times 8^0 = 8 \times 1 = 8$

$2 \times 8^1 = 2 \times 8 = 16$

$2 \times 8^2 = 2 \times 64 = 128$

$= 8 + 16 + 128$

$= (152)_{10}$

$= (228)_8 = (152)_{10}$

b. $(7014)_8 = (?)_{10}$

(7 0 1 4)

$4 \times 8^0 = 4 \times 1 = 4$

$1 \times 8^1 = 1 \times 8 = 8$

$0 \times 8^2 = 0 = 0$

$7 \times 8^3 = 7 \times 512 = 35874$

$= 4 + 8 + 0 + 3584$

$= 3596$

## Hexadecimal Number System

The hexadecimal number system consists of 16 digits, i.e., (0–9) and (A–F). Since, this system uses sixteen digits, it has the base 16, where A stands for 10, B for 11, C for 12, D for 13, E for 14 and F for 15.

Some examples of hexadecimal number system are — $(ACD)_{16}$, $(A14)_{16}$, $(986)_{16}$, etc.

## Conversion of Decimal into Hexadecimal Number System

To convert a decimal number system into hexadecimal number system, follow these steps:

**Step 1.** Divide the decimal number by 16.

**Step 2.** Write the remainder on the right side and divide the quotient again by 16.

Write the remainder in alphabetic form, i.e., (10 – 15) into (A – F) wherever needed.

**Step 3.** Repeat step 2 till the quotient is Zero.

**Note:** Always write the remainder in reverse order from bottom to top or from the Most Significant Digit to the Least Significant Digit, to form a hexadecimal equivalent of the decimal number.

a. $(9983)_{10} = (?)_{16}$

| | | |
|---|---|---|
| 16 | 9983 | |
| 16 | 623 | 15 (F) |
| 16 | 38 | 15 (F) |
| 16 | 2 | 6 |
| | 0 | 2 |

$= (9983)_{10} = (26FF)_{16}$

b. $(317547)_{10} = (?)_{16}$

| | | |
|---|---|---|
| 16 | 317547 | |
| 16 | 19846 | 11 (B) |
| 16 | 1240 | 6 |
| 16 | 77 | 8 |
| 16 | 4 | 13 (D) |
| | 0 | 4 |

$(317547)_{10} = (4D86B)_{16}$

## Conversion of Hexadecimal into Decimal Number System

To convert hexadecimal number system into decimal number system, follow these steps:

**Step 2.** Multiply each hexadecimal number with its positional value which, in terms of powers of 16, is starting from the extreme right digit.

Write the alphabets (A–F) as digits (10–15) wherever needed.

**Step 2.** Increase the power one by one of 16.

**Step 3.** Now, sum up all the products to get the decimal number.

a. $(1253)_{16}$

(1 2 5 3)

$3 \times 16^0 = 3$

$5 \times 16^1 = 80$

$2 \times 16^2 = 512$

$1 \times 16^3 = 4096$

b. $(4D63)_{10} = (?)_{10}$

$(4\ D\ 6\ 3)_{16}$

$3 \times 16^0 = 3 \times 1 = 3$

$6 \times 16^1 = 6 \times 16 = 96$

$D \times 16^2 = 13 \times 256 = 3328$

$4 \times 16^3 = 4 \times 4096 = 16384$

= 3 + 80 + 512 + 4096

= 4691

Therefore, $(1253)_{16} = (4691)_{10}$

= 3 + 96 + 3328 + 16384

= 19811

$(4D63)_{16} = (19811)_{10}$

## COMPUTER ETIQUETTES

Do not pull any wire of the computer.

# Let's Implement

**A. Tick (✓) the correct answer.**

1. Which number system consists of 0 and 1?
   a. binary number system ◯ b. decimal number system ◯
   c. octal number system ◯ d. hexadecimal number system ◯
2. Which number system has '10' as its base?
   a. binary number system ◯ b. decimal number system ◯
   c. octal number system ◯ d. hexadecimal number system ◯
3. Which number system has '16' as its base?
   a. hexadecimal number system ◯ b. decimal number system ◯
   c. binary number system ◯ d. octal number system ◯
4. Which number system consists of 8 digits?
   a. binary number system ◯ b. octal number system ◯
   c. hexadecimal number system ◯ d. decimal number system ◯
5. To convert a decimal number into a binary number, divide the number by ___________.
   a. 2 ◯ b. 8 ◯ c. 10 ◯ d. 16 ◯

**B. Fill in the blanks.**

1. The base of the binary number system is ______________.
2. The base of ______________ number system is 8.
3. ______________ number system is used by all digital computers.
4. ______________ number system uses 16 symbols to represent numbers.
5. Hexadecimal number system has ______________ as its base.

**C. Write True or False.**

1. The numbers used in hexadecimal number system are 0 to 15. ☐
2. The octal number system consists of 8 digits, i.e., 0 to 7. ☐

3. The digits 0 and 1 are known as binary digits or bits.
4. To convert a decimal number into an octal number, divide the number by 10.
5. To convert a decimal number into a hexadecimal number, divide the number by 10.

**D** **Answer the following questions.**

1. Explain number system. Name different types of number system.
2. Briefly explain the binary number system.
3. Briefly explain the hexadecimal number system.
4. Write a short note on octal number system.
5. What are the rules to convert a decimal number into an octal number?

## Refreshment Corner

**Solve the following.**

1. Convert the following decimal numbers into binary octal and hexadecimal numbers.
   a. $(432)_{10}$ b. $(99887)_{10}$ c. $(3689)_{10}$ d. $(7763)_{10}$
2. Convert the following binary numbers into decimal numbers.
   a. $(111100001)_2$ b. $(11000010)_2$
   c. $(10011010)_2$ d. $(11111110)_2$
3. Convert the following octal numbers into decimal numbers.
   a. $(7766)_8$ b. $(3407)_8$ c. $(3322)_8$ d. $(77001)_8$
4. Convert the following hexadecimal numbers into decimal numbers.
   a. $(ACE)_{16}$ b. $(ABCDE)_{16}$ c. $(A638)_{16}$ d. $(BBCF2)_{16}$

## RACKING YOUR BRAIN

Life Skills and Values

Find out if there are applications to convert numbers into various bases.

## Just Have a Discussion

Communication

Divide the class into small groups and conduct a discussion on— 'Binary Number System and Decimal Number System'.

### Teacher's Note

The students should be given sufficient number of questions to practise.

# MORE ON EXCEL 2016

## Outlines

- Freeze Panes
- Filtering Data
- Adding Subtotal
- Goal Seek
- Sorting Data
- Data Validation
- PivotTable
- Conditional Formatting

## Freeze Panes

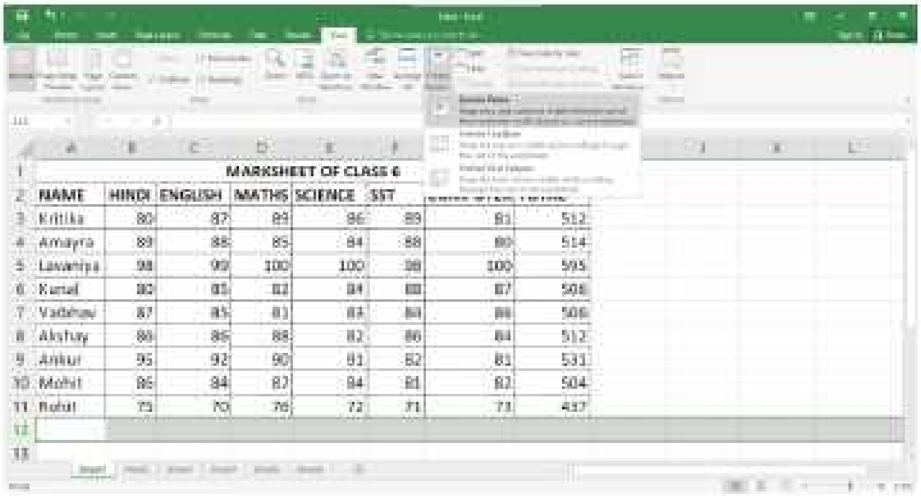
Whenever you're working with a lot of data, it can be difficult to compare information in your workbook. Fortunately, Excel includes several tools that make it easier to view content from different parts of your workbook at the same time, including the ability to freeze panes. You may want to see certain rows or columns all the time in your worksheet, especially header cells. By freezing rows or columns in place, you'll be able to scroll through your content while continuing to view the frozen cells.

### To Freeze Row

To freeze row, follow these steps:

1. Select the row below the row(s) you want to freeze.
2. On the View tab, select the Freeze Panes command, then choose Freeze Panes from the drop-down menu.

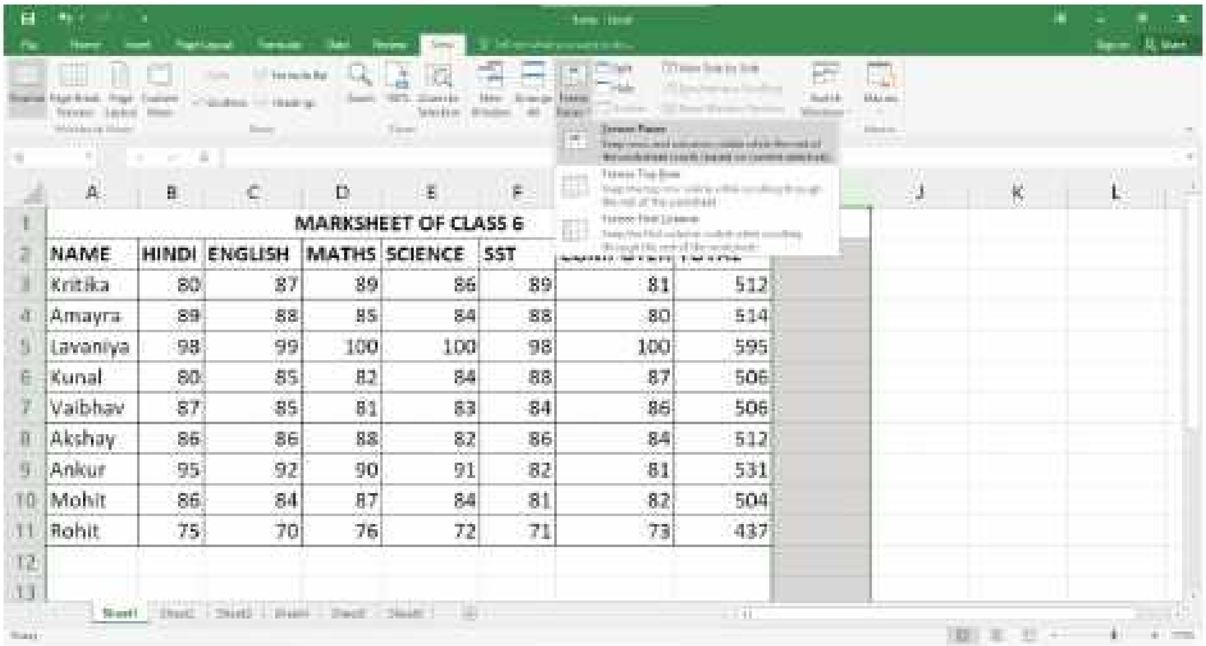
   The rows will be frozen in place, as indicated by the gray line.



### To Freeze Column

To freeze column, follow these steps:

1. Select the column to the right of the column(s) you want to freeze.
2. On the View tab, select the Freeze Panes command, then choose Freeze Panes from the drop-down menu.

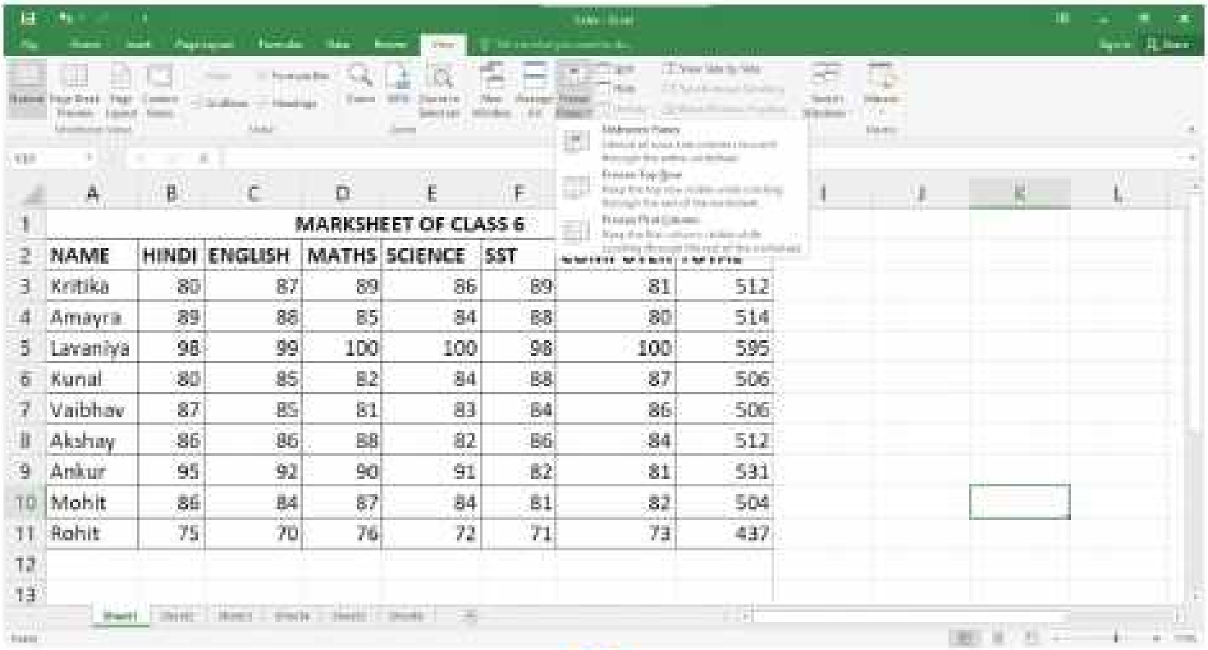


| | | | | | | | |
|---|---|---|---|---|---|---|---|
| MARKSHEET OF CLASS 6 | | | | | | | |
| NAME | HINDI | ENGLISH | MATHS | SCIENCE | SST | | |
| Kritika | 80 | 87 | 89 | 86 | 89 | 81 | 512 |
| Amayra | 89 | 88 | 85 | 84 | 88 | 80 | 514 |
| Lavaniya | 98 | 99 | 100 | 100 | 98 | 100 | 595 |
| Kunal | 80 | 85 | 82 | 84 | 88 | 87 | 506 |
| Vaibhav | 87 | 85 | 81 | 83 | 84 | 86 | 506 |
| Akshay | 86 | 86 | 88 | 82 | 86 | 84 | 512 |
| Ankur | 95 | 92 | 90 | 91 | 82 | 81 | 531 |
| Mohit | 86 | 84 | 87 | 84 | 81 | 82 | 504 |
| Rohit | 75 | 70 | 76 | 72 | 71 | 73 | 437 |

The column will be frozen in place, as indicated by the gray line.

### To Unfreeze Panes

If you want to select a different view option, you may first need to reset the spreadsheet by unfreezing panes. To unfreeze rows or columns, click the Freeze Panes command, then select Unfreeze Panes from the drop-down menu.



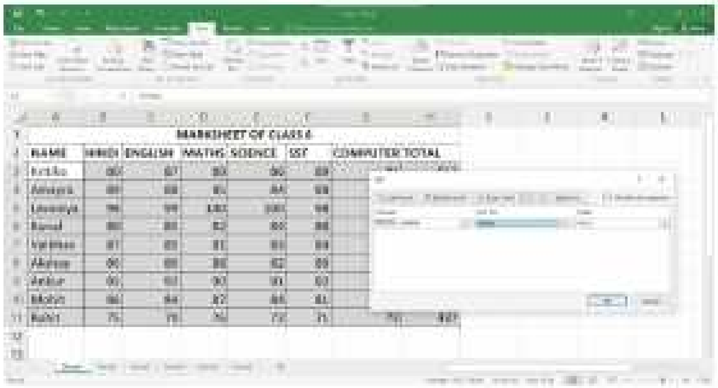
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| MARKSHEET OF CLASS 6 | | | | | | | |
| NAME | HINDI | ENGLISH | MATHS | SCIENCE | SST | | |
| Kritika | 80 | 87 | 89 | 86 | 89 | 81 | 512 |
| Amayra | 89 | 88 | 85 | 84 | 88 | 80 | 514 |
| Lavaniya | 98 | 99 | 100 | 100 | 98 | 100 | 595 |
| Kunal | 80 | 85 | 82 | 84 | 88 | 87 | 506 |
| Vaibhav | 87 | 85 | 81 | 83 | 84 | 86 | 506 |
| Akshay | 86 | 86 | 88 | 82 | 86 | 84 | 512 |
| Ankur | 95 | 92 | 90 | 91 | 82 | 81 | 531 |
| Mohit | 86 | 84 | 87 | 84 | 81 | 82 | 504 |
| Rohit | 75 | 70 | 76 | 72 | 71 | 73 | 437 |

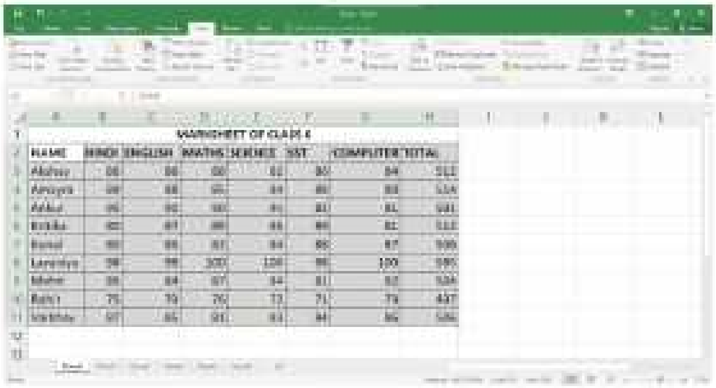
## Sorting Data

Sorting means arranging data either in an ascending or in a descending order in a worksheet. Data can be sorted in rows on the basis of text, numbers, combination of text and numbers or dates. Once the data is organised, it becomes easy to work with.

To sort the data shown in the given figure, follow these steps:

1. Select the range of cells containing the data to be sorted.
2. Click on the Sort option in the Sort & Filter group on the Data tab. The Sort dialog box appears.
3. Select the Column on the basis of which you want to sort, from the Sort by the drop-down list.
4. Select the Values option from the Sort on drop-down list.
5. Select the order of sorting from the Order drop-down list.
6. Click on the OK button.





**Info Zone**

To sort the list based on one column quickly, select any cell in the column and click the Sort A to Z button for ascending or Sort Z to A button for descending order in the Sort & Filter group on the DATA tab. These options are also available in the Editing group on the Home tab.

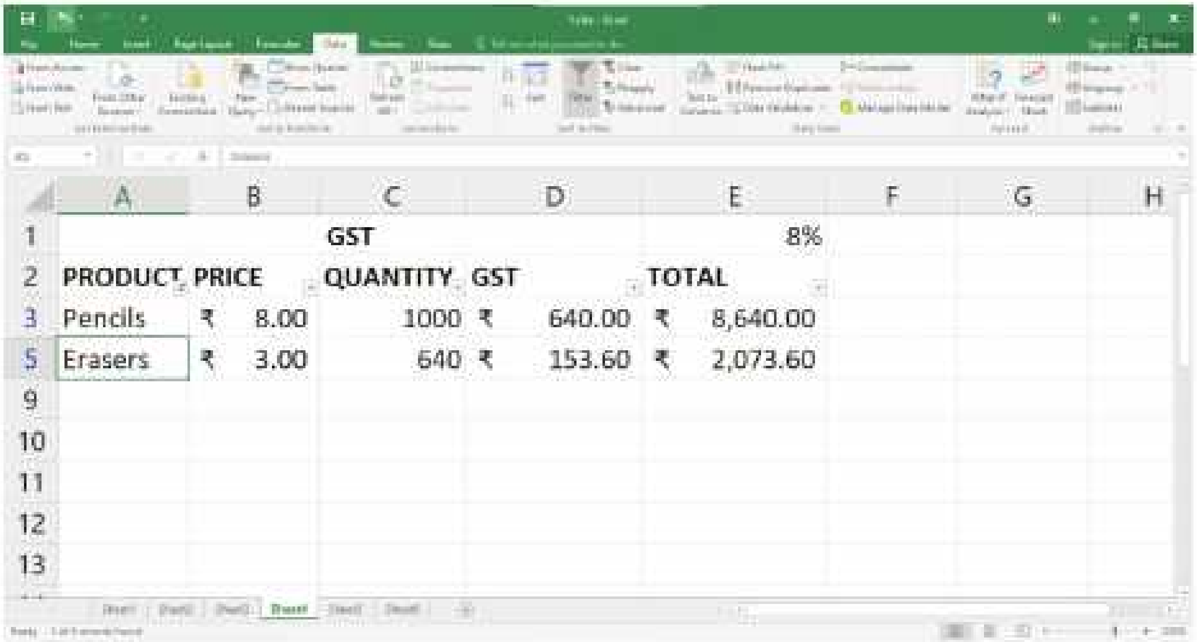
## Filtering Data

The Filter feature allows you to see only those records that you want to display, temporarily hiding the rest of the data from view. We can say that viewing only selective data that satisfies a given condition is known as filtering data.Filtering is different from sorting. Unlike sorting, filtering does not rearrange the data. To filter the data, follow these steps:

1. Type the data as shown.
2. Click on any cell in the range you want to filter (for example, A5).

3. Click on the Filter option in the Sort & Filter option in the Data tab.
4. The Filter arrows appear next to the column headings. Click on the arrow next to the column heading to specify the condition(s).
5. The drop-down list provides many options for filtering the data. Select the check boxes of the values on the basis of which you want to filter and clear the other check boxes.
6. Click on the OK button.

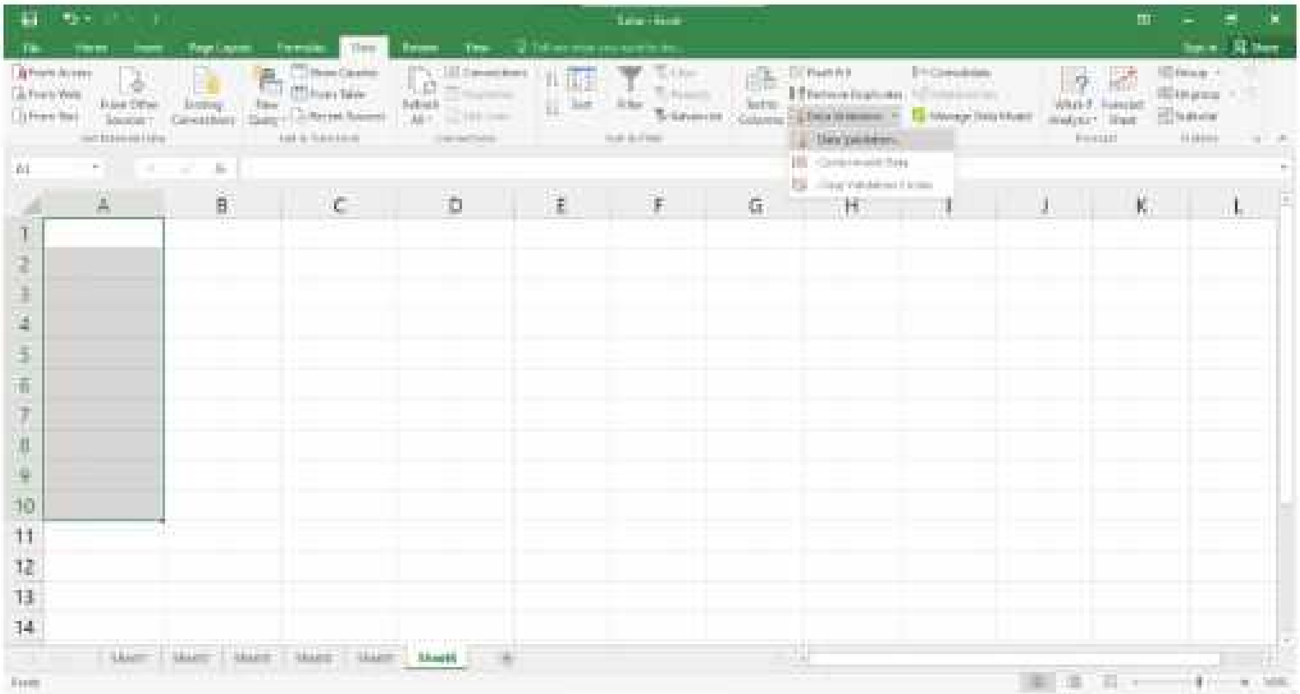
The list is filtered according to the chosen criteria.



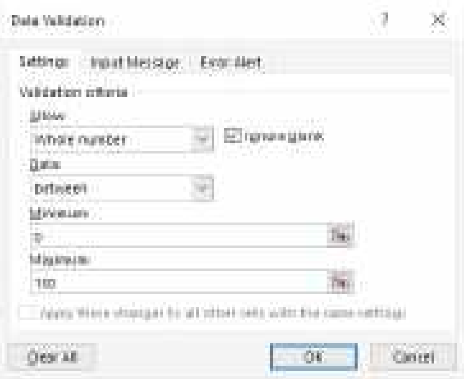

## Data Validation

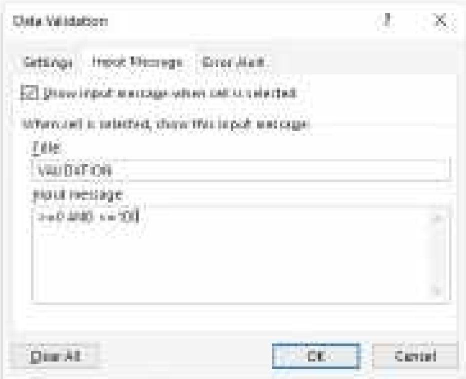
Data Validation is used to restrict the cell entries within a specified range. To apply data validation, follow these steps:

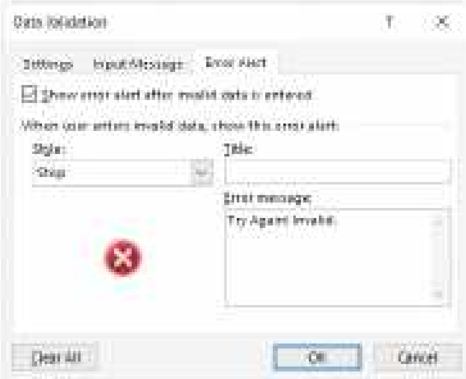
1. Select the range A1 : A10.
2. Click on the Data Validation option in the Data Tools on the Data tab.
3. A drop- down list appears. Click on the Data Validation... option.



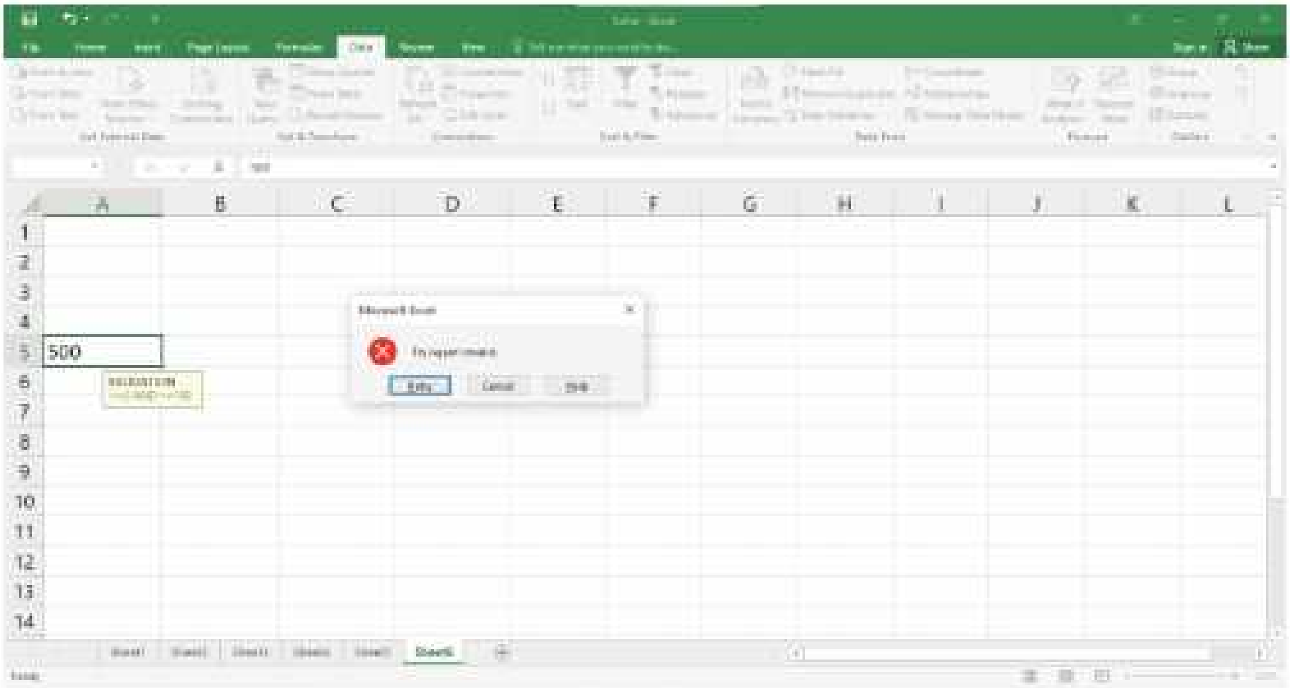
4. The Data Validation dialog box appears. Click on the Settings tab.
5. In Allow: drop-down box, select Whole number option.
6. In Data: drop-down list box, between option is selected by default.
7. In Minimum: text box, type 0 and in Maximum: text box type 100.
8. Click on the Input Message tab.
9. In the Title: text box, type VALIDATION. In Input Message: box, type > = 0 AND < = 100.
10. Click on the Error Alert tab.
11. Type 'Try Again! Invalid'. in Error Message : box.
12. Click on the OK button.




If you try to enter data beyond the specified limit in the selected range, an error message will be displayed. Click on Retry to enter another value in the cell.

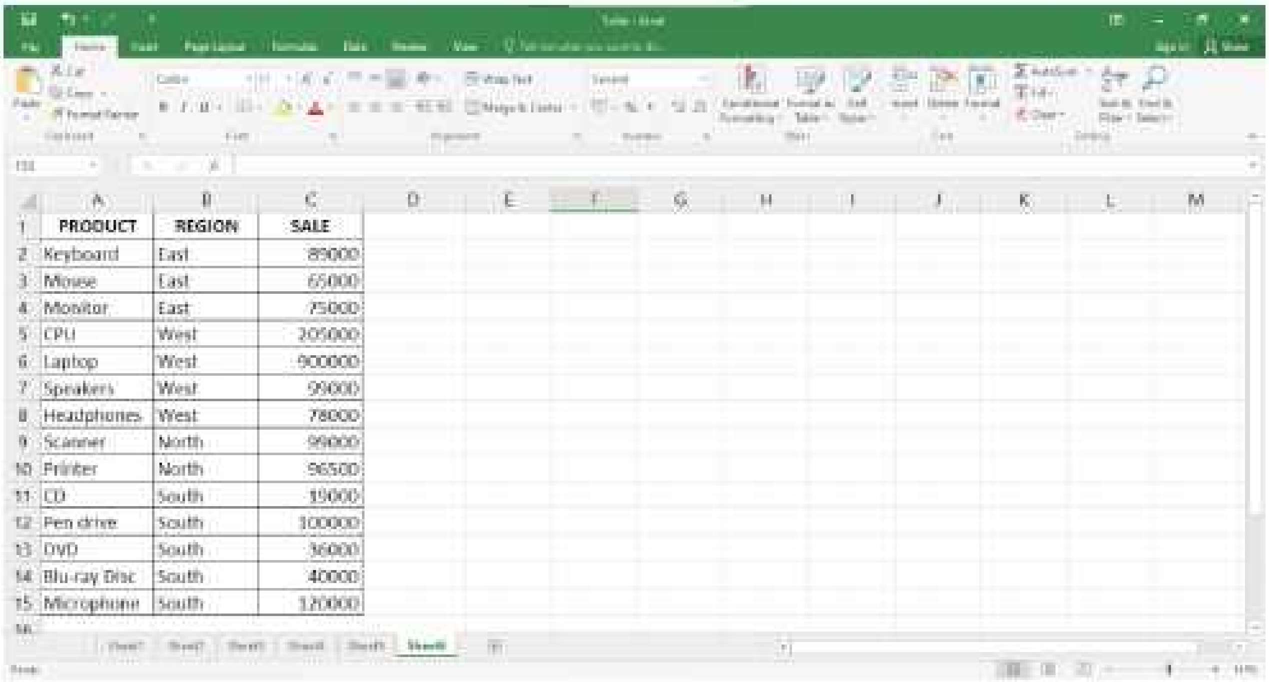



## Adding Subtotal

Subtotal helps us to manage and analyse the data. To apply subtotals, the database must be sorted.

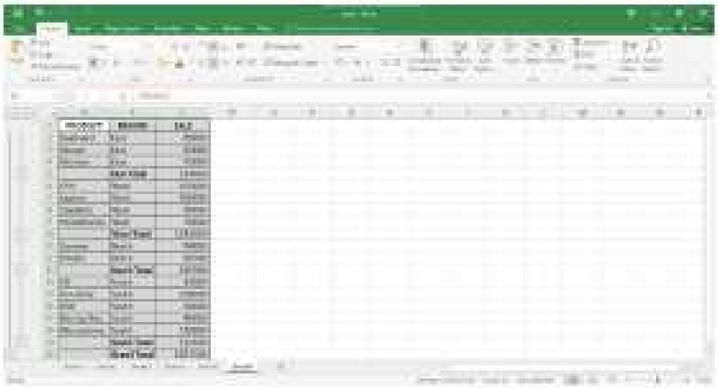
To add subtotal, follow these steps:

1. Type in the data as shown and select the range.




2. Click on the Subtotal button in the Outline group on the Data tab. The Subtotal dialog box appears.

3. Select Region from At each change in: drop-down list.
4. To calculate the sum of Sale, select Sum function from Use Function: drop-down list.
5. In Add subtotal to: drop-down list, click on Sale.
6. Deselect the Replace current subtotals option by clicking on its check box. This option overwrite the existing subtotal if already calculated.
7. The Summary below data check box is already marked. This option will place the total below the data of each.
8. Click on the OK button.

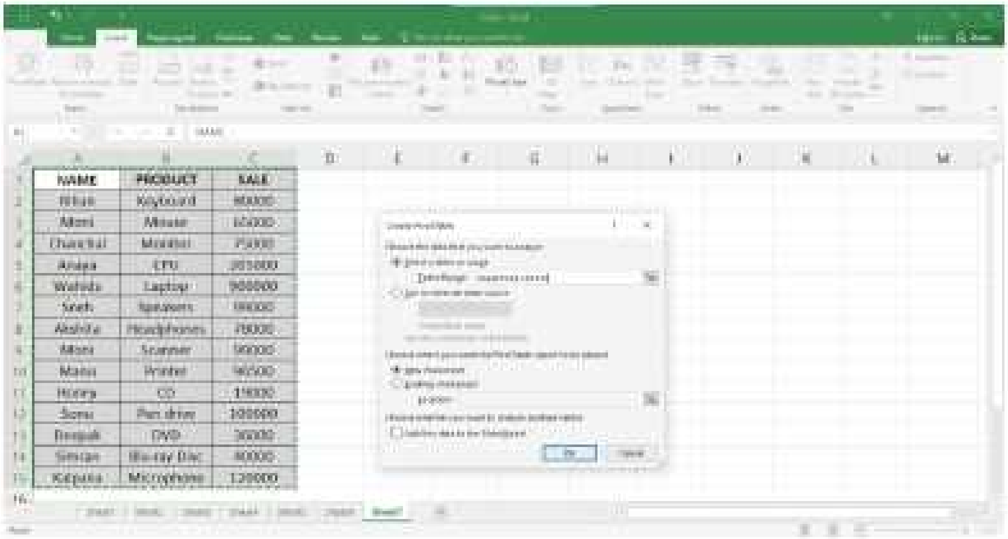



The subtotal will be displayed as shown in the figure given above.

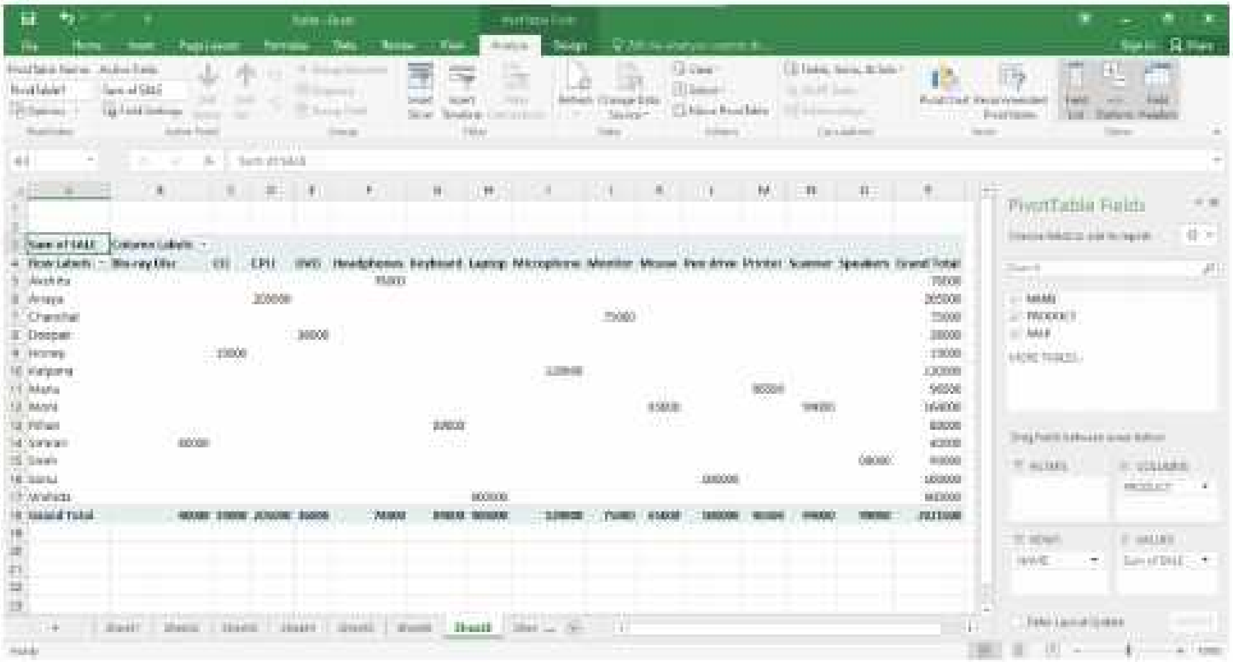
## PivotTable

PivotTable is a powerful tool for consolidating, summarising and presenting data. To create PivotTables, follow these steps:

1. Enter the data in a new worksheet and select the data range.
2. Click on the PivotTable option in the Tables group on the Insert tab.
3. The Create PivotTable dialog box appears. In the Table/Range: text box, the range is displayed that you have selected for the PivotTable. The New Worksheet radio button is selected by default. Click on the OK button.



4. The PivotTable layout is displayed on the new worksheet. The PivotTable Fields List task pane appears on the right side of the screen. Click and drag the Name field from the Choose fields to add to report pane and drop it into the ROWS quadrant.
5. Drag the Product field into the COLUMNS quadrant.
6. Drag the Sale field into the VALUES quadrant.

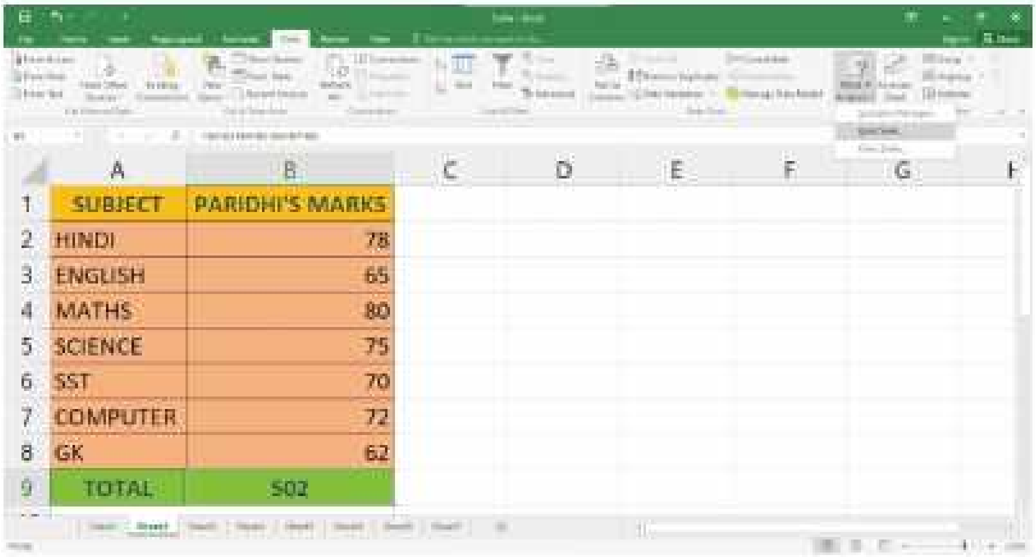



You can re-arrange the data in any way you like. You can also change the place of buttons here. If you do not like their placement, click the Undo button.

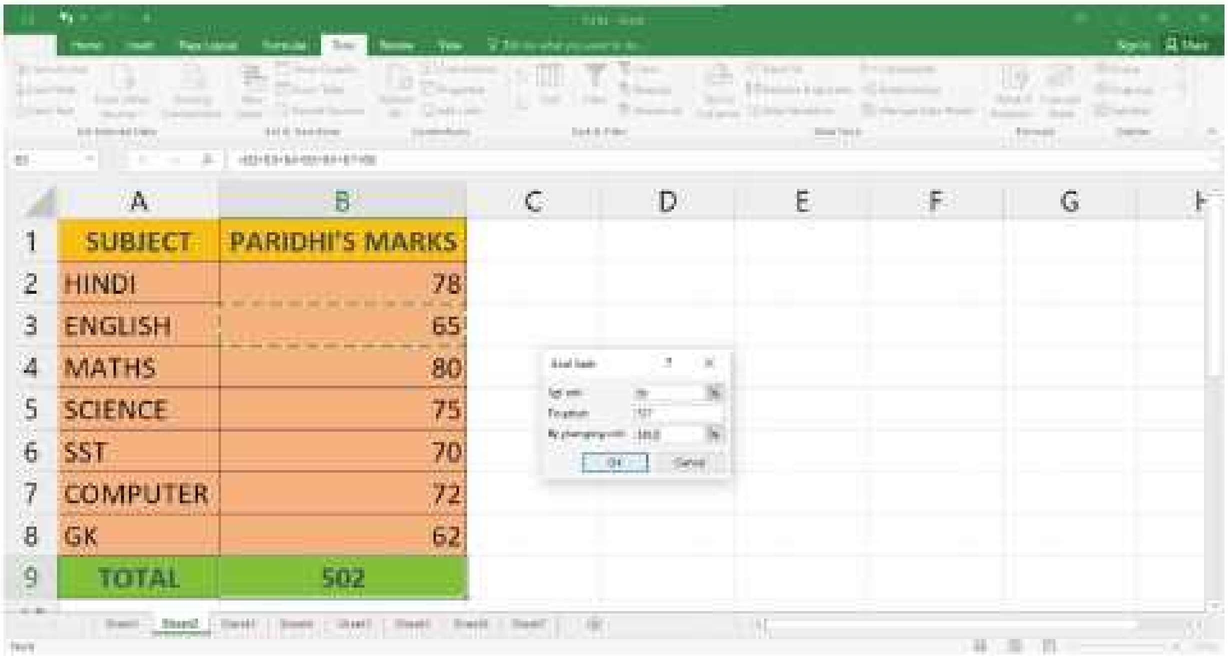
## Goal Seek

Goal Seek is a wonderful feature for fixing a specific result for one cell by adjusting a value in another cell. Follow the steps given below:

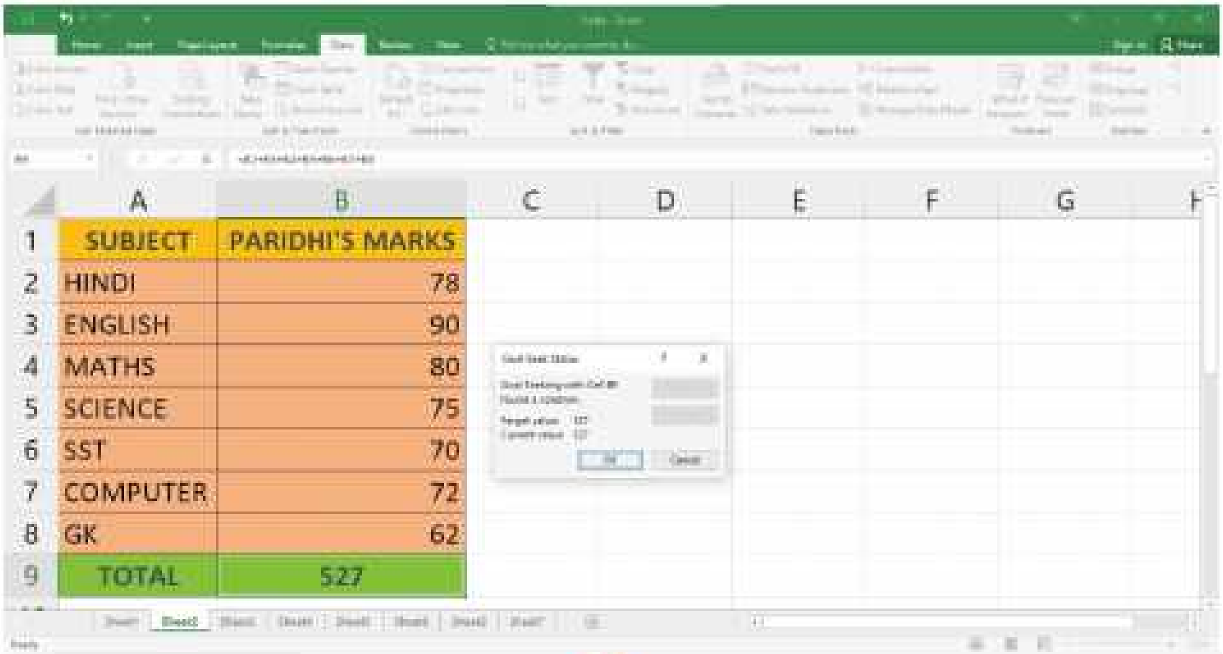
1. Enter data in a worksheet as shown.
2. Click on the What-if Analysis option in the Data Tools group on the Data tab.
3. A drop-down list appears. Select Goal Seek... option.



4. The Goal Seek dialog box appears. In Set cell: text box, define the cell address B9, on which the goal seek is to be applied. In this box, always refer the cell that contains formula.
5. Type the new value 527 as a result in To value: text box. In this box, always refer the cell that contains formula.
6. Press the Tab key and click on the cell B3 in a worksheet. The address will appear in the By changing cell: text box.
7. Click on the OK button.




8. The Goal Seek Status dialog box appears on the screen displaying the Target value 527 and Current value 527. Click on the OK button. Observe the change in the cell B3.



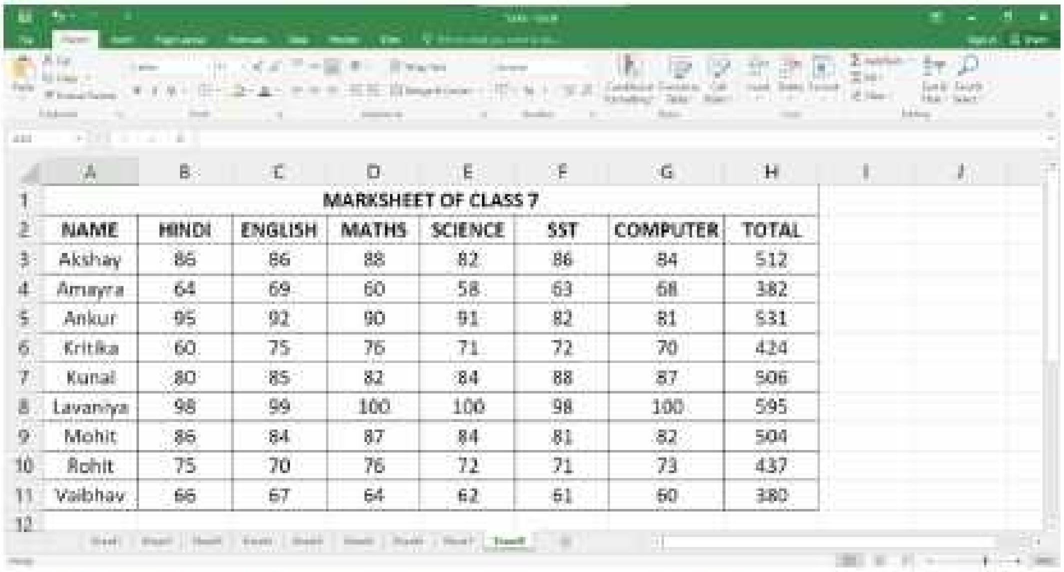
## Conditional Formatting

Conditional formatting is a feature in Excel that lets you apply formatting such as cell shading or font colour to a cell only when the cell contents satisfy a given condition. When the value of the cell meets the format condition, the selected formatting is applied to the cell. If the value of the cell does not meet the format condition, the default formatting is used.

Conditional formatting is a feature that allows you to set a cell's format according to conditions that you specify.

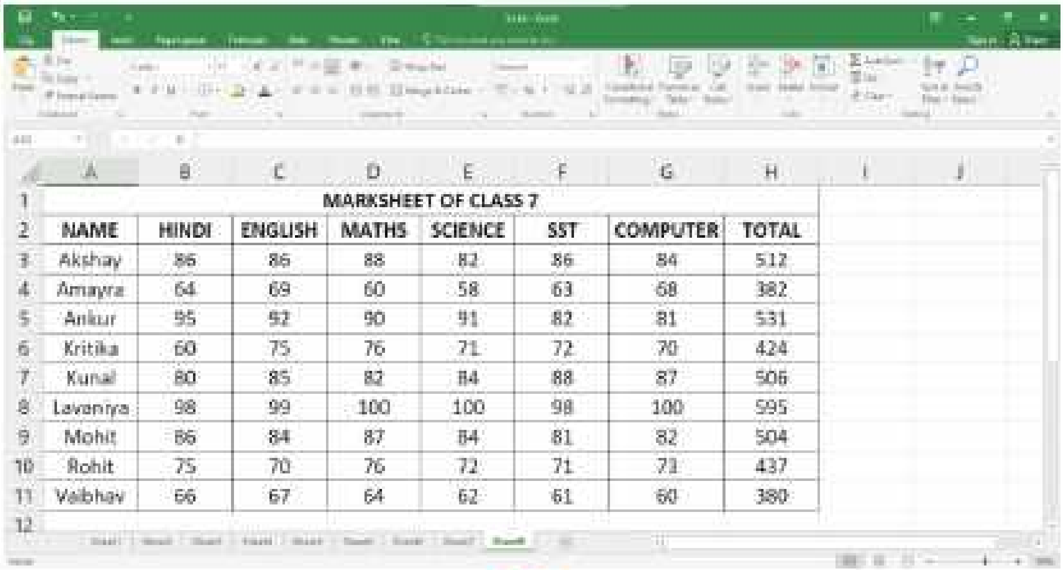
To apply conditional formatting, follow these steps:

1. Enter the data in a new worksheet as shown and select the range of cells to be formatted (for example, B3: G11).



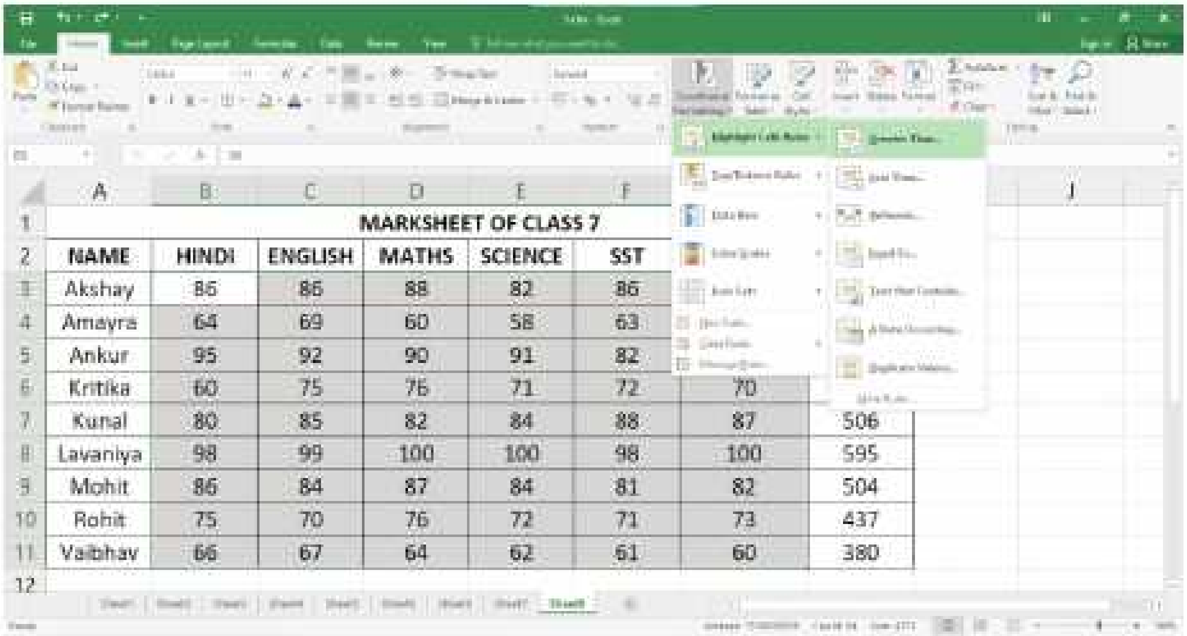
| | A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|---|
| 1 | MARKSHEET OF CLASS 7 | | | | | | | |
| 2 | NAME | HINDI | ENGLISH | MATHS | SCIENCE | SST | COMPUTER | TOTAL |
| 3 | Akshay | 86 | 86 | 88 | 82 | 86 | 84 | 512 |
| 4 | Amayra | 64 | 69 | 60 | 58 | 63 | 68 | 382 |
| 5 | Ankur | 95 | 92 | 90 | 91 | 82 | 81 | 531 |
| 6 | Kritika | 60 | 75 | 76 | 71 | 72 | 70 | 424 |
| 7 | Kunal | 80 | 85 | 82 | 84 | 88 | 87 | 506 |
| 8 | Lavaniya | 98 | 99 | 100 | 100 | 98 | 100 | 595 |
| 9 | Mohit | 86 | 84 | 87 | 84 | 81 | 82 | 504 |
| 10 | Rohit | 75 | 70 | 76 | 72 | 71 | 73 | 437 |
| 11 | Vaibhav | 66 | 67 | 64 | 62 | 61 | 60 | 380 |

2. Click on the Conditional Formatting option in the Styles group on the Home tab. A list of options is displayed.



| | A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|---|
| 1 | MARKSHEET OF CLASS 7 | | | | | | | |
| 2 | NAME | HINDI | ENGLISH | MATHS | SCIENCE | SST | COMPUTER | TOTAL |
| 3 | Akshay | 86 | 86 | 88 | 82 | 86 | 84 | 512 |
| 4 | Amayra | 64 | 69 | 60 | 58 | 63 | 68 | 382 |
| 5 | Ankur | 95 | 92 | 90 | 91 | 82 | 81 | 531 |
| 6 | Kritika | 60 | 75 | 76 | 71 | 72 | 70 | 424 |
| 7 | Kunal | 80 | 85 | 82 | 84 | 88 | 87 | 506 |
| 8 | Lavaniya | 98 | 99 | 100 | 100 | 98 | 100 | 595 |
| 9 | Mohit | 86 | 84 | 87 | 84 | 81 | 82 | 504 |
| 10 | Rohit | 75 | 70 | 76 | 72 | 71 | 73 | 437 |
| 11 | Vaibhav | 66 | 67 | 64 | 62 | 61 | 60 | 380 |

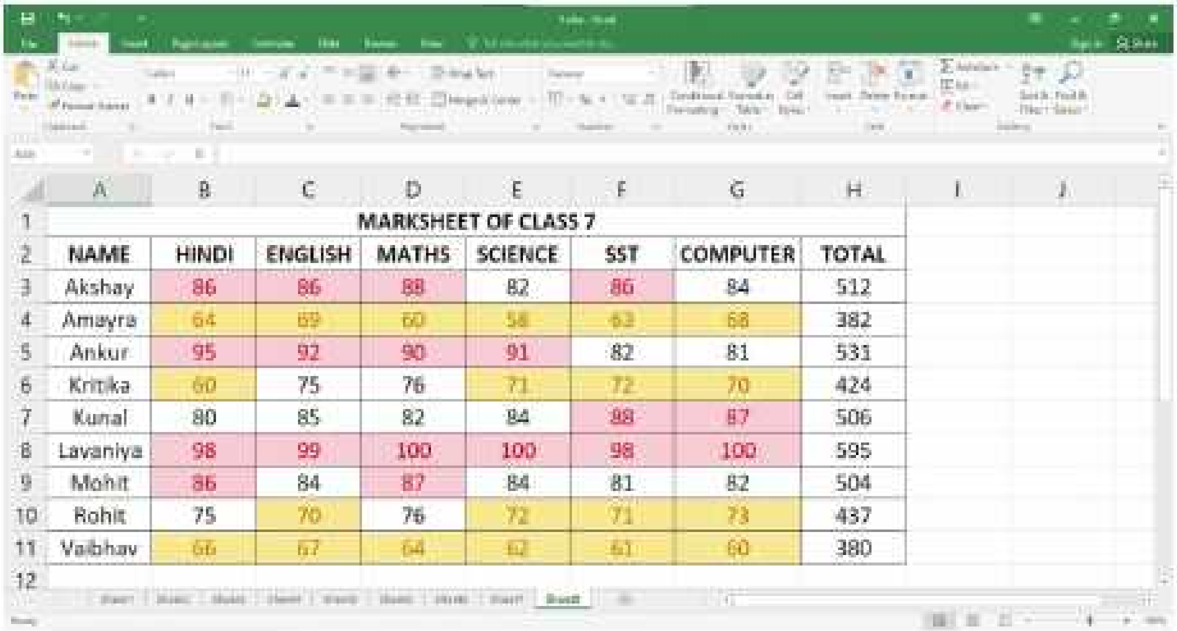
3. Select the required option as per the criteria you want to apply. Here we select Highlight Cells Rules>Greater Than....




4. A dialog box appears. Enter the value and specify the formatting to be applied. In this example, enter the value as 85 and choose Light Red Fill with Dark Red Text as the formatting option.


5. Click on the OK button.
6. Similarly, you can choose Highlight Cells Rules > Less than... .
7. A dialog box appears. Enter the value and specify the formatting to be applied. In this example, enter the value as 75 in Yellow Fill with Dark Yellow Text as the formatting option.
8. Click on the OK button.


The cells are formatted according to the specified conditions as shown in the figure given below.



| | A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|---|
| 1 | MARKSHEET OF CLASS 7 | | | | | | | |
| 2 | NAME | HINDI | ENGLISH | MATHS | SCIENCE | SST | COMPUTER | TOTAL |
| 3 | Akshay | 86 | 86 | 88 | 82 | 86 | 84 | 512 |
| 4 | Amayra | 64 | 69 | 60 | 58 | 63 | 68 | 382 |
| 5 | Ankur | 95 | 92 | 90 | 91 | 82 | 81 | 531 |
| 6 | Kritika | 60 | 75 | 76 | 71 | 72 | 70 | 424 |
| 7 | Kunal | 80 | 85 | 82 | 84 | 88 | 87 | 506 |
| 8 | Lavaniya | 98 | 99 | 100 | 100 | 98 | 100 | 595 |
| 9 | Mohit | 86 | 84 | 87 | 84 | 81 | 82 | 504 |
| 10 | Rohit | 75 | 70 | 76 | 72 | 71 | 73 | 437 |
| 11 | Vaibhav | 66 | 67 | 64 | 62 | 61 | 60 | 380 |

## COMPUTER ETIQUETTES

Never eat or drink near a computer.

## Let's Implement

**A** **Tick (✓) the correct answer.**

1. ______________ feature is used to set a specific result for one cell by adjusting a value in another cell.
   - a. Filter ◯ b. Goal Seek ◯
   - c. Sort ◯ d. Data Validation ◯
2. ______________ feature allows us to arrange the given data according to a particular field either in an ascending or in a descending order.
   - a. Goal Seek ◯ b. Freeze panes ◯
   - c. Data Validation ◯ d. Sort ◯
3. ______________ helps us to manage and analyse the data.
   - a. Subtotal ◯ b. Data Validation ◯ c. Sort ◯ d. Filter ◯
4. ______________ does not rearrange the data.
   - a. Sorting ◯ b. Filtering ◯
   - c. Freezing ◯ d. None of these ◯

5. Click the sort ______________ button for ascending the data.

   a. A to Z ◯ b. Z to A ◯ c. A to M ◯ d. S to A ◯

## B Fill in the blanks.

1. To apply subtotals, the ______________ must be sorted.
2. Once the data is ______________ , it becomes easy to work with.
3. ______________ is a powerful tool for consolidating, summarising and presenting the data.
4. ______________ is used to restrict the cell entries within a specified range.
5. ______________ option is in the sort & filter group on the Data tab.

## C Write **True** or **False**.

1. Filtering means sorting data. ☐
2. You can sort data only in ascending order in a worksheet. ☐
3. The subtotal button is in the outline group. ☐
4. You can see certain rows all the time in your worksheet. ☐
5. We cannot use data validation to restrict the cell entries. ☐

## D Answer the following questions.

1. What is the use of the sorting feature in Excel?
2. What is the use of Subtotal?
3. State the use of PivotTable.
4. How is sorting different from filtering data in Excel?
5. What is the use of conditional formatting feature in Excel?

## Refreshment Corner

**Write the use of the following features of Excel 2016.**

1. Freeze Panes : ______________________________
2. Sorting : ______________________________
3. Filtering : ______________________________
4. Data Validation : ______________________________
5. Subtotal : ______________________________
6. PivotTable : ______________________________
7. Goal Seek : ______________________________
8. Conditional Formatting : ______________________________

## Practical Session



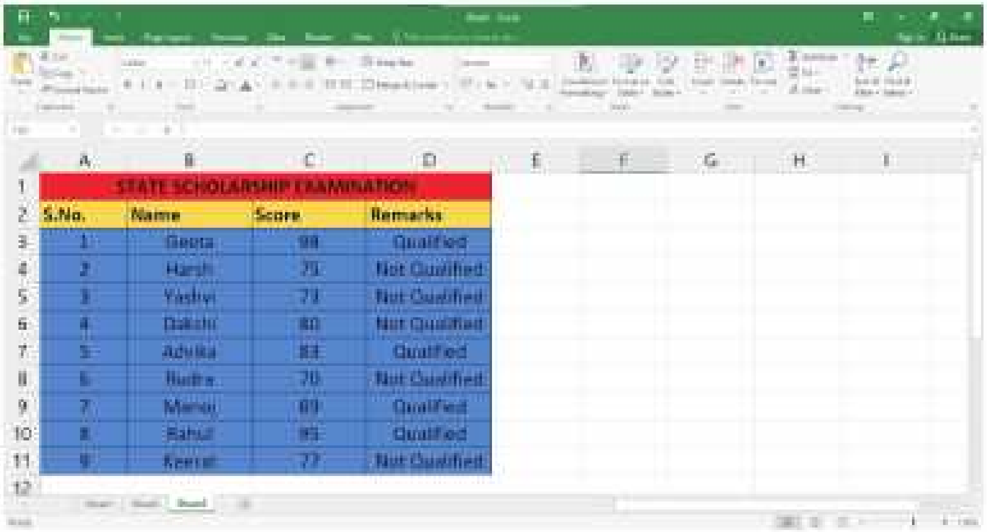
1. Create the worksheet given below and apply PivotTable feature on it.

| | A | B | C | D |
|---|---|---|---|---|
| 1 | DETAILS OF EMPLOYEES OF JP COMPANY | | | |
| 2 | NAME | DEPARTMENT | DESIGNATION | SALARY |
| 3 | Sumit | Production | Manager | 60000 |
| 4 | Amkur | Purchase | Manager | 70000 |
| 5 | Pankaj | Sale | Executive | 54000 |
| 6 | Atikur | Store | Store Keeper | 30000 |
| 7 | Nitin | Sale | Manager | 68000 |
| 8 | Ritik | Production | Executive | 60000 |
| 9 | Vineet | Purchase | Executive | 59000 |
| 10 | Deepak | Development | Executive | 55000 |
| 11 | Sparsh | Development | Manager | 78000 |
| 12 | Harsh | Store | Store Keeper | 32000 |

2. Jiya and Riya collected information on the most preferred means of transportation from different students. 19 students preferred train, 50 students preferred car and 24 students preferred airplane. 12 students chose ship, while 38 students chose bus as their preferred means of transport.
   a. Draw a table to show how many students prefer which means of transport.
   b. Calculate the total number of students surveyed.
3. Create the worksheet given below and apply conditional formatting as —
   a. Sale>8000 should appear with Light Red Fill with Dark Red Text.
   b. Sale<4000 should appear with Red Fill with Dark Red Text.

| | A | B | C | D |
|---|---|---|---|---|
| 1 | Sales Report from Jan to Mar 2023 | | | |
| 2 | Salesperson | Jan | Feb | Mar |
| 3 | Pawan | ₹ 2,536.00 | ₹ 7,500.00 | ₹ 9,800.00 |
| 4 | Rehman | ₹ 2,900.00 | ₹ 9,400.00 | ₹ 9,700.00 |
| 5 | Abhishek | ₹ 3,500.00 | ₹ 2,300.00 | ₹ 4,800.00 |
| 6 | Irshad | ₹ 7,500.00 | ₹ 4,500.00 | ₹ 6,500.00 |
| 7 | Prabhat | ₹ 9,500.00 | ₹ 3,620.00 | ₹ 6,200.00 |
| 8 | Prateek | ₹ 4,500.00 | ₹ 8,520.00 | ₹ 6,100.00 |
| 9 | Sumit | ₹ 7,800.00 | ₹ 9,540.00 | ₹ 6,320.00 |
| 10 | Shashank | ₹ 9,630.00 | ₹ 6,540.00 | ₹ 8,520.00 |
| 11 | Shanky | ₹ 9,420.00 | ₹ 7,850.00 | ₹ 4,850.00 |
| 12 | Ashok | ₹ 10,000.00 | ₹ 6,520.00 | ₹ 6,852.00 |

4. The list of students who have appeared for the first level of the state scholarship examination, is shown in the worksheet given below. Display the names of those students who have qualified for the second level of the examination.



| | A | B | C | D |
|---|---|---|---|---|
| 1 | STATE SCHOLARSHIP EXAMINATION | | | |
| 2 | S.No. | Name | Score | Remarks |
| 3 | 1 | Geeta | 98 | Qualified |
| 4 | 2 | Harsh | 75 | Not Qualified |
| 5 | 3 | Yashvi | 73 | Not Qualified |
| 6 | 4 | Dakshi | 80 | Not Qualified |
| 7 | 5 | Advika | 83 | Qualified |
| 8 | 6 | Rudra | 70 | Not Qualified |
| 9 | 7 | Manoj | 89 | Qualified |
| 10 | 8 | Rahul | 95 | Qualified |
| 11 | 9 | Keerat | 77 | Not Qualified |

**Observe the figure and answer the questions.**

a. Which feature of Excel will you use to arrange the data in ascending order of column F?

b. Which feature of Excel has been used to highlight the cells having marks less than 75 in the cells range B3:F8?

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | MARKSHEET OF CLASS 7 | | | | | |
| 2 | NAME | HINDI | ENGLISH | MATHS | SCIENCE | SST |
| 3 | Rajeev | 86 | 86 | 88 | 82 | 86 |
| 4 | Rajat | 64 | 69 | 60 | 58 | 63 |
| 5 | Mohit | 89 | 92 | 90 | 91 | 82 |
| 6 | Mayank | 60 | 66 | 86 | 71 | 72 |
| 7 | Kunj | 80 | 85 | 82 | 84 | 88 |
| 8 | Ajeet | 78 | 89 | 70 | 77 | 88 |

## RACKING YOUR BRAIN

Life Skills and Values

Find out if there are applications to convert numbers into various bases.

## Just Have a Discussion

Communication

Divide the class into small groups and conduct a discussion on— 'Sorting and Filtering the Data'.

### TEACHER'S NOTE

Discuss sorting, filtering, data validation, subtotal, pivot table, goal seek and conditional formatting by taking some practical examples.

3

Scan Me

# INTRODUCTION TO MS ACCESS 2016

## Outlines

- Data
- Features of Database
- Database Management System (DBMS)
- Objects in MS Access
- Parts of the MS Access 2016 Window
- Creating a Database
- Creating a Table using Templates
- Database
- Designing a Database
- Introduction to MS Access 2016
- Starting MS Access 2016

You must have seen your teachers carrying class lists having data of all the students of your class arranged in tabular form with headings, rows and columns containing their marks, phone numbers and other details. Such lists or records are maintained by almost all organisations. But these details need to be updated frequently and when these details or records exist in huge numbers, it becomes difficult to keep them in manual form and update them on paper. It is then that we need a computerised record maintenance system called the DBMS or the Database Management System. Let's begin by understanding the terms 'data' and 'database'. In this chapter, you will learn about databases and MS Access 2016.

## Data

Facts related to any object in consideration are collectively known as data. For example, your name, age, height, weight, etc. are examples of data that is related to you. An image, a file, a pdf, etc. can also be considered as data.

**Info Zone**

The attendance register of your class, the common dictionary at our homes are examples of databases.

## Database

Database is a collection of inter-related data which helps in efficient retrieval (taking out information) insertion and deletion of data from the database and organises the data in the form of tables, views, reports, etc. Let's discuss a few examples:

- An online telephone directory would use a database to store data pertaining to people, phone numbers, other contact details, etc.
- Your electricity service provider is using a database to manage billing, client related issues, to handle faulty data, etc.
- A school's database organises data about students, teachers and admin staff, etc.

## Features of Database

Look at the following example of database. It has four fields, namely Roll No., Name, Class, and Marks.

| Roll No. | Name | Class | Marks |
|---|---|---|---|
| 1 | Divyanshu | 7 | 92 |
| 2 | Priyanshu | 8 | 94 |
| 3 | Ashish | 10 | 66 |

Let us now discuss each component of a database:

- **Field :** It is the column heading and contains a set of values for a single type.
- **Record :** It is the row containing values under each field for one entry.
- **Value :** It is the data item which is the smallest unit in a database. It can be numeric, character or alphanumeric.

## Designing of Database

The designing process of a database consists of the following steps:

1. Determine the purpose of your database.
2. Find and organise the information required.
3. Divide the information into tables.
4. Turn the information items into columns.
5. Specify the primary keys.
6. Set up the table relationships.

7. Refine your design.
8. Apply the normalisation rules.

## Database Management System (DBMS)

The software used to manage a database is called a Database Management System (DBMS). A DBMS serves as an intermediary between the user and the database. The database structure is stored as a collection of files. So, we can access the data in those files through the DBMS. A DBMS that stores data in the form of tables and enables you to join the tables by creating a link between them is called Relational Database Management System (RDBMS). Some of the examples of DBMS are MS Access, Open Office database, Oracle, and Microsoft SQL server. In this chapter, we shall learn about Microsoft Access 2016. DBMS allows users to perform the following tasks:

### Data Definition

It helps in organising data into various types so that a relation can be set for accessing information.

### Data Updation

It helps in insertion, modification and deletion of the data in the database.

### Data Retrieval

It helps in taking out relevant information from the stored data in an organised manner.

### User Administration

It also helps in data security by monitoring different users using a single database system.

## Introduction to MS Access 2016

Microsoft Access is the most popular RDBMS that comes as a part of the Microsoft Office suite. It is used to organise and manipulate data. It organises the data in the form of tables. It provides the facility to create a relationship between these tables, using common fields. MS Access provides a graphical user interface for managing the data. Databases in MS Access 2016 are composed of four main objects which allow you to enter, store, analyse and compile the data.


## Objects in MS Access

The main objects that can be created in Access are as follows:

**Tables :** Tables are the building blocks of a database. They store the complete data in a structured manner, i.e., in the form of rows and columns. Every table has a finite number of columns and rows.


**Queries :** A query is a special view of a table corresponding to some desired information to be extracted. A query can display chosen records or chosen fields within records. A query can apply to one table or to multiple tables, if they are linked by common data fields.

**Forms :** They provide a user interface that lets the users enter and change data in the tables.

**Reports :** If forms are the input, then reports are for output. Reports are used to display the data stored in database tables in a professional format, for printing purposes.

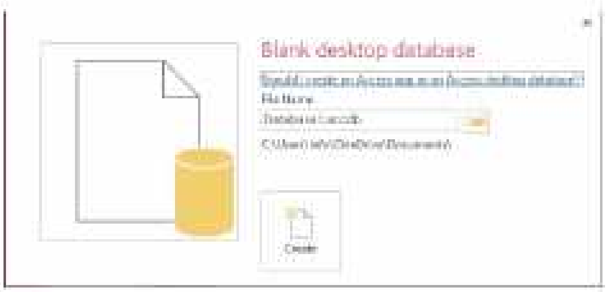
Even if you have a good idea of how each object can be used, it can initially be difficult to understand how they all work together. It helps to remember that they all work with the same data. Every piece of data a query, form or report is stored in one of your database tables.

## Starting MS Access 2016

To start MS Access 2016, follow these steps:

1. Click on the Start button.
2. Select Access 2016.
3. The Microsoft Access start-up window opens as shown in the figure. Click on the Blank desktop database.
4. The Blank desktop database dialog box appears. Click on the Create button.

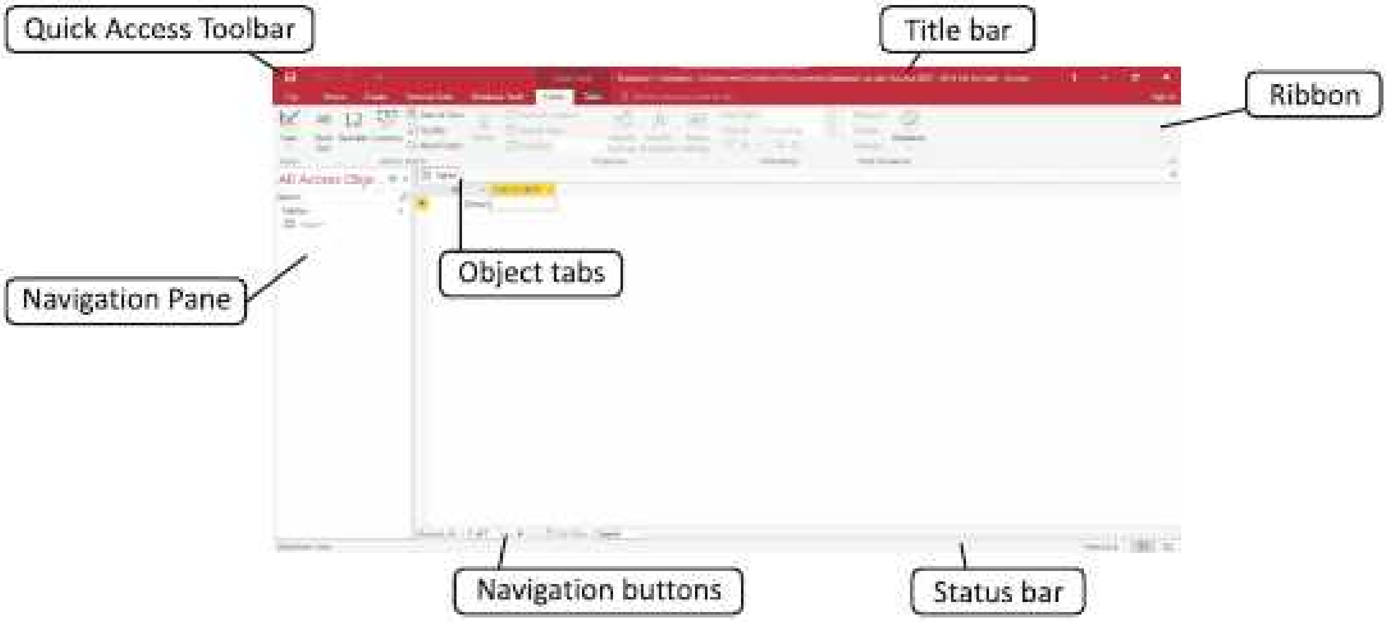





The MS Access 2016 window appears.

## Parts of the MS Access 2016 Window

The various parts of the MS Access 2016 window are as follows:




**Title Bar:** It appears at the top of the window and displays the name of the active database, and by default, displays the path to the folder where it is stored.

**Quick Access Toolbar:** It appears at the top left corner of the Access window. It has buttons for commands that are used frequently. By default, the following buttons are present on it.

1. **Save Button :** to save your work
2. **Undo Button :** to undo the previous action/roll back the action done
3. **Redo Button :** to redo the action that was undone

You can customise the Toolbar and add more buttons to it by clicking on the arrow next to the Redo button.

**Ribbon :** This is the area below the Title bar. The top of the ribbon has a set of tabs. Clicking the tab displays the associated set of commands. Commands are organised into groups. The main tables are File, Home, Create, External Data and Database Tools.

Some tabs appear only when you work with certain objects like forms. These tabs are called contextual tabs.


**Navigation Pane :** This is the area of the left of the window, and it displays your database objects. When you open a database or create a new one, the names of the database objects appear in the Navigation pane.

**Object Tabs :** The objects that you have opened in a database appear right above the work area of a tabbed form. When you click on an object tab, the contents of that object are displayed in the work area. Click on the cross button × on the right end of the bar to close the object displayed in the current tab.


**Navigation Buttons :** The Navigation buttons display the current record number in an object.


**Status Bar :** This bar is located at the bottom of the window. On its extreme left, it displays the name of the current view, and on its right, it displays four view buttons – Datasheet View and Design View.

## Creating a Database

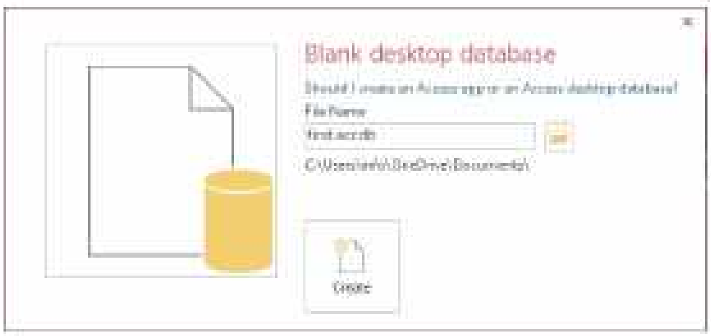
MS Access lets you create a new database using the following options:

- **Blank Database :** You can create a new database from scratch using this option.
- **Installed Templates :** Access comes with several pre-installed templates. These templates create a complete database application. They contain tables, queries, reports and other objects that you need to start working.

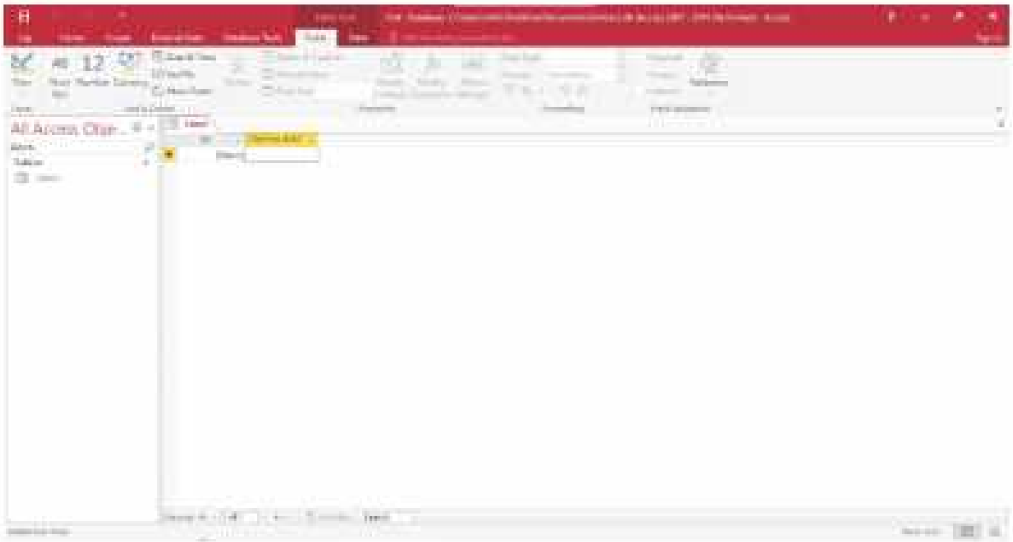
### Creating a Blank Database

To create a blank database in Access, follow these steps:

1. Click on the File tab.
2. Click on the New option.
3. Click on the Blank desktop database.
4. Type a name for the database in the File Name text box.
5. Click on the Create button.


A blank database gets created. Access creates the database with an empty table named Table1.



Access 2016 is .accdb.

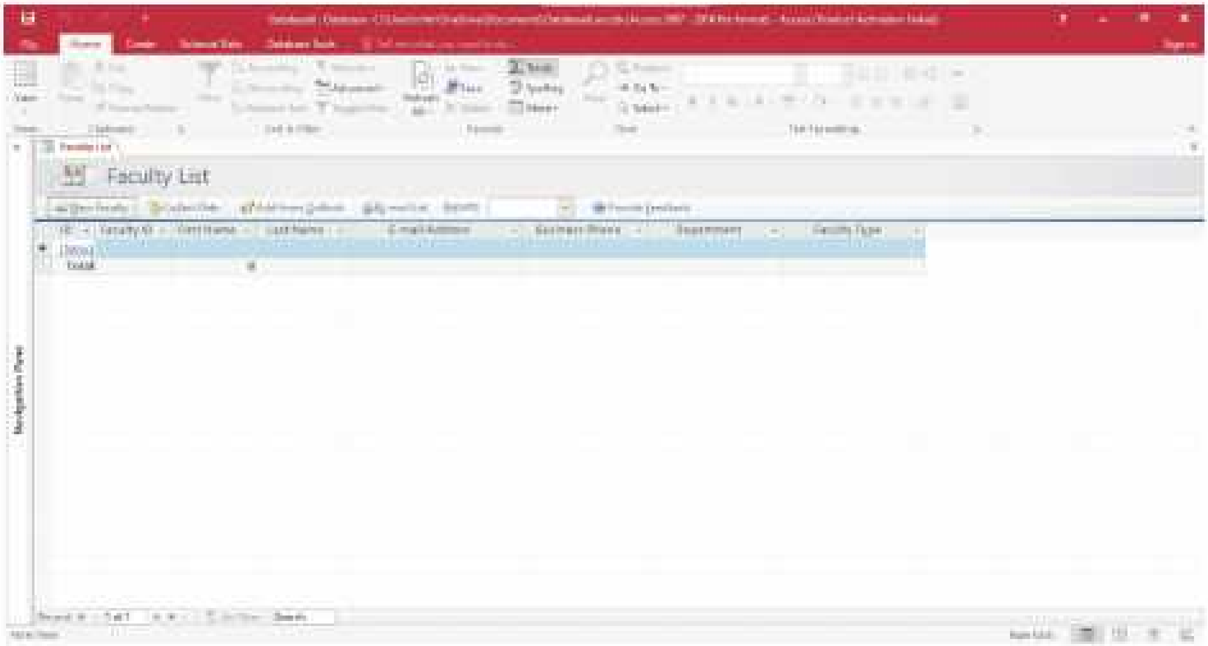
## Creating a Table using Templates

One of the easiest ways to create a table is to use a template. A template is a ready-to-use database that contains various types of tables, queries, forms, etc., needed to perform a specific task.

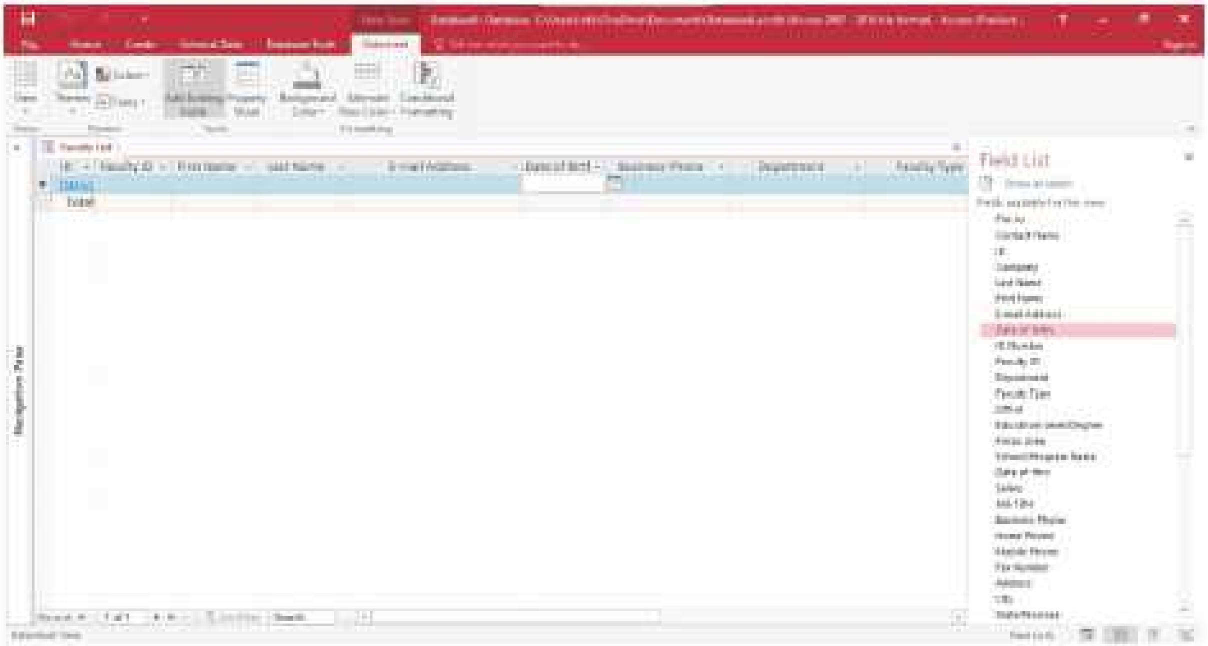
To create a Table using Templates, follow these steps:

1. Click on the File tab.
2. Click on the New option.
3. Click on the Database link.
4. Now, select the templates category, say Education.
5. Access will display the list of all the available templates. Select any template from the displayed list. Here, we have selected the Faculty template.
6. Type a name for the database in the File Name text box.
7. Click on the Create button.

Access creates the table and opens it in the Form View. It contains the fields which are relevant to the Faculty List.



8. Click on the Datasheet View button on the right side of the status bar.
9. A new tab Datasheet under Form Tools appears. Click on the Add Existing Fields option in the Tools group.
10. The Field List task pane appears on the right side. Select the fields from the list that you want to include in the Student List. Drag the selected field to the table and drop it where you want to position it when the insertion point appears.



Now, you can enter data into the table.

## COMPUTER ETIQUETTES

Always press the keys of the keyboard softly.

# Let's Implement.

**A** **Tick (✓) the correct answer.**

1. The standard file name extension for database in MS Office Access 2016 is __________.
   a. .accb  b. .accdb  c. .acc  d. .asd
2. We use the ______________ to save our work.
   a. Redo button  b. Undo button  c. Save button  d. None of these
3. ______________ is the column heading and contains a set of values for a single type.
   a. Field  b. Record  c. Both a and b  d. None of these
4. ______________ can be numeric, character or alphanumeric.
   a. Record  b. Value  c. Field  d. None of these
5. ______________ store the complete data in a structured manner.
   a. Tables  b. Queries  c. Both a and b  d. None of these

**B** **Fill in the blanks.**

1. ______________ is the row containing values under each field for one entry.
2. A DBMS servers as an intermediary between the ______________ and the database.
3. Databases in MS Access 2016 are composed of ______________ main objects.
4. ______________ are the building blocks of a database.
5. If forms are the input, then reports are for ______________ .

**C** **Write True or False.**

1. DBMS stands for Data Manage System.
2. Facts related to any object in consideration are collectively known as data.
3. Value is the data item which is the smallest unit in a database.
4. Data retrieval helps in insertion, modification and deletion of the data in the database.
5. Title bar appears at the bottom of the window.

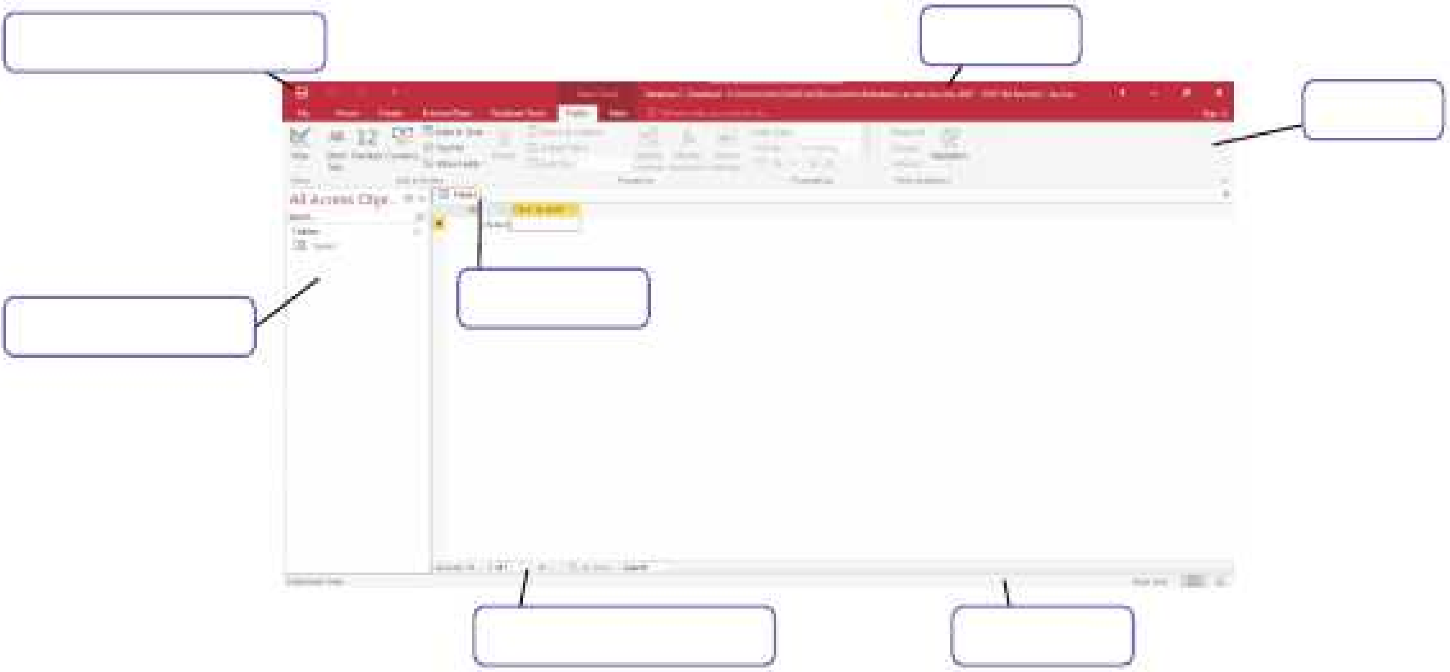
**D** **Answer the following questions.**

1. What is a database?
2. What do you understand by the term DBMS?
3. Explain RDBMS.

4. Define tables.
5. How can we start MS Access 2016?

## Refreshment Corner

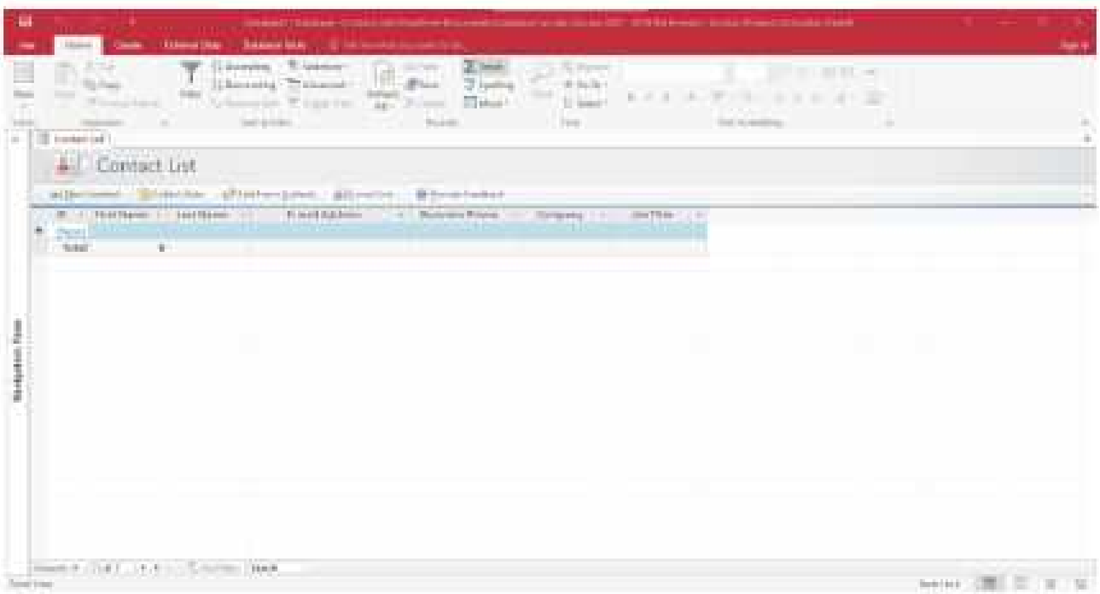
- **Label the figure given below.**



## Practical Session

Experiential Learning

- Open MS Access 2016 and create a table using template- Personal contact manager under the category– Contacts as given in the figure.



- Now, Click on the Datasheet View button on the right side of the status bar.
- A new tab Datasheet under Form Tools appears. Click on the Add Existing Fields option in the Tools group.
- The Field List task pane appears on the right side. Select the fields from the list that you want to include in the Student List. Drag the selected field to the table and drop it where you want to position it when the insertion point appears.
- Now, enter the data into the table.

### Competency Based Questions

1. Goldie creates a table using template. Her table is opened in form view. Now, she wants to enter data in a table. Can you suggest her the view in which she can enter data in her table?
2. Suman wants to display the data stored in database tables in a professional format, for printing purposes. Which object in MS Access 2016 should she use?

## RACKING YOUR BRAIN

Life Skills and Values

What will you do, if your friend is not able to create a database in MS Access 2016? Justify your answer.

### Just Have a Discussion

Communication

Conduct a group discussion on the topic 'Advantages of DBMS'.

### Teacher's Note

1. Discuss parts of the MS Access 2016 window with the students.
2. Explain the difference between Excel and Access.

# 4

# INTRODUCTION TO KRITA

## Outlines

- What is Krita?
- Starting Krita
- Tools in Krita
- Drawing Tools
- Selection Tools
- Downloading and Installing Krita
- Components of Krita
- Brush Tools
- Inserting an Image

## What is Krita?

Krita is a photo-editing software. With a photo-editing software, a simple looking picture can be enhanced to an extent that it looks as if captured by an expert photographer.

The word 'Krita' comes from the Swedish language; where 'Kri' means 'crayons' and 'Rita' means 'to draw'. Krita is an open-source software designed for digital drawing, photo editing and 2D animation. Some of the popular features of Krita are as follows:

- It is an open source software, which means that the Krita software code is available for public collaboration. You can download Krita for free from its official website, Krita.org.
- The user interface of Krita is easy to understand and use. It consists of menus, buttons, panels, etc.
- Krita interface consists of different panels that can be docked anywhere on the screen. This helps you to change the layout of the screen according to your needs and convenience.
- Brushes are the most important tools in drawing. In Krita, there are a total of hundred different brush colours that will give you an amazing drawing experience. These brushes are used to give different effects to the drawings.
- Krita has a great feature, the Brush Stabilizer, which is used to enhance the smoothness in the brush strokes.

- It has a huge built-in gallery of vectors that can be used for drawing comic strips.

## Downloading and Installing Krita

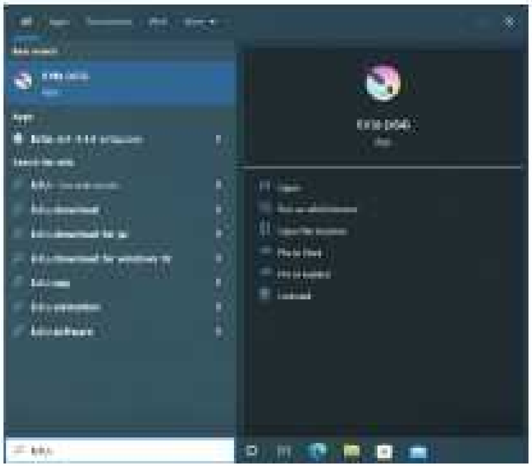
To download and install Krita, follow these steps:

1. Open any web browser.
2. Go to https://Krita.org/en/download/Krita-desktop/.
3. Now, click on the installer button. The setup of Krita will be downloaded.
4. Double-click on the installed setup to install Krita.

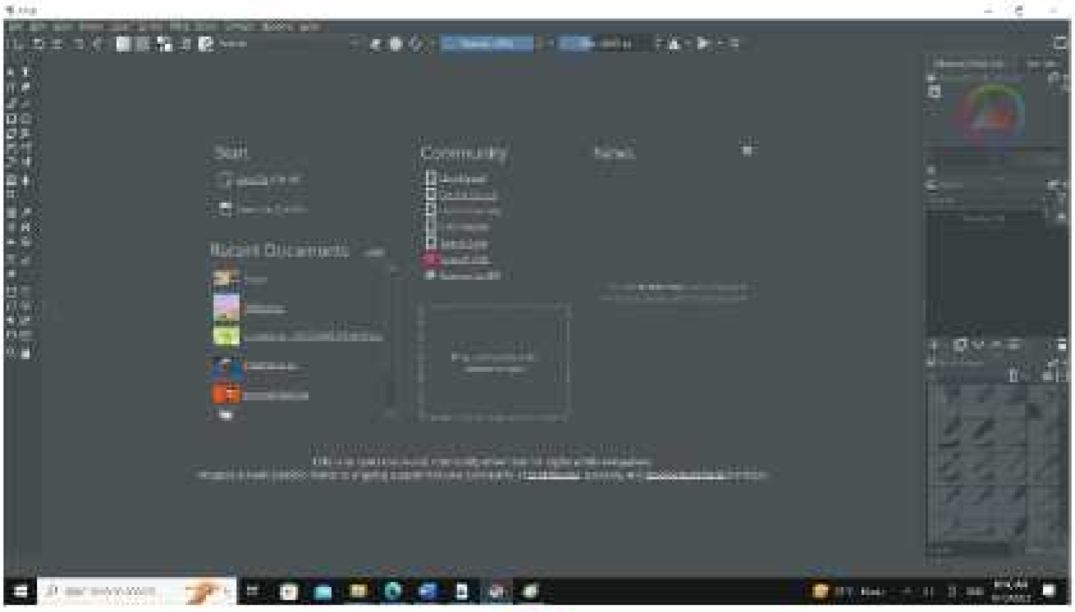
## Starting Krita

To start Krita, follow these steps:

1. Click on the Start button.
2. Scroll the list to find Krita and click on Krita.



The Krita window will appear.



3. Select the New option. This will open the Create new document dialog box.
4. Now, enter the required details, like Name, Width, Height, Orientation and Resolution of the image file you wish to create.
5. Finally, click on the Create button.

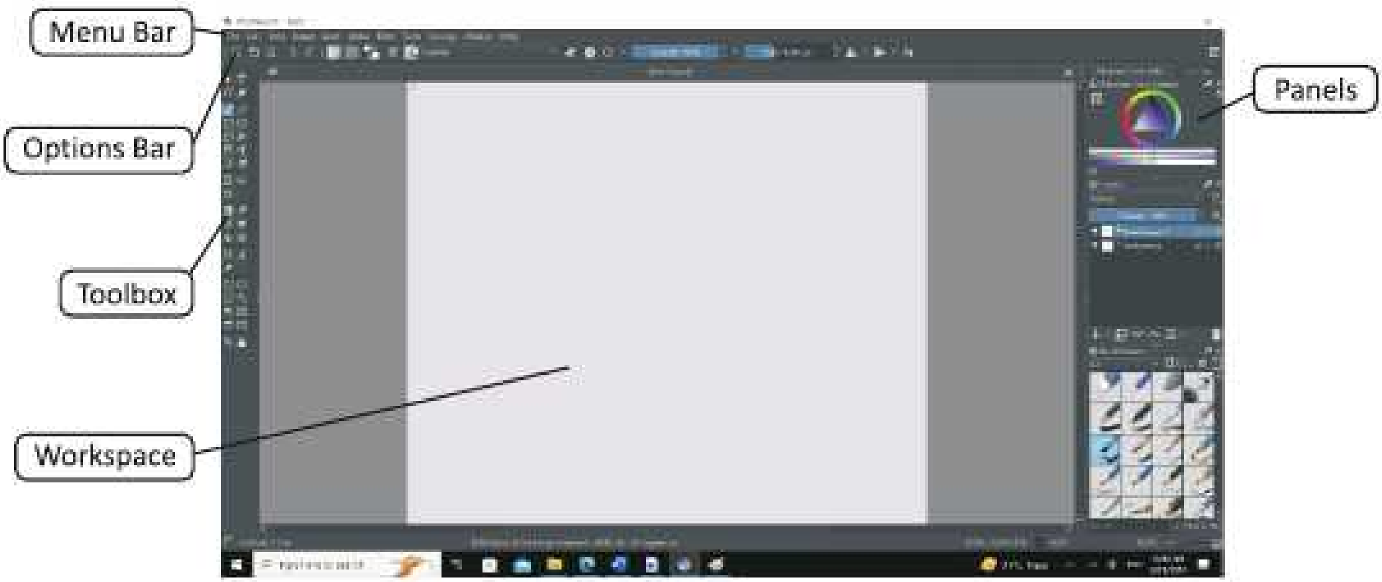
The window with blank canvas will appear.

**Smart Tip**

In the Search box, type Krita and then click on the Krita option.

## Components of Krita

The various components of Krita are as follows:



### Menu Bar

It is the top-most horizontal bar in Krita that contains all the menus, such as File, Edit, View, Layers, Filters, Tools, etc.

### Options Bar

It is present below the menu bar. It displays the options related to the currently selected tool in the toolbox that you are working with.

### Workspace

It is the area that displays the image file that you want to edit. The name of the image file appears as a tab at the top of the workspace. It is also known as Canvas.

### Panels

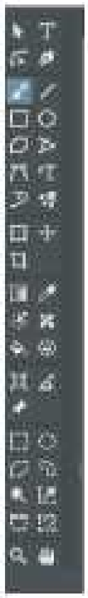
On the right side of Krita window, there is a separate area for the panels. Krita offers various panels that can either be grouped, stacked or docked. Panels can be placed anywhere on the screen. A panel that can be moved from one place to another, is called floating panel or docker.

### Toolbox

It contains the most commonly used tools. By default, toolbox docker appears on the left side of Krita window. You can resize and move it as you desire.

## Tools in Krita

The different types of tools present in Krita are as follows:



## Brush Tools

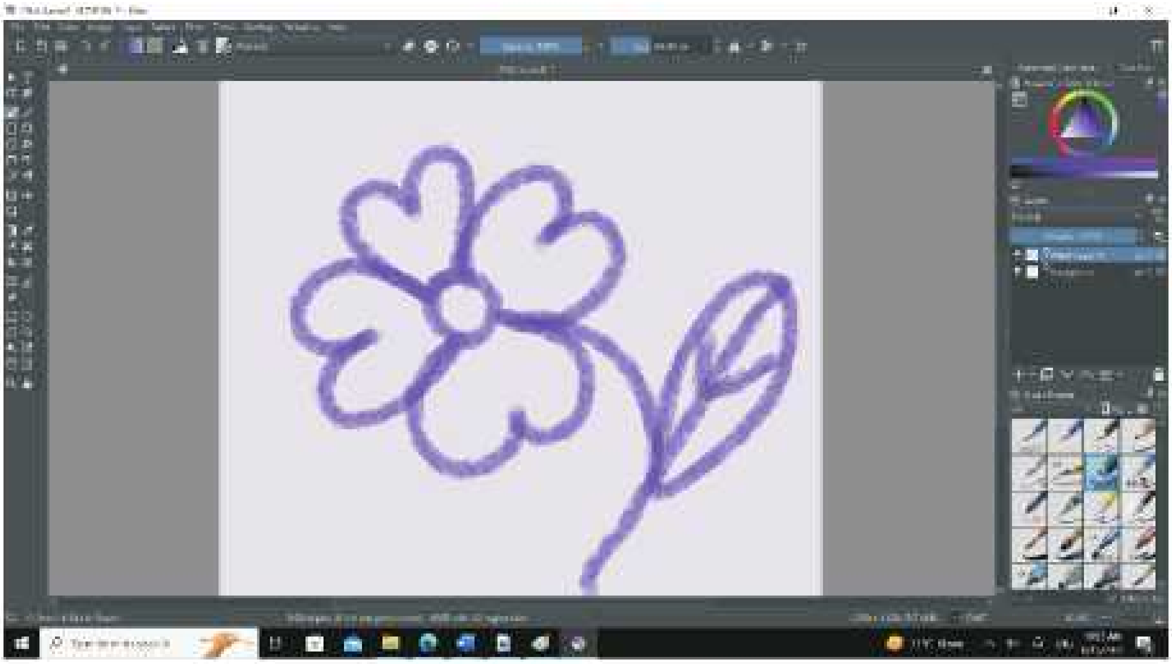
Brush tools are used to give different colouring effects to drawing. There are two types of Brushes tools present in Krita. They are:

- Freehand brush tool
- Dynamic brush tool

### Freehand Brush Tool

This tool is used to draw brush strokes to give an effect of painting to the image. To use this tool, follow these steps:

1. Click on the Freehand Brush tool.
2. Set the opacity in the options bar to make the colour opaque (intense) or transparent (weak).
3. Set the size of the brush in the options bar.
4. Now, press and hold down the left mouse button and drag the mouse to draw.



Info Zone

To erase some portion of an image or workspace, use the Set eraser mode option present in the options bar.

### Dynamic Brush Tool

This tool adds the custom smoothing dynamics to your brush. This will give you similar smoothing results as the normal freehand brush. There are a couple options that you can change.

- **Mass :** It averages our movement to make it appear smoother. Higher values will make your brush move slower.
- **Drag :** It is a rubber band effect that will help our lines come back to our cursor. Lower values will make the effect more extreme.

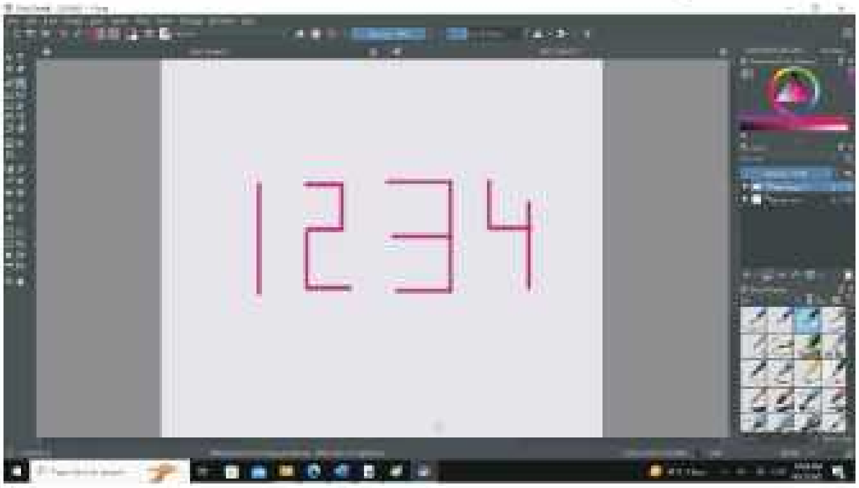
## Drawing Tools

There are various drawing tools in Krita. Let us learn about them.

### Line Tool

This tool is used to draw a straight line. To draw a line, follow these steps:

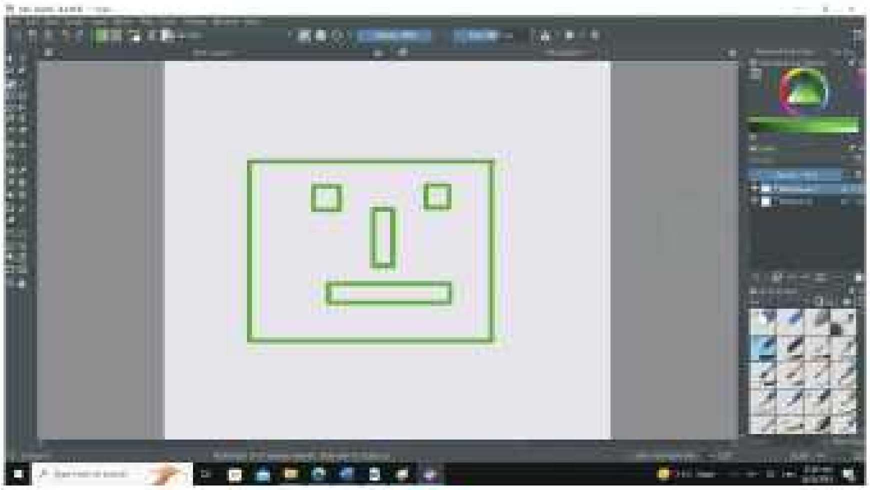
1. Select Line tool from the toolbox.
2. Set the opacity and size of the line.
3. Select the colour of your choice from the Advanced color selector in the panel.
4. Now, press and hold down the left mouse button and drag the mouse to draw a line.



### Rectangle Tool

This tool is used to draw a rectangle or a square. To draw a rectangle, follow these steps:

1. Select Rectangle tool from the toolbox.
2. Set the opacity and size of the line.
3. Select the colour of your choice from the Advanced color selector in the panel.
4. Now, press and hold down the left mouse button and drag the mouse to draw a rectangle.



## Ellipse Tool

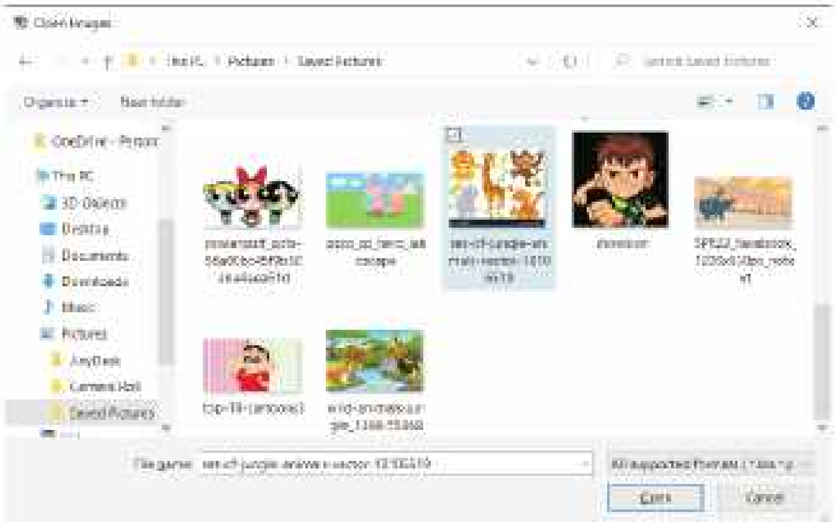
This tool is used to draw a circle or an oval. To draw a circle or an oval, follow these steps:

1. Select Ellipse tool from the toolbox.
2. Set the opacity and size of the line.
3. Select the colour of your choice from the Advanced color selector in the panel.
4. Now, press and hold down the left mouse button and drag the mouse to draw a circle or an oval.

## Inserting an Image

To open an existing image, follow these steps:

1. Click on the File menu.
2. Select the Open option. The Open Images dialog box appears.
3. Select the image you want to insert.



The image will be added in the workspace area. Once the image is added, you can edit it using the various tools available in the toolbox.

## Selection Tools

The Selection tools are used to alter a section of the image or paint it, without affecting the complete image. After selecting an area of the image, you can make the changes that you want in the selected area.

### Rectangular Selection Tool

The Rectangular Selection tool is used to select an image or a part of it as a rectangle for retouching or to perform cut, copy or edit operations.

1. Select the Rectangular Selection tool from the toolbox.
2. Place the cursor on the image. Click and hold the left mouse button and then drag out a marquee to select the required part of the picture.

You will notice that a dashed border has covered the selected object as shown in the figure. Now, you can perform any operation that you want on the selected area.

### Elliptical Selection Tool

The Elliptical Selection tool allows you to select the elliptical (circular) area of an image.

1. Select the Elliptical Selection tool from the toolbox.
2. Drag with the left mouse button on the image. You can observe an elliptical marquee (dashed border) around the image.

### Freehand Selection Tool

The Freehand Selection tool works as a freehand selection tool. This tool is useful when you are working with a portion of an image that has smooth edges and curves.

1. Open any image. Click on the Freehand Selection tool in the toolbox.

2. Press the left mouse button and drag around the part of the image to be selected. This will mark the selection.

3. Release the mouse at the end point.

### Similar Color Selection Tool

Using the Similar Color Selection tool, you can make selections by selecting any point of a colour in an image. In simple words, this tool is used for quick selection of areas that have similar colours. Follow these steps to select a part of the image using the Similar Color Selection tool.

1. Open any image.
2. Click on the Similar Color Selection tool in the toolbox.
3. Press the left mouse button. Click on any point of colour in the image.

Observe that all the objects with similar colour will be automatically selected. The changes you will make in one object, will have similar effect on the objects with the

## COMPUTER ETIQUETTES

Always clean the computer with soft and dry cloth.

# Let's Implement

**A** Tick (✓) the correct answer.

1. The ______________ tool is used to draw a square or rectangle.

   a. Ellipse ◯  b. Square ◯  c. Polygon ◯  d. Rectangle ◯

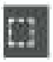
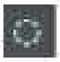
2. Which of the following is the Dynamic Brush tool?

   a.  ◯  b.  ◯  c.  ◯  d.  ◯

3. ______________ tool is used to draw a circle.

   a. Ellipse ◯  b. Line ◯  c. Rectangle ◯  d. None of these ◯

4. Which of the following is the Freehand Selection tool?

   a.  ◯  b.  ◯  c.  ◯  d.  ◯

5. ______________ is the top-most horizontal bar in Krita.

   a. Menu bar ◯  b. Options bar ◯  c. Toolbox ◯  d. None of these ◯

**B** Fill in the blanks.

1. ______________ is an open-source software.
2. The hundred different ______________ in Krita are used to give different colouring effects to the drawings.
3. ______________ Brush tool adds the custom smoothing dynamics to your brush.
4. The ______________ Selection tool works as a freehand selection tool.
5. The ______________ is the area that displays the image file that you want to edit.

**C** Write **True** or **False**.

1. By default, toolbox docker appears on the right side of Krita window. ☐
2. The Options bar is present above the menu bar. ☐
3. The workspace is also known as canvas. ☐
4. Krita offers various panels that can either be grouped, stacked or docked. ☐
5. The Ellipse tool in Krita is available in the Brush Presets panel. ☐

**D** Name the following.

1. Two components of Krita ______________ ______________
2. Two Brush tools ______________ ______________
3. Two Drawing tools ______________ ______________
4. Two Selection tools ______________ ______________

**E** Answer the following questions.

1. What is the use of Freehand Brush tool?

2. What is the use of Rectangle tool and Ellipse tool?
3. What is the use of Freehand Selection tool?
4. Explain the utility of the Dynamic Brush tool in Krita.
5. What is the use of Similar Color Selection tool?

## Refreshment Corner

- **Write down the steps to use the Freehand Selection tool in Krita.**

________________________________________

________________________________________

________________________________________

________________________________________

## Practical Session

Experiential Learning

- **Draw the following pictures in Krita.**

## RACKING YOUR BRAIN

Life Skills and Values

Your friend is trying to select an image or a part of it as a rectangle for retouching or to perform cut, copy or edit operations. But he/she uses the Similar Color Selection tool and his/her project is spoiled. What will you do — laugh on him or help him? Justify your answer.

### Just Have a Discussion

Communication

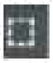
Discuss the difference between Freehand Brush tool and Dynamic Brush tool in the class.

### Teacher's Note

The teacher should give ample hands— on practice in Krita on the various tools taught in this chapter.

# QUICK REVIEW 1

**Tick (✓) the correct option.**

1. Which number system consists of 0 and 1?
   - a. binary number system
   - b. decimal number system
   - c. octal number system
   - d. hexadecimal number system
2. ______________ feature allows us to arrange the given data according to a particular field either in an ascending or in a descending order.
   - a. Goal Seek
   - b. Freeze panes
   - c. Data Validation
   - d. Sort
3. ______________ is the column heading and contains a set of values for a single type.
   - a. Field
   - b. Record
   - c. Both a and b
   - d. None of these
4. Which of the following is the Freehand Selection tool?
   - a. 
   - b. 
   - c. 
   - d. 
5. ______________ does not rearrange the data.
   - a. Sorting
   - b. Filtering
   - c. Freezing
   - d. None of these
6. Which number system has '10' as its base?
   - a. binary number system
   - b. decimal number system
   - c. octal number system
   - d. hexadecimal number system
7. ______________ helps us to manage and analyse the data.
   - a. Subtotal
   - b. Data Validation
   - c. Sort
   - d. Filter
8. ______________ can be numeric, character or alphanumeric.
   - a. Record
   - b. Value
   - c. Field
   - d. None of these
9. ______________ is the top-most horizontal bar in Krita.
   - a. Menu bar
   - b. Options bar
   - c. Toolbox
   - d. None of these
10. The ______________ tool is used to draw a square or rectangle.
    - a. Ellipse
    - b. Square
    - c. Polygon
    - d. Rectangle

5

Scan Me

# HTML BASIC

**Outlines**

- HTML Command Tags
- New Web Page Creation
- HTML Attributes
- Creating your HTML Document

HTML stands for Hypertext Markup Language. HTML is the language of internet. By using this language, you can create a website. It has all the features of a basic word processing program and is capable of handling graphics too. If you remember that in the earlier class it was taught that there are languages which need to be compiled or assembled. In this language, nothing of that sort has to be done. Just type, save and run.

### HTML Features

- HTML is a computer language, related to computer programming languages. HTML has its own syntax, slang and rules for proper communication.
- HTML is used to define the different parts of a web page.
- HTML is a language used to design the layout of a document and to specify the hyperlink.
- HTML tells the browser how to display the contents of a hypertext document, *i.e.*, a document including text, images and other support media.
- HTML is designed to work on a wide variety of platforms.

## HTML Command Tags

HTML tags are in-built coded commands that convey a special meaning which defines the structure and appearances of a web page.

HTML tags are written within the less than (<) and greater than (>) sign also known as angle brackets. These signify the opening and closing of the commands. The command for opening and closing are the same but, for one exception, the closing one has a slash sign (/) attached to it.

For example, The bold command is like this

<B> Bold text </B>

Info Zone

- Tags are also known as keywords.
- Tags are used in pairs, i.e., starting tag <HTML> and
- closing tag </HTML>, etc.

Tags are of two types : Container tag and Empty tag

### Container Tag

Container tag requires to pair tags, i.e., a starting as well as an ending tag, *e.g.*, <Title>...</Title>.

Notice that an ending tag is similar to that of a starting, except that it begins with a slash (/) symbol. Some examples of container elements are:

<HTML>...</HTML>

<HEAD>...</HEAD>

<TITLE>...</TITLE >

<BODY>...</BODY>

The Container tags affect everything that comes between their starting and ending tag.

### Empty tag

Empty tag requires just a starting tag and not an ending tag, e.g., <BR>, <BASE>. Empty elements just carry out their specific job, e.g., <HR> inserts a horizontal rule and <BR> breaks a line. Some examples of empty elements are: <BR>, <BASE>, <BASEFONT>, <HR>, <IMG>, <LINK>, etc.

## HTML Attributes

An attribute is used to define the characteristics of an element and is placed inside the element's opening tag. All attributes are made up of two parts: a name and a value.

- The name is the property you want to set. For example, the <font> element carries an attribute whose name is face, which you can use to indicate which typeface you want the text to appear in.
- The value is what you want the value of the property to be. The first example was supposed to use the Arial typeface, so the value of the face attribute is Arial.

The value of the attribute should be put in double quotation marks, and is separated from the name by the equals sign. You can see that a color for the text has been specified as well as the typeface in this <font> element:

<font face = “arial” color = “yellow”>

### Quotation Marks

Mostly all values should be enclosed in straight quotation marks "------". However, you can omit the quote mark if the value only contain letters (A-Z of a-z), digits (0-9), a hyphen (-) or period (.).

### Spacing

There is no restriction on the number of spaces given by you. HTML always finds out the extra spaces and ignores them. In addition to that, there is the concept of entering tag while beginning each paragraph. Each new paragraph would begin on a new line. It goes without saying that the tag <P> would always have to be followed up with a </P>.

### Special Symbols

Besides the usual ASCII set of characters, HTML supports other characters that can be used as it is and will be displayed as it is, in the browser. There are four symbols that have special meaning in HTML documents. These are the greater than (>), less than (<), straight double quotation marks (" "), and the ampersand (&). If you simply type them in your HTML document, the browser may attempt to interpret them.

## New Web Page Creation

Certain things are required to work with HTML and these are:

**TEXT Editor :** Text Editor such as Notepad (Usually found on windows under All programs Accessories menu), WORDPAD, etc., which is used to type the code of HTML language, for creating HTML document.

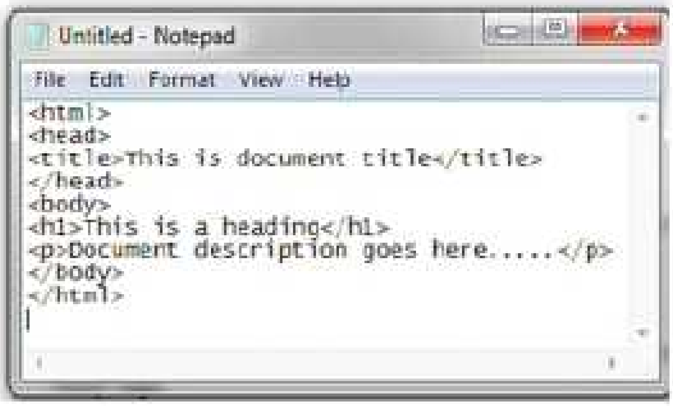
**Web Browser :** Web Browser is a software program which is used to access the webpages created in HTML language., e.g., Microsoft Internet Explorer, Netscape Navigator, etc. for viewing the document created in HTML language.

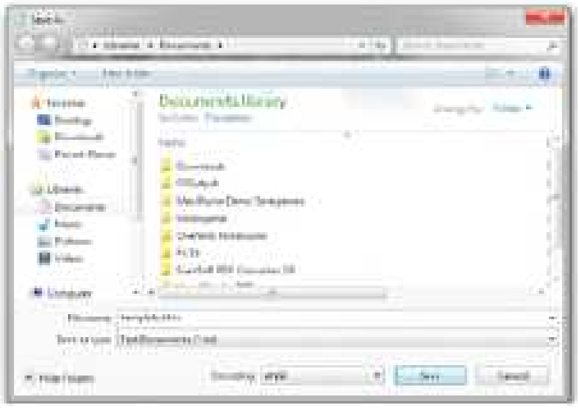
Type the text in any text editor like notepad, wordpad, etc.

Just follow the steps for creating a new document .

- Open Notepad or another text editor.
- At the top of the page, type <html>.
- On the next line, indent five spaces and now add the opening header tag: <head>.
- On the next line, indent ten spaces and type <title> </title>.
- Go to the next line, indent five spaces from the margin and insert the closing header tag: </head>.
- Five spaces from the margin on the next line, type<body>.
- Now drop down another line and type the closing tag right below its matter : </body>.

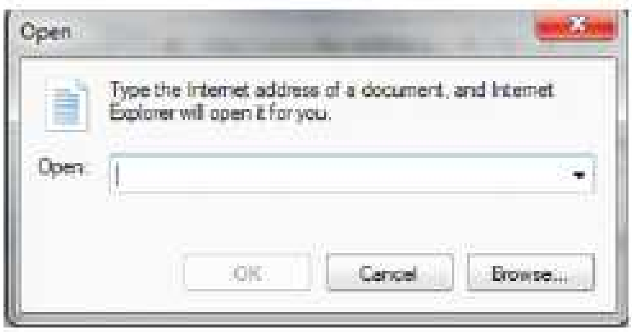
- Finally, go to the next line and type </html>.
- In the File menu, choose Save As.
- In the Save as Type option box, choose All Files.
- Name the file template.htm or .html.
- Click Save.



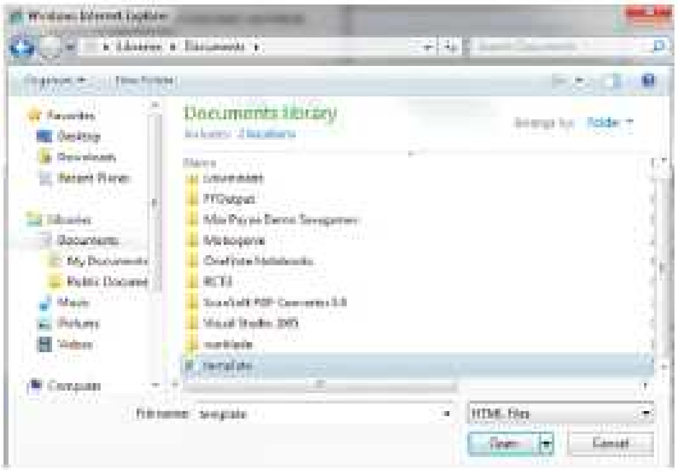

You have basic HTML document now. To see some result put the following code in title and body tags. For viewing the web page in the web browser:

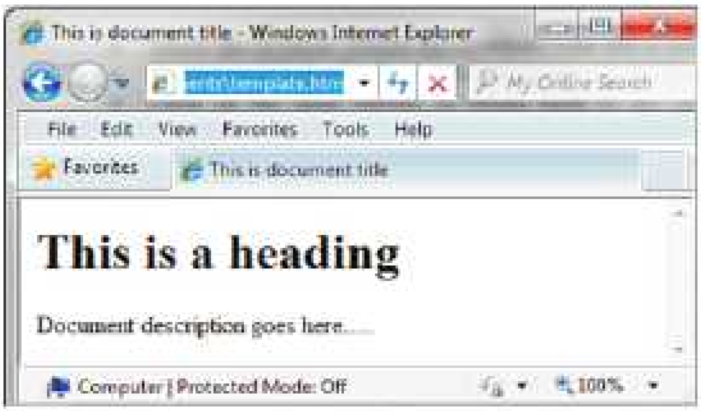
- Open your web browser (Internet Explorer)
- Click on File>Open to see the Open dialog box.


- Enter the complete path of the HTML document file (name and location) here.
- If you do not know the path, click on the Browse button.
- The Windows Internet Explorer dialog box opens.

- Select the location (folder/drive) where your file is located.
- Select the HTML file that you want to open in the browser.
- Click on the open button, to return back to the open dialog box. The path of the HTML document will appear in the open field. Click on the OK button.
- The HTML file will be displayed in the web browser as a web page.




## Creating Your HTML Document

In this topic, we will cover the explanation of the layout of HTML document and some basic tags. The basic tags of HTML are as follows:

<HTML> </HTML>

<TITLE> </TITLE>

<BODY></BODY>

<HEAD> </HEAD>

These are the required tags and all HTML documents must have these basic tags. Here is the explanation of these tags.

### <HTML> </HTML>

These tags tell your browser what kind of document it is. In fact, this is the starting tag of your document, if the browser does not find <HTML> tag in the document. Notice that this pair one is open <HTML> and the other is close </HTML> tag. The document begins with the open <HTML> tag and closes with the close </HTML> tag. Between these two tags, all other tags and information are placed. Remember to close the tag, otherwise your page will not load property.

### <TITLE> </TITLE>

These tags tell the browser about the title of the page. This text is displayed in the title bar of the browser on the top. Whatever is written between these tags, is considered as the title of the page. When open <title> is encountered, the browser understands what is written next in the title until the close </TITLE> is found.

An HTML document contains two distinct parts, the head and the body. The head contains information about the document that is not displayed on the screen. The body then contains everything else that constitute the displayed part of the web page. To define both the sections, we have the tags <head> and <body>.

### <HEAD> </HEAD>

This is the heading of your document. This will tell your browser that this is the heading. Again, take care to close the tag.

### <Body> </Body>

It should be placed right after the <Head> or the <HTML> tag. All the tags related to the different items of HTML, formatting, background colour or image and font color for text, click and visited click, etc. are enclosed within this tag pair.

So the basic structure of any HTML page is:

<HTML>

<HEAD>

Header information

<TITLE>

FIRST APPLICATION

<TITLE>

</HEAD>

<BODY>

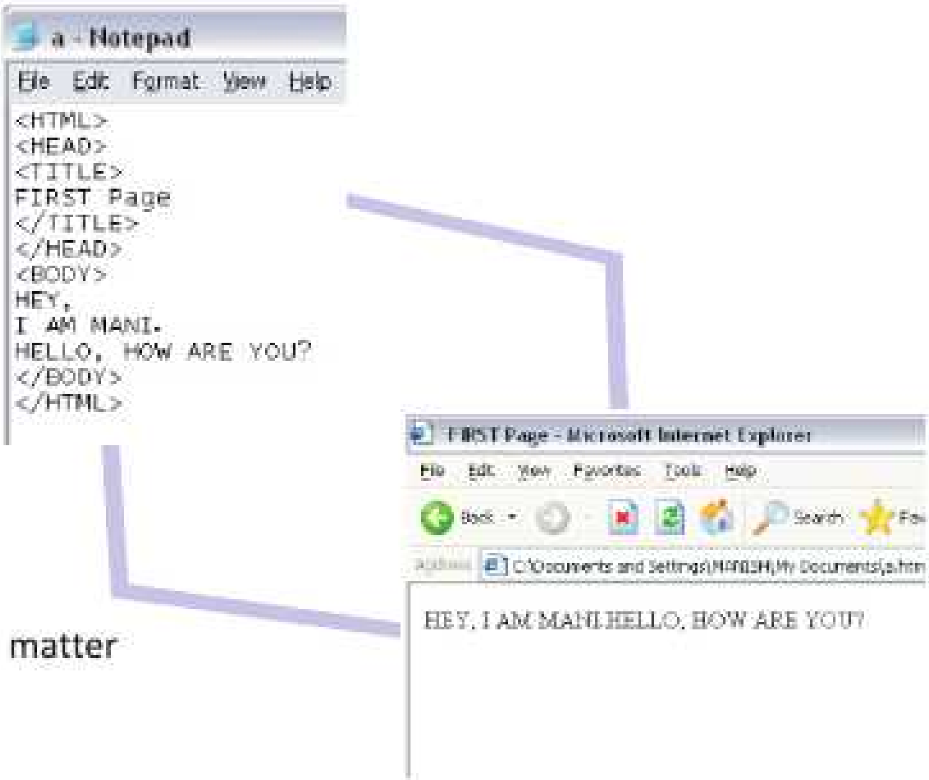
......................................

...................................... matter

......................................

</BODY>

</HTML>




Now, using all these tags, a very simple HTML page can be displayed with some title and some matter displayed in the browser window.

The code when executed via a web browser will create a page as shown in the above figure.

## COMPUTER ETIQUETTES

Do not change the settings in the computer.
Do not move any equipment from its original position.

# Let's Implement

**A Tick (✓) the correct answer.**

1. Which tag defines the contents of a web page :
   a. <HEAD> ◯ b. <BODY> ◯ c. <TITLE> ◯ d. <HTML> ◯
2. Which of the following is a Text Editor?
   a. PowerPoint ◯ b. Internet Explorer ◯
   c. Notepad ◯ d. MS-QBASIC ◯
3. Which of the following is a web browser?
   a. MS-Word ◯ b. Internet explorer ◯
   c. Notepad ◯ d. MS-QBASIC ◯
4. Which tag defines the page header?
   a. <TITLE> ◯ b. <HEAD> ◯
   c. <BODY> ◯ d. <HTML> ◯
5. Which of the following is a Container tag?
   a. <HR> ◯ b. <BR> ◯ c. <HEAD> ◯ d. All of these ◯

**B Fill in the blanks.**

1. HTML is the language of ______________ .
2. HTML tags are written within the ______________ and ______________ signs.
3. ______________ supports other characters except usual ASCII set of characters.
4. ______________ is a text editor.
5. ______________ is a program used to access the webpages.

**C Write True or False.**

1. HTML is the language of Internet. ☐
2. HTML is used for website creation. ☐
3. HTML pages are written within the "{" and "}" brackets. ☐
4. HTML document is saved using an extension .htm or .html. ☐
5. Container tag holds other attributes inside it. ☐

## D Answer the following questions.

1. Define HTML.
2. What do you understand by HTML tags?
3. Define special symbols in HTML.
4. Write the steps to make a web page.
5. Write short notes on given tags:
   a. <HTML> b. <Body>
6. Define the following Tags.
   a. <HTML> ... </HTML> b. <HEAD> ... </ HEAD>
   c. <TITLE> ... (</TITLE> d. <BODY> ... (</BODY>

## Refreshment Corner

- **Write the structure of the HTML.**

## Practical Session

Experiential Learning

- **Do the following.**

1. Type the following in Notepad.

```
<HTML>
<HEAD>
<TITLE>  MY FIRST WEB PAGE </ TITLE>
</ HEAD>
<BODY>
```

A web page is a collection of Text, Images, sound, movie, etc.

</ Body>

</HTML>

b. Save the file as "Webpage. html" in MY DOCUMENTS folder.

c. Open Internet Explorer software.

d. Open the file which you have created and saved in MY DOCUMENTS folder.

2. Write the HTML coding of the given text.

a. Twinkle twinkle, little star,
How I wonder what you are!
Up above the world so high,
Like a diamond in the sky.

b. Rain, rain, go away,
Come again another day.
Little Johny wants to play,
Rain, rain, go away.

**Competency Bases Questions**

1. Chetan wants to type the code of HTML language, for creating HTML document. What should he use?
2. Monkia wants a software program which is used to access the web pages created in HTML language. She has to view the document in HTML language. What should she use?

## RACKING YOUR BRAIN

Life Skills and Values

Your teacher asks you to create a web page. Will you create it by yourself or copy tags from the Internet? Justify you answer.

## Just Have a Discussion

Communication

Discuss the uses of HTML in the class in detail.

## Teacher's Note

The teacher should demonstrate to the students how to create the web pages.

# 6

# INTRODUCTION TO APP DEVELOPMENT

## Outlines

- Mobile App
- Types of Mobile Apps
- Downloading Apps (Installing/Deleting Apps)
- Advantages of Mobile Apps
- Categories of Apps
- Working of Apps
- Simple App Development

## Mobile App

These days most of the people use smart phones, tablets, etc. because of their multi-usage features. The applications that run on these devices are known as mobile apps. Some of the apps can come preloaded on the device, while others can be downloaded from the app store or the internet. Some may require internet connection, i.e., are online apps, while some can even work in an offline mode. For example, Gmail, WhatsApp, Instagram, Facebook, etc.

## Advantages of Mobile Apps

Mobile Apps provide the users with various features and advantages which are responsible for their increased popularity among the users. Some of the advantages of these mobile apps are as listed below:

1. **Multipurpose :** Mobile apps can have access to various device apps such as camera, contacts, microphone, etc. These, in addition to their specific features, provide a better experience for the users and also serve as a multipurpose platform.
2. **Sharing Features :** Mobile apps provide better and faster sharing of information. Data such as document files, images, clips, etc. stored on a device, can be shared instantly via mails, WhatsApp, Facebook, etc.
3. **Personalisation :** It is one of the important features of mobile apps. It takes into consideration the user's choices, likings, behaviours, culture, etc., hence providing a userspecific personalised experience.

4. **Safe and Secure :** Mobile apps enable the users to have a safe access for modifying and sharing of data. Thus, the apps ensure a high level of security beyond the built-in security system of the mobile.
5. **App Notifications :** The push notifications play an important role in communicating with the users. Various updates and promotional notifications can be provided to the users through push notifications.
6. **Can be Accessed Offline :** Apps provide both online and offline access to the users. Some features such as making payments, banking, uploading or downloading data may require an internet access, but other features can be made available to the users in offline mode.

## Types of Mobile Apps

Mobile apps can be classified into three types. They are as follows:

1. Native App 2. Web App or Online App 3. Hybrid App

### Native App

Native app is developed for a particular platform or an operating system. These apps are coded in specific programing language for a specific platform, such as Java for Android operating system and C and C++ for iOS operating system. They directly interact with the device hardware and hence can access various features such as cameras, address book, Bluetooth, microphone, etc. These apps can be downloaded through Play Store on the user's devices. Some of these apps can be used without internet connection such as calendar, camera, contacts, calculator and gallery, while some may require an active internet connection. Since these apps are platform dependent, they are faster and more reliable in terms of their performance.

**Info Zone**

Active Internet connection means that the system is currently connected to internet to access that information by using that browser.

### Web App or Online App

The web applications or web apps are the programs which are stored on a remote server and are delivered to the user's device with the help of the Internet browser interface. These are accessed through a web browser on the user's mobile phone. These applications are developed on HTML, CSS and JavaScript.

Since these apps are browser based, they can work effectively on devices having different platforms or operating systems. So, these are not developed for a specific platform or operating system, i.e., are platform independent. These apps can also be accessed on computers and laptops. But these webpages have a different look when used on them.

For example, shopping carts, online forms, video and photo editing, email programs (Gmail) social networking sites, etc.

Here are some examples of web apps (online apps) with their addresses:

- **Facebook :** http:// www.facebook.com
- **Flipkart :** http:// www.flipkart.com
- **OLX :** http:// www.olx.com

### Hybrid App

The combination of the native apps and the web apps is referred to as the hybrid Apps. Like native apps, these can be installed from the app store, can access device features and can also be accessed offline, but their internal working is like that of a web app. These are coded via Java Script, HTML and CSS. Hybrid apps provide an easy approach together with sophisticated support, for the three major mobile platforms, viz. iOS, Android and Windows Phone. Some of these hybrid apps are discussed as below:

**Gmail :** Gmail is a popular mailing system among users. It has been using HTML for internet mailing services for a long time and can be installed from the app store. It enables the users to access the documents offline, however it needs internet connection to provide the user with fresh mails. Hence, it is classified as a hybrid app.

**Baskin-Robbins :** This application uses HTML5 to provide the best offline access to the users on various platforms. The network connection is required only for communicating with the ice-cream stores.

**Differences between Web App and Native App**

| Web App | Native App |
|---|---|
| 1. The web app runs on the browser.<br>2. A web application needs active internet mode.<br>3. The web applications are independent of environments and can be adopted to all the devices.<br>4. The web apps face the threat of hacking.<br>5. It can update itself.<br>6. It does not need to be downloaded. | 1. The native app runs on the device itself.<br>2. The native app can work with/without the internet.<br>3. The native apps once made for a particular environment can't be used for other environments.<br>4. The native applications are more secured.<br>5. It needs to be constantly updated by the user.<br>6. It can either be already installed or downloaded on the device. |

**Website and Web App:** A website is a compilation of interlinked web pages which are present under a single domain name. It contains the information about a particular subject published by a single person or an organisation.

The websites are designed using HTML, CSS or JavaScript. Some of the websites are composed of static webpages, *i.e.*, the webpages that are used to provide the information only. While today, some have dynamic webpages, *i.e.*, these allow the user to communicate with the server. Each website uses a URL (Universal Resource Locator) or web address to access its web pages.

Just like the website, web applications are also developed with the help of HTML, CSS and JavaScript. However, these apps are interactive websites. The user gets involved as an integral part of the web application. These web apps are dynamic and ever changing since they rely on the audience/viewers' interaction.

**Web App is Different from a Website**

In general, a website is just informational with a set of pages starting with its home page. Whereas, a web app is a website which is designed stylishly and responds well, when viewed on a smart phone. There are many such web apps which are responsive and have a great deal of interactivity. These web apps necessarily load in browsers like the Chrome, Safari or Firefox. These apps don't take up any memory or storage on the user's device. They function as an app and don't need to be downloaded like the mobile apps.

There are a few online applications, viz. Flipkart, OLX which can be viewed as a Web App as well as website. It provides a different look when viewed as a Web App and a Website.

## Categories of Apps

As discussed earlier, there are numerous apps available catering to user's needs. These apps can be categorised as discussed below:

### Gaming Apps

These are one of the most popular apps among users. It has been found that people spend more time on games, irrespective of their age group. Mobile gaming has always been entertaining and thrilling for users. Today, many games with their impressive graphics, audio and visual treats, provide a real-life gaming experience to its users. This is the reason as to why the app developers invest more time and resources into creating new games and mobile versions of well-known stationary games.

For example, Clash of Clans, Candy Crush Saga, Angry Birds Go, Solitaire, etc.

### Business Apps

Apps have also penetrated the business domain of the world. Many tasks involved in smooth functioning of a business can be carried out using apps such as billing, accounting, booking, sending mails, tracking work progress, etc. Since, these increase efficiency and productivity of a person or an organisation, these are also referred to as productivity apps. These apps vary depending upon the needs of the business or an individual.

For example, Adobe Acrobat Reader, Walkie-Talkie Messenger, Facebook Pages Manager, Point of Sale (POS) machine, etc.

### Educational Apps

Apps can also be developed to help an individual learn and grow. These are educational apps that enable the user to access informative content in the form of videos, visuals, games, pictures, etc. These apps serve as interactive and innovative learning tools and have gained much popularity among students, teachers and parents.

For example,

- **Duolingo :** Learn Languages for Free
- **Photomath :** Learn Mathematical solution
- **Quizlet :** App to Study and Learn
- **Lumosity :** Brain Training
- **Coursera :** Take general and specialised courses from various universities around the world

### Entertainment Apps

While working on mobile most of the time, we are busy with messaging, chatting, searching events, watching videos online, posting photos and so on. The Apps used to carry out these tasks are referred to as the communication apps. Apps also serve as a companion in our leisure time. It is quite easy now a days to entertain ourselves with our mobile. Such apps that are developed for entertainment purposes, are known as entertainment apps. The user can catch up on latest movies, trends, watch TV shows, listen to songs, make videos and much more by these apps.

For example, Netflix, Dubsmash, Talking Tom Cat, Amazon Prime Video, etc.

### Travel Apps

Now a days, travelling to a new place has become a lot easier and comfortable, thanks to our smart phones. Our smart phones have also taken on the role of our travel guides. They help us in booking trains, flights, accommodations, explore activities and places to visit at our destination, inform us about weather conditions, provide us with reviews and ratings and much more. They offer us a customised experience where a user can explore and plan his journey according to his desire.

For example, Google Maps is used for navigation; sky scanner, ixigo, IRCTC Mobile apps, Kiwi, etc. for booking train and flights; trivago, booking.com, OYO, Airbnb for accomodation bookings.

### Commonly used Mobile Apps

Some of the commonly used mobile apps are discussed below:

**Google Maps :** This app helps in navigation. When you open Google Maps, it will show your current location on your device. On setting the destination, it will show various routes (by default, the shortest route is selected) and their estimated time arrivals (ETAs). By selecting the desired route, it will start directing you to reach the location. It also guides the user by using voice commands. It constantly updates the route and alerts the user of any congestion/traffic.

**Facebook :** It is one the most famous social networking sites. It is a platform where the users can share their thoughts, opinions, messages, audios and videos. This shared content can be made available to the public or can be shared privately with a group of friends that the user chooses. It is a user-friendly app. Along with a close connection with friends and family, it has now become a platform to reach a much wider audience where people market their products and also create awareness.

**WhatsApp :** It is another social networking app which allows the users to communicate with each other instantly. Text messages, audios, videos, images, documents, links, etc. can be shared among the users. It can be accessed through a mobile app and a computer as long as the user's mobile phone is connected to the computer via internet connection. This is known as WhatsApp Web. It also provides instant voice and video calling throughout the globe.

**Skype :** Skype was developed by Microsoft Inc. It is a communication application that allows the user to interact through voice calls and video chat by using the mobile devices, tablets, laptops and desktop computers. It is also used for its following features:

- It enables one-to-one, group video calling and conferencing among people.
- It facilitates screen sharing.
- It allows the users to converse in different languages as with Skype Translate, *i.e.*, it translates each side of the conversation in real-time.

With all these features, it has emerged as an important app in business and educational institutes.

Nowadays, various other such apps have also emerged, such as Zoom, Google Meet, etc.

**YouTube :** The YouTube is one of the most popular mobile apps, which allows video sharing among people. Previously, viewing YouTube videos on a personal computer required the Adobe Flash Player plug-in to be installed on the browser. But now, with upgraded web browsers (HTML5), you can view the videos without Adobe Flash Player.

You can upload and watch the videos instantly over internet connection on an app or on a web browser. It allows the user to download and save videos enabling offline access. You can also post comments and share them instantly over other platforms.

**Google+ :** The Google+ (pronounced as Google Plus), G+ or G-Plus is an internet based social networking website, owned and operated by the Google Inc. To join Google+, you need to have a Google account. Once you activate Google+, you can now share messages, link with everyone or with only those within designated circles which is a concept similar to 'groups' on Facebook. It also provides a multi-person instant text messaging and video chatting services(called Hangouts) for the users.

**Twitter :** The Twitter is a social networking service which enables the users to communicate as among themselves with the help of short messages called tweets. It is a free networking site that allows a registered user to post, like and retweet tweets, while an unregistered user can only read the tweets. It can be accessed online through various devices like mobiles, computers, tablets, etc.

**Instagram :** It is a social networking service owned by Facebook Inc. It is a photo and video sharing networking app. It can be accessed via an Instagram or a facebook account. Through this account, the user can connect with other networking sites as well. It provides the user with offline access. However, to upload content and view update posts, internet connection is required.

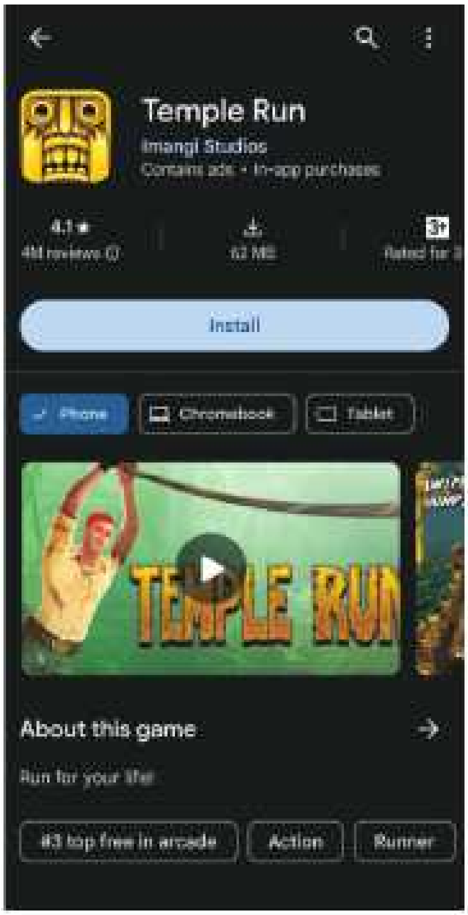
**Some Inventors of Mobile Apps**

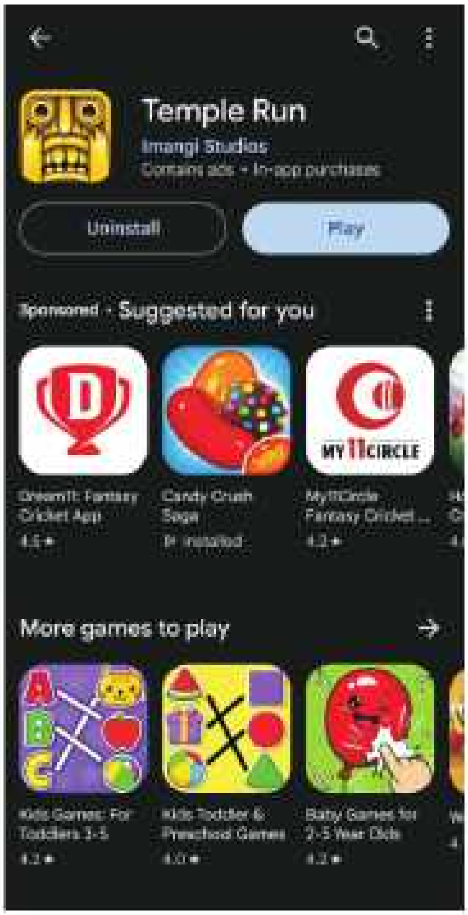
| Mobile Apps | Developers/Inventors |
|---|---|
| Google Maps | Incorporated by Google |
| Facebook | Mark Elliot Zuckerberg |
| WhatsApp | Brian Acton and Jan Koum |
| Twitter | Jack Dorsey, Noah Glass, Bizstone and Evan Williams |
| YouTube | Steve Chen, Chad Hurley, Jawed Karim |
| WeChat | Incorporated by Tencent Holdings Ltd. |
| Skype | Niklas Zennstrom, James Fraiis |
| Instagram | Kevin Systrom, Mike Krieger |

## Downloading Apps (Installing/Deleting Apps)

You can always download the apps from the respective app store. The apps are either free or need to be purchased. If you have an Android device, it comes pre-installed with Play Store. Tap the Play Store icon.

The Google Play window appears as shown in the figure. You can use the search feature to search for a specific application. Let us say, search for games, a list of games appears. To download a game (e.g., Temple Run), tap it and then tap the INSTALL button as shown in the figure. To delete an app, go to Settings and then the App. The App information screen appears. Tap UNINSTALL.




## Working of Apps

Apps work on Client-Server model. Here, the application program or the browser on the device is the client. The server is the program that manages all processes and stores all data. The client sends the request to the server using HTTP protocol. The server sends the response back to the client using HTTP protocol. When the two programs are communicating, they have to follow some rules called protocols.

Sends request using http

Mobile device runs the application → ← Program that manages data and processes

Sends response using http

## Simple App Development

App Development (or Application Development) refers to the process of developing app that can run on desired mobile platforms. You can create an application for various purposes such as education, business, result tabulation, project work, etc. The apps are developed using programming language, such as C, C++ or Java. Even if, you are not well-versed with these languages, you can create apps using free app development software available on the internet. For example, MIT App Inventor, appypie, nativ, kinetise, etc.

Here, we will learn creating apps using MIT App Inventor.

### MIT App Inventor

App Inventor is a free app development tool, provided by Google. With the help of App Inventor, we can create apps for smart phones. The working of App Inventor is similar to Scratch programming language, in which we drag and drop blocks for making an app. App Inventor has two basic views:

- **The Design View :** This view contains all the components required to design an application.
- **The Block Editor View :** This is the place where we code the components added in the Design Window and instruct them the actions to be performed by them.

### COMPUTER ETIQUETTES

The lab should be kept clean and tidy at all times.

## Let's Implement

**A** Tick (✓) the correct answer.

1. Mobile apps can be classified into ______________ types.

   a. two ◯ b. three ◯ c. four ◯ d. five ◯

2. Gmail is a ______________ app.

   a. native ◯ b. web ◯ c. hybrid ◯ d. None of these ◯

3. It is not an educational app.

   a. Duolingo ◯ b. Lumosity ◯ c. Photomath ◯ d. Temple Run ◯

4. It is a photo and video sharing networking app.

   a. Gmail ◯ b. Instagram ◯ c. Both a and b ◯ d. None of these ◯

5. Which of the following programming languages is not used in developing mobile apps?

a. Java ◯ b. C++ ◯ c. C ◯ d. HTML ◯

## B Fill in the blanks.

1. Twitter allows short messaging called ______________ .
2. The ______________ software allows free app development.
3. ______________ app is developed for a particular platform or an operating system.
4. ______________ apps enable the users to have a safe access for modifying and sharing of data.
5. ______________ is a social networking service owned by Facebook Inc.

## C Write **True** or **False**.

1. Google Maps is incorporated by Google.
2. The client is the program that manages all processes and stores all data.
3. The server sends the request to the server using HTTP protocol.
4. App Development refers to the process of developing apps that can run on desired mobile platforms.
5. App Inventor is a free app development tool, provided by Google.

## D Answer the following questions.

1. What are the advantages of Mobile apps? Explain.
2. How are the different Mobile apps categorized? Give two examples of each.
3. What is the difference between Web app and Native app?
4. Write a short note on:

   a. Facebook b. Instagram c. WhatsApp d. Twitter
5. How is the Google Maps app useful in searching the destination for journey?

## Refreshment Corner

### Name the following.

1. Three types into which Mobile Apps are classified:

   a. ______________________________

   b. ______________________________

   c. ______________________________

2. Two examples of Native apps:

a. ______________________________

b. ______________________________

3. Two examples of Hybrid apps:

a. ______________________________

b. ______________________________

4. Two softwares used for free App development:

a. ______________________________

b. ______________________________

## Practical Session

Experiential Learning

- You have to download some apps on your mobile phone.

**Do it using the given steps :**

- Tap the Play Store icon to open the Play Store.
- Now use the search feature to search for a specific app.
- A list of apps appears.
- To download, tap the app and then tap the INSTALL button.
- The app will be downloaded.
- Do the same to download all apps.

## RACKING YOUR BRAIN

Life Skills and Values

Your friend wants to use some Apps, but he/she is unable to use them. What will you do? Justify your answer.

## Just Have a Discussion

Communication

Discuss the different categories of apps in the class in detail.

## Teacher's Note

- Ask the students to share their experiences (for those of any other member of their family) of using mobile apps.
- Demonstrate the working of some apps on a mobile phone or on a computer connected to a projector.
- Free app development softwares and sites may also be demonstrated in the classroom using a computer and a projector.

7

# CYBERCRIME THREATS AND SECURITY

## Outlines

- Internet
- Disadvantages of Internet
- Unethical Practices
- Safety Measures While Using Computer and Internet
- Advantages of Internet
- Computing Ethics
- Cybercrime
- Digital Footprint

## Internet

Internet is one of the largest information databases. It has become a vital part of our life. Internet stands for International Network of computers. It is a global system of inter-connected computer networks that enables the users to share the information and various resources with each other. It uses the common communication standards and interfaces to provide the physical backbone for many interesting applications.

Tasks that were usually performed through personal interaction, such as banking, shopping or communication, can now be done online. We can share the data and information across the globe instantly. Indeed! Internet has transformed the world into a smaller and a convenient place to live in. Today, the buzzword is only INTERNET. The network of Internet is getting wider with each passing day. To analyse how fast the use of Internet is growing, pick any newspaper or magazine, and carefully observe the advertisements given therein. You would realise that almost every enterprise has its website listed in addition to its address and telephone numbers. Every organisation is making its presence felt on the Internet.

Internet is a boon to the world. Almost every aspect of our life is touched by the Internet. Interestingly, Internet provides 24x7 services.

## Advantages of Internet

### Education

Internet has changed the traditional learning system. It is widely used for educational purposes by scholars, students and teachers, to learn anytime and anywhere. It has shown dramatic impact on higher education as more universities offer online courses. IGNOU is one of the universities which offers this facility. Internet is a vast storehouse of information and is also used to publish papers and articles.

### Convenient Mode of Communication

Internet is a convenient and economical mode of communication. Users can communicate with each other using various services available on Internet, like e-mail, chat, video conferencing, instant messenger, etc.

### Business

The Internet has brought new opportunities for businesses to offer goods and services online, which has proved to be beneficial. Meetings can be conducted inexpensively through the video conferencing facility, without physically visiting the place. Financial transactions can be made online from the office or home.

### E-commerce

It is the concept used for any type of commercial or business deal that involves the transfer of information across the globe by using the Internet. Nowadays, many companies have their own portals/websites, which are used for E-commerce.

E-commerce industry has become one of the biggest industries in the world. We can now purchase every possible goods and get services from different shopping portals around the world. It also includes booking tickets for bus, flight, train and hotel rooms.

### Media and Entertainment

Internet adds value to media and entertainment. Downloading games, videos, songs, visiting chat rooms or surfing the web, are some of the uses, which attract people towards the Internet. Internet also provides the facility to read different newspapers, magazines and books online.

### Social Networking

It is the latest way to communicate. It offers a platform, where an individual can connect with friends and strangers on the basis of shared interests or views. Some examples of social networking sites are: Facebook, Twitter, LinkedIn, My Space, etc.

### Forum

In Internet, Forum is an online discussion site where like-minded people can hold the conversations in the form of posted messages.

### Health and Fitness

Internet is now playing a healthy role in providing the extensive information on health and fitness. Using various medical sites, one can read more about various diseases, their cures and precautionary measures.

## Disadvantages of Internet

Internet has several advantages for every individual and it is extremely popular in this modern age. However, it is a tool, which must be handled with caution, as it's misuse can be hazardous. Some of the threats of Internet are as follows:

### Virus Threat

Computers that are connected to the Internet are more prone to virus attacks, which can result in the crashing of system, data loss, and hardware malfunctioning.

### Spamming

It refers to unwanted e-mails, that are of commercial nature, sent indiscriminately to multiple mailing lists, individuals or newsgroups. It unnecessarily uses the system's resources and can also be used to spread computer virus or other malicious software.

### Theft of Personal Information

Internet is the main source of piracy, plagiarism and other evil issues, relating to the privacy of information of an individual or an organisation. One should be careful enough while handling one's personal information, such as name, address, credit card number, etc., as they can be accessed by hackers or crackers for personal monetary gains. Such elements can also disturb system applications by spreading virus and changing program logics and outputs.

### Pornography

This is a major issue posed by the Internet. There are many pornographic sites on the Internet that can be found easily and be hazardous to the mental health of children.

### Cyber Terrorism

It causes a serious threat to the world nowadays. Modern terrorist organisations perform target attacks on power plants, banks, commercial areas or buildings of national importance. This can be conducted remotely with the help of a mobile phone and the Internet, which are less expensive than traditional terrorist methods.

### Time Wastage

Although Internet provides various services to its users, yet people generally waste time and energy in surfing the net. This can result in the loss of our personal interaction with people and it can also affect both of our socialising skills and health.

## Computing Ethics

Ethics is a set of moral principles that govern the behaviour of a group or an individual. Similarly, Computing Ethics is a set of procedures, moral principles, and ethical practices that regulate the use of computer. It focuses on ethical implementation and use of computing resources and includes new issues that are raised by new technologies. It basically aims at encouraging IT users to be responsible in order to utilise the technology tools judiciously, respect views of others, and acknowledge the rights and properties of people on the Internet.

Some of the common ethical guidelines which should be followed while using a computer, are given below:

- Do not use computer technology to cause disruption or interference in other users' work.
- Do not spy on another person's computer data.
- Do not use computer technology to steal the information.
- Do not contribute to the spread of wrong information using the computer technology.
- Avoid buying pirated software. Pay for software unless it is free.
- Do not use someone else's computer resources without an authorisation.
- It is wrong to claim ownership on a work, which is the output of someone else's intellect.
- Before developing a software, think about the social impact it can have.
- Be respectful and courteous with the fellow members while communicating on the Internet.

## Unethical Practices

In the coming section, we will discuss some of the common unethical practices which are prevalent in the society:

### Plagiarism

Plagiarism is the usage or imitation of the language and thoughts of another person projected as one's own original work. It is considered a crime or fraudulent act.

- The Merriam-Webster Dictionary describes 'plagiarism' as follows: To steal and pass off (the ideas or words of another) as one's own.
- To use (another's production), without crediting the source.
- To commit literary theft.
- To present as new and original, an idea or product derived from an existing source.

International Copyright Laws also state that, "expression of original ideas is considered as Intellectual Property and is protected by copyright laws", just like original inventions. Almost all forms of expressions fall under the copyright protection as long as they are recorded in some manner (such as a book or a computer file).

Plagiarism has become very easy with technological advances, like Web Searching and Copy/Paste options. Most critics have linked carelessness, laziness, lack of interest and arrogance, as the major reasons of Plagiarism.

**Steps to Prevent Plagiarism :** The simplest way to prevent plagiarism is the 'Citation'. Acknowledging the original writer and the source from where the material has been taken, is called 'Citation'. It includes the following steps:

- All sources are neither accurate, nor they validate the information as accurate. Therefore, one must specify the source, which informs the readers regarding the basis of one's ideas and the extent of one's research. Citation gives strength to your resource.
- To avoid plagiarism, one can rephrase the material.
- Use quotation marks around all the quoted words of another person to avoid plagiarism.
- In the education sector, students should be encouraged to present their original and innovative ideas. Even if they wish to refer to the resources, they should be made aware of the pattern of references. In this manner, plagiarism can be avoided and students could be motivated to work hard and bring out their originality.

### Cyberbullying

It is an act of harming, harassing or targeting a person by another person using Internet, in a deliberate manner. This includes insulting remarks and threatening messages sent by email, spreading rumours about the person either by e-mails or social networking sites, posting embarrassing photos and videos to hurt the person, derogatory remarks against gender, race, religion or nationality.

**Steps to Prevent Cyberbullying :** The steps to prevent cyberbullying are as follows:

- Do not disclose your identity to unknown persons.
- Safeguard your password and all private information from inquisitive peers. Do not give bullies any opportunity to misuse the information on the social media platform.
- Search the whereabouts of the person you meet online and know about him fully before extending your handshake.
- Restrict your online profile to be checked by the trusted friends only.
- If you are cyberbullied by someone, do not remain silent but take an action against cyberbullying. Do not be repulsive in your language, rather block the communication with the cyberbully and share this problem with your parents.

## Phishing

It is an act of sending an e-mail to a user misleading him to believe that it is from a trusted person or organisation. The user is asked to visit a website in which he is supposed to update or validate his personal details, such as user name, password, credit card details, etc. In this way, the operators of the fake websites steal your identity and commit crimes in your name. This could damage your reputation and cause heavy financial losses.

**Steps to Prevent Phishing :** The steps to prevent phishing are as follows:

- Identify suspected phishing e-mails that come from unrecognised senders. Guard against scam.
- Do not click on links, download files or open attachments in e-mails, from unknown senders.
- Beware of links in e-mails that ask for the personal details.
- Enhance the security of your computer by protecting it with a firewall, spam filters, antivirus and anti-spyware software.
- Do not divulge personal information over the phone unless you initiate the call. Enter your sensitive data in secured websites only. The URL of a secured website starts with https//: and your browser displays a green coloured icon of the closed lock.
- Check your online accounts and bank statements regularly to ensure that no unauthorised transactions are made.

## Hacking

Hacking refers to an illegal intrusion into a computer system or network. Hackers are highly technical people who secretly break into computers to steal important data or sometimes destroy it. Hackers not only steal important data but also hack applications and websites to change program logics. Hackers usually tamper with data for unethical purposes, just for an obsession to break the system securities. Although hacking is done without any consent from the user, yet it is not always destructive. Hacking which is done for a good cause, such as national security, etc., is known as Ethical hacking.

There is an equivalent term to hacking, *i.e.*, cracking. Crackers are technical people, who are experts in breaking into systems, to steal important data, such as financial details or passwords, etc. Sometimes they use key loggers for this purpose. Crackers also cause harm to computers by destroying data. Apart from this, they disturb applications by spreading malwares, changing program logics and outputs.

**Steps to Prevent Hacking :** The steps to prevent hacking are as follows:

- Keep your passwords secret and change them periodically.
- Update your operating systems and other software like antivirus, frequently.

- Avoid using open Wi-Fi as it makes easy for hackers to steal the information and download inappropriate content.
- Always make use of secure websites by looking at the green lock sign or https on the address bar. This ensures that the sensitive data, which is passed from your device, is encrypted, making it difficult for the hacker to decipher the information.
- Always check permissions and authenticity on the apps before installing them. You should also ensure that the apps are not accessing unnecessary information. For example, a drawing app should not have access to your contact list or your network information.
- Never store your credit card information on a website.

### Spamming

Spam are unwanted bulk e-mails that come from strange sources. Spam is generally sent in large numbers for commercial advertising. In spamming, millions of copies of the same message are sent to e-mail users worldwide. Spam is a serious security concern as it can be used to deliver Trojan Horses, Viruses, Worms, Spyware, and organise targeted phishing attacks.

**Smart Tip**

Sometimes spam attackers keep sending bulk mails till the mail server gets full. This practice is known as Mail Bombing.

**How to Identify Spam?**

- Messages that do not include your e-mail address in the TO: or CC: fields are common forms of Spam.
- Some Spam can contain offensive language or links to websites with inappropriate content.

**Steps to Prevent Spamming :** The steps to prevent spamming are as follows:

- Install Spam filtering/blocking software.
- If you suspect that an e-mail is Spam; do not respond, just delete it.
- Consider disabling the e-mail client's preview pane and reading e-mails in plain text.
- Reject all instant messages from persons who are not in your contact list.
- Do not click on the URL links within Instant Messenger, unless it is received from a known source.
- Keep the software and security patches up-to-date.

In this digital era, there has been a significant increase in the use of technology. As the technology is progressing, so are the privacy parameters diminishing. Internet is commonly used for various purposes, such as for personal communication, accessing information on any topic, and for financial transactions, etc. The use of Internet can affect the privacy rights and personal data of a person.

Privacy is one of the major concerns in one's life. The right to privacy may be defined as the claim of individuals, groups or institutions, to determine when, how and to what extent, the information about them could be communicated to others. Privacy issues come into picture, when personal information is collected, used, and shared, without the consent of the authorized person/business concerned. So, certain preventive measures should be taken to avoid infringement of privacy.

**Steps to Protect Privacy :** The steps to protect privacy are as follows:

- Be careful while sharing your personal profile on the social networking sites.
- Never share your log-in details with anyone.
- Delete the sensitive data when not in use.
- Never click on a link sent to you via text or e-mail from a stranger.
- Protect your online privacy by disabling the cookies in the web browser being used. A Cookie is a small text file stored on your computer's browser directory that collects the information of your online activity and reports back to the host.
- Install security software, like antivirus, anti-spyware or firewall software, to protect your system against intrusions and infections.
- Lock up your system when not in use.

## Software Piracy

When software is copied and distributed illegally, it is called Software Piracy. A license or copyright is required to copy the genuine programs of someone else. You encourage software piracy when:

- You purchase a single licensed copy of the software and load it onto several computers contrary to the license's terms. This is called Softlifting.
- You make unauthorised copies of the copyrighted software and distribute them.
- You install unauthorised copies of software on the hard disks of personal computers.
- You make unauthorised selling of software for temporary use or sell or buy the software CDS/DVDs on rent.

**Steps to Prevent Software Piracy :** The steps to prevent software piracy are as follows:

- Buy the licensed copy of software. Follow the terms and conditions of the license for each software product you purchase. Never make copies and circulate them.

- Purchase software CDs from a reputed seller to ensure that it is genuine. Avoid using illegal CDs given by your friends to listen to your favorite music or watch movies.
- If software is available online, download it directly from the manufacturer's website.
- Register your software to avoid Softlifting.

### Intellectual Property Rights

Intellectual Property is a term that refers to the legal property rights of a person over his/her creations of mind, both artistic and commercial. Under Intellectual Property Law, owners are granted certain exclusive rights over the use of their creations, such as musical, literary and artistic works for a certain period of time. These rights ensure that the creator's hard work is safe and protected from unauthorised copying and piracy.

### Steps to Protect Intellectual Property Rights

**Patent Your Inventions :** A patent is a right relating to the new inventions that grant the inventor the sole right to make, use, and sell that invention, for a specific period of time. The validity period for patent in India is 20 years from the date of filing of patent application.

**Copyright Your Art and Publications :** A copyright is a legal right granted by law to the creator for his original work. The rights provided under Copyright law prohibit others to create a copy of the work, translation of the work and distribution of the work to the public. Legal action can be taken if someone copies the work without the permission of the author or the organisation. It covers artistic and literary creations, such as novels, plays, poems, musical compositions and architectural designs. The copyright of a work lasts for more than seventy years, even after the death of its author.

**Register Your Trademarks :** This protects your brand names, logos, packaging and the other distinctive designs, that identify and distinguish the products of one manufacturer or seller from that of other's. The validity period for a trademark is ten years and can be renewed.

## Cybercrime

A Cybercrime is any illegal activity done through Internet, *e.g.*, identity theft; where somebody can steal your e-mail id or password and use it to send fake e-mails to people containing false information about the product or winning a lottery, etc.

Then there are credit card account thefts, Internet frauds, like, ordering goods in your name, extracting the mobile phone contacts, etc., forgery *i.e.*, imitating documents, currency and objects of other people with bad intentions, harassing others and mischief mongering, by sending threatening messages, all of which come under the jurisdiction of Indian Penal Code (IPC). Currently BNS (Bharatiya Nyaya Sanhita)

Cybercrimes can be divided into three main categories:

- Crime against an Individual person

- Crime against property
- Crime against an organisation/society

## Safety Measures While Using Computer and Internet

Let us look at the various safety measures which should be undertaken while using computer or Internet:

- Minors should always surf Internet under the supervision of their parents.
- Parents should decide and suggest age appropriate websites to their children.
- To minimise the chances of the attack of hackers and crackers, one should use strong passwords. A strong password consists of at least six characters (and the more the characters, the stronger the password) that are a combination of letters, numbers and symbols (@, #, $, %, etc.), if allowed.
- To avoid losing data privacy while transmitting the data from one site to another, encryption can be implemented. Encryption is the process of transforming data into an unreadable code. The result is an encrypted file, which is transferred over the network. At the receiving end, it undergoes a process called Decryption, which converts the encrypted information into readable form.
- Set up your computer for automatic software and operating system updates to make your system robust.
- Taking regular data backup is the primary and the most reliable method of data protection.
- Ignore unwanted and strange e-mails and be cautious of attachments, links, and forms in e-mails that come from unknown people.
- Make use of Firewalls. Firewall is a security system that prevents unauthorised people to access your system and network. It can be either hardware or software or a combination of both. It is implemented on the gateway of a network and follows a specific set of rules defined by the user or the network administrator. Based on these rules, it controls the incoming and outgoing network traffic.

## Digital Footprint

As you know footprint is the impression left by a foot or shoe on the ground or a surface, similarly a digital footprint is the impression or information about a person that exists on the Internet because of his online activity. This is the information transmitted online like filling a registration form, e-mail attachments, online shopping, commenting on a social networking site, uploading videos, digital images, etc., all of which leave a trail of personal information available to others. This information gets logged in a database. A digital footprint is also known as digital dossier.

Digital footprints can be categorised into Active and Passive data traces. The active ones are those data traces, which the user leaves intentionally. The e-mail, chats, Facebook and blog posts are some of the ways, with which a user can create active digital footprints.

Passive data traces, connected to an individual, are generated by others or gathered through the user's online activities without his knowledge. Few activities which create passive data traces, include surfing the sites for searching the information, online shopping, etc.

### Implications of Digital Footprint

As discussed above, all the online activities leave behind the footprints or information traces. These information traces could be used positively or negatively. So, before indulging in any activity, one should think over its outcomes. Follow the guidelines given below while being online:

- Never post any silly or abusive comment.
- Tune the privacy settings of your social media account in a way that only your family or friends can see your posts and pictures.
- Do not log on to an inappropriate website.
- Do not upload any inapt picture.
- Avoid accepting the friend requests from strangers.

Be vigilant and extra cautious while being online as your carelessness or ignorance may generate a negative digital footprint, which can spoil your reputation. It can even affect your chances of gaining admission in a prestigious college or getting a job.

### COMPUTER ETIQUETTES

Never use pen drive in your computer without taking the permission of your teacher.

## Let's Implement

**A** **Tick (✓) the correct answer.**

1. The unwanted bulk e-mails that come from strange sources are called as __________ .

   a. Spam ◯ b. Trash ◯ c. Junk ◯ d. None of these ◯

2. ____________ is a small text file stored on your computer's browser directory that collects the information of your online activity and reports back to the host.

   a. Register ◯ b. Cookie ◯ c. Browser ◯ d. Junk ◯

3. A ______________ is a legal right granted by law to the creator for his original work.

a. Copyright ◯ b. Patent ◯ c. Trademark ◯ d. None of these ◯

4. ______________ is the process of transforming data into an unreadable code.

a. Encoding ◯ b. Encryption ◯ c. Decryption ◯ d. None of these ◯

5. ______________ is an act of sending an e-mail to a user misleading him to believe that it is from a trusted person or organisation.

a. Phishing ◯ b. Spamming ◯ c. Hacking ◯ d. None of these ◯

## B Fill in the blanks.

1. ______________ is a set of procedures, moral principles, and ethical practices that regulates the use of computer.
2. The simplest way to prevent plagiarism is the ______________
3. ______________ is an act of harming, harassing or targeting a person by another person using Internet, in a deliberate manner.
4. An illegal intrusion into a computer system or network is known as ______________ .
5. ______________ is a term referring to the legal property rights of a person over his/her creations of mind, both artistic and commercial.

## C Write True or False.

1. Spam are unwanted bulk e-mails that come from strange sources. ▭
2. Hacking is the usage or imitation of the language and thoughts of another person projected as one's own original work. ▭
3. You can identify the unsecured websites by looking at the green lock sign or https on the address bar. ▭
4. A digital footprint is also known as digital dossier. ▭
5. A patent grants the inventor the sole right to make, use and sell that invention, for an unlimited period. ▭

## D Answer the following.

1. What are the advantages of Internet?
2. Write down any four ethical guidelines which should be followed while using a computer.
3. Define plagiarism. What steps can be taken to prevent it?
4. How can you protect Intellectual Property Rights from getting violated?
5. Write a short note on:

a. Spamming b. Cybercrime c. Digital Footprint

d. Software Piracy e. Hacking f. Phishing

## Refreshment Corner

- **Write down th steps to prevent hacking.**

**Competency Based Questions**

1. Neha is confused in deciding whether the mail which she has received, is a genuine one or a spam. Can you help her in identifying the spam by specifying any indicative feature?
2. Ruby is surfing internet to prepare herself for the Civil Services Examination. She wants to know whether the websites which she is referring, are secured or not. Suggest her the way with which she can clear her doubt.

### RACKING YOUR BRAIN

Life Skills and Values

What will you do if you receive a spam mail? Justify your answer.

### Just Have a Discussion

Communication

Conduct a group discussion on: Hackers vs Crackers and Cyberbullying vs Cyber Trafficking.

### TEACHER'S NOTE

- Discuss the various Intellectual property rights with the students. Tell them why it is unethical to copy someone's intellectual or creative property.
- Guide the students on how to deal with cyberstalking and cyberbullying.
- Demonstrate to the students how to scan a computer using antivirus software.

# 8

# MORE ON PYTHON

## Outlines

- What are Operators?
- Precedence of Operators
- Types of Control Structures
- Types of Operators in Python
- Introduction to Control Structures
- Conditional Statements

## What are Operators?

Look at this expression: 45 + 62 – 23 * 2. Here, the numbers 45, 62, 23 and 2 are called operands, whereas the symbols +, – and * are called operators. Operators are symbols that represent arithmetic and logical operations on operands and provide a meaningful result. For performing any operation, the operator requires one or more operands. The valid combination of operators and operands make an expression which returns a computed result. Examples of some operators are +, -, *, /, % and so on. In the Python programming environment, you often need to use operators for the purpose of arithmetic or logical calculations.

## Types of Operators in Python

There are different types of operators that you can use to perform mathematical and logical operations in Python, such as arithmetic, relational and logical.

Arithmetic Operators

These operators are used for performing basic mathematical calculations, that is, addition (+), subtraction (-), multiplication (*) and division (/).

The arithmetic operators are further classified into the following:

- **Unary Operator :** These are the operators which operate only on one operand. Look at the example of unary operators as shown below:

```
IDLE Shell 3.11.5
File  Edit  Shell  Debug  Options  Window  Help
Python 3.11.5 (tags/v3.11.5:cce6ba9, Aug 24 2023, 14:38:34) [MSC v.1936 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> i = -100
>>> j = + 500
>>> print(i,j)
-100 500
>>>
                                                                    Ln: 7  Col: 0
```

- **Binary Operator :** These are the operators which operate on two operands. For example, x + y. Here, '+' is a binary operator working on x and y. Binary operators are further classified as shown below:

| Operator | Name | Description | Example (x = 20 and y = 3) | Output |
|---|---|---|---|---|
| + | Addition | Adds values on either side of the operator | x + y | 23 |
| – | Subtract | right hand operand from left hand operand | x – y | 17 |
| * | Multiply | Multiplies values on either side of the operator | x * y | 60 |
| / | Divide | Divides left hand operand by right hand operand | x / y | 6.66 |
| % | Modulus | Divides left hand operand by right hand operand and returns remainder | x % y | 2 |
| ** | Exponentiation | Performs exponential (power) calculation on operands | x**y | 8000 |

| | | | | |
|---|---|---|---|---|
| // | Floor or Integer Division | Divides and cuts the fractional part from the result | x // y | 6 |

**Program:** To show all the arithmetic operators' functions

```
*untitled*
File  Edit  Format  Run  Options  Window  Help
x=10
y=5
print ("Sum= ", x+y)
print ("Subtraction= ", x-y)
print ("Multiplication= ", x*y)
print ("Division= ", x/y)
print ("Floor Division= ", x//y)
print ("Remainder after division= ", x%y)
print ("Exponential = ", x**y)

Ln: 10  Col: 0
```

The result of the above program is:

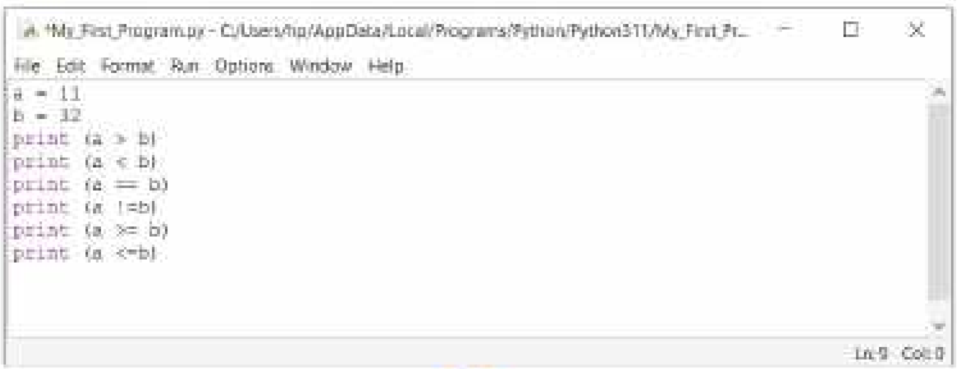
```
IDLE Shell 3.11.5
File  Edit  Shell  Debug  Options  Window  Help
Python 3.11.5 (tags/v3.11.5:cce6ba9, Aug 24 2023, 14:38:34) [MSC v.1936 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> x=10
>>> y = 5
>>> print ("Sum= ", x+y)
    Sum=  15
>>>
    ======= RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python311/f.py ======
    Sum=  15
    Subtraction=  5
    Multiplication=  50
    Division=  2.0
    Floor Division=  2
    Remainder after division=  0
    Exponential =  100000
>>>

Ln: 16  Col: 0
```

### Relational Operators

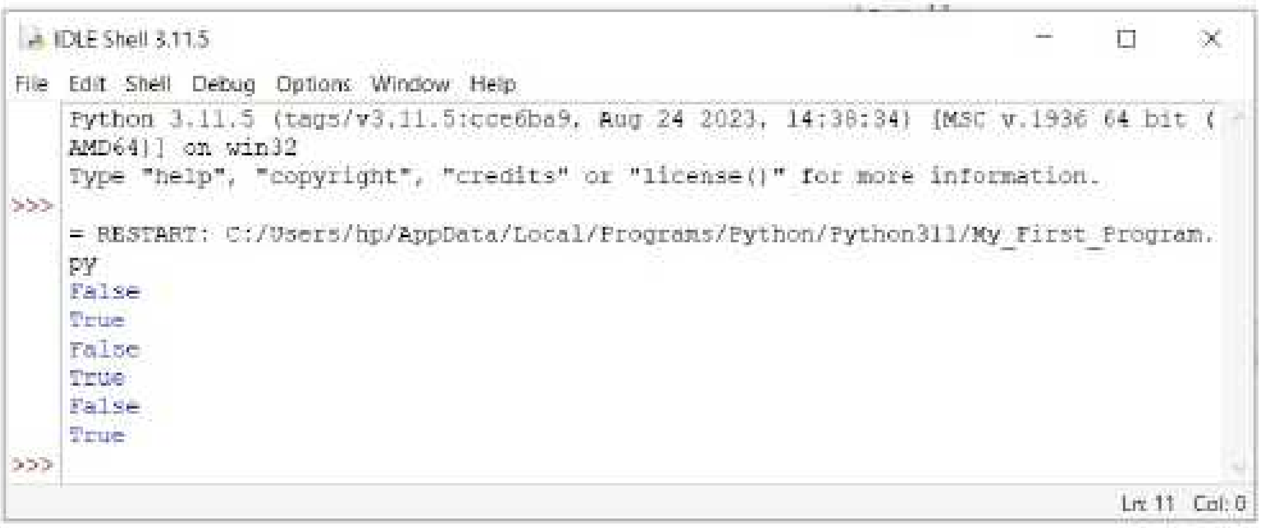
These operators are used for depicting the relationship between operands. These operators compare two values and determine the result in Boolean expression, which can be either True or False. There are six types of relational operators in Python.

| Operator | Name | Description | Example (x = 8 and y = 6) | Output |
|---|---|---|---|---|
| = = | Equal | It checks if the values of two operands are equal or not. If yes, then the condition becomes true. | x == y | False |
| != | Not equal | It checks if the values of two operands are equal or not. If the values are not equal, then the condition becomes true. | x!=y | True |
| > | Greater than | It checks if the value of left operand is greater than the value of right operand. If yes, then the condition becomes true. | x>y | True |
| < | Less than | It checks if the value of left operand is less than the value of right operand. If yes, then the condition becomes true. | X<y | False |
| <= | Greater than or equal to | It checks if the value of left operand is greater than or equal to the value of right operand. If yes, then the condition becomes true. | x >= Y | TRUE |
| <= | Less than or equal to | It checks if the value of left operand is less than or equal to the value of right operand. If yes, then the condition becomes true. | x <= y | FALSE |

**Program :** To show all the relational operators' functions



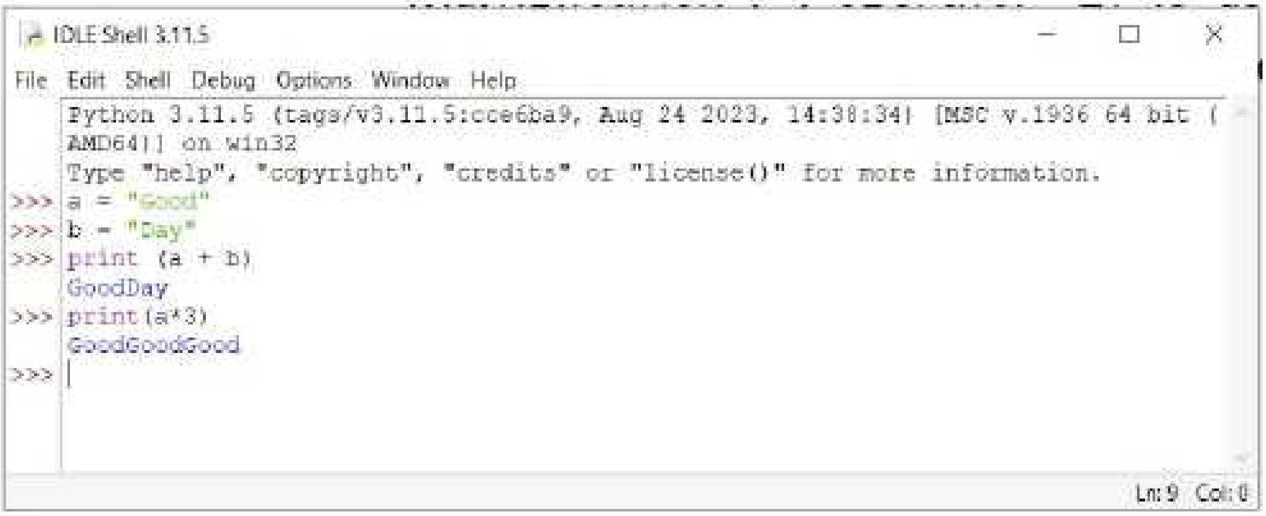
The result of the above program is:




## String Operators

If you are working with the string values, then you have to perform operations on them. Only two kinds of operations are performed on string data types. You can either join two strings or replicate a string multiple time. '+' and '*' operators are used on strings.

- **Plus (+) operator :** It is known as concatenation operator when you use it with strings. It is used for joining two or more strings.
- **Multiplication (*) operator :** It is used for replicating a given string a number of times. It is also known as replication operator.




## Precedence of Operators

Precedence of operators determines the order in which the operators are executed. The operator precedence in Python is listed in the given table. The highest precedence is at the top.

The following table shows the precedence of operators in Python:

| Operator | Name |
|---|---|
| () | Parenthesis |
| ** | Exponent |
| *, /, %, // | Multiplication, Division, Modulo, Floor Division |
| +, - | Addition, Subtraction |
| ==, !=, >, <, >=, <= | Comparison |
| =, +=, -=, *=, /=, %=, **=, //= | Assignment |
| and, or, not | Logical |

## Introduction to Control Structures

In the world of programming, you often need to perform tasks based upon a condition which is either True or False.

Consider the example of a program that prints the greatest of two numbers. If Number 1 is greater than Number 2, then it prints Number 1, else it prints Number 2. This is called a Conditional statement. Every decision involves an option between two alternatives'Yes' and 'No'.

Consider another example, a man needs to print his name 100 times. How should the program be controlled? A few lines of the program will have to be performed repeatedly for a specific number of times till the given condition is true. If the condition is false, the control comes out of the loop and the repetition stops. This is called Iteration.

Sometimes, a program may need to take decisions or repeat a block of code. For this, Python provides control structures. In a program, a control structure determines the order in which 106 statements are executed. Let us learn about different types of control structures in detail.

## Types of Control Structures

In Python, there are different types of constructs or statements which govern the flow of the control of a program. These are:

1. **Sequential Statements :** In sequence construct means the statements are being executed sequentially. This represents the default flow of statement. Every Python program begins with the first statement of program. Each statement in turn is executed. When the final statement of program is executed, the program is done. Sequence refers to the normal flow of control in a program and is the simplest one.

2. **Conditional Statements :** In programming languages, conditional statements (also known as decision-making statements) cause the program control to transfer to a specific location depending on the outcome of the conditional expression. Every decision involves a choice between the two alternatives 'Yes' and 'No'. If a conditional statement is true, then one set of statements is executed, otherwise, the other set of statements is executed.

3. **Iterative Statements :** These statements also known as looping statements, enable the execution of a set of statements to be repeated till the condition is true. As soon as the condition becomes false, the control comes out of the loop and the repetition stops.

   In this chapter, you will be studying conditional statements in detail.

## Conditional Statements

Decisions in a program are used when the program has conditional choices to execute code block. Let's take an example of traffic lights, where different colours of lights lit up at different intervals based on the condition of the road or any specific rule.

It is the prediction of conditions that occur while executing a program to specify actions. Multiple expressions get evaluated with an outcome of either TRUE or FALSE. Python provides decision-making statements to make decisions within a program for an application based on the user requirement.

Python provides various types of conditional statements:

- if statement
- if...else statement
- if...elif...else ladder

### The if Statement

The if statement is the simplest conditional statement. Here, a statement or a collection of statements within the if block is executed only if a certain condition or expression evaluated to be True. If the condition evaluates to be False, then the control of execution is passed to the next statement after the if block. The syntax of the if statement is given below:

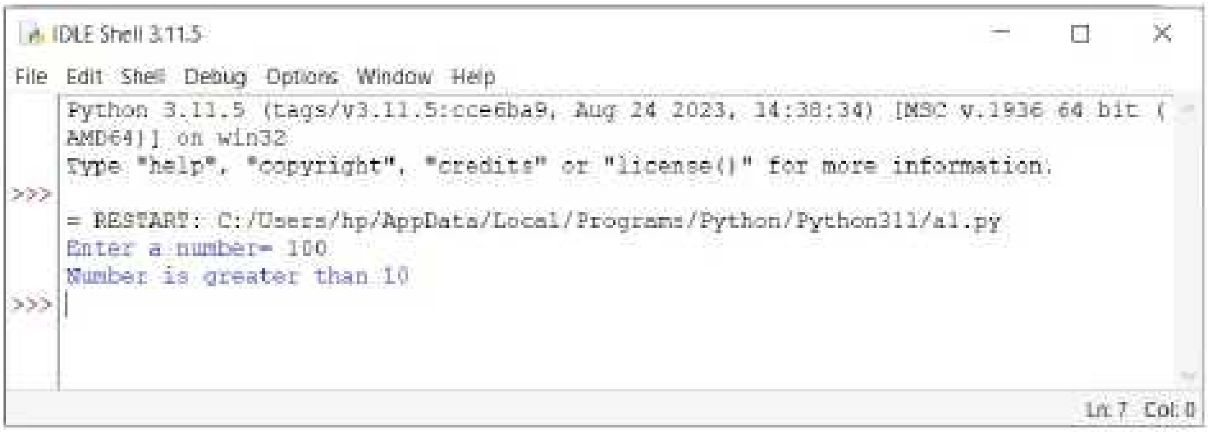
**Syntax :**

```
if (Test Expression):
    Indented statement block
# if block ends here
```

**Info Zone**

In Python, the non-zero value is interpreted as true. None and 0 are interpreted as false.

**Program :** To check whether a given number is greater than 10 or not.


## The if...else Statement

The if...else statement checks for a condition. If the condition evaluates to be True, the indented block following the if statement is executed, otherwise the indented block after the else statement is executed. The syntax of the if... else statement is given below:

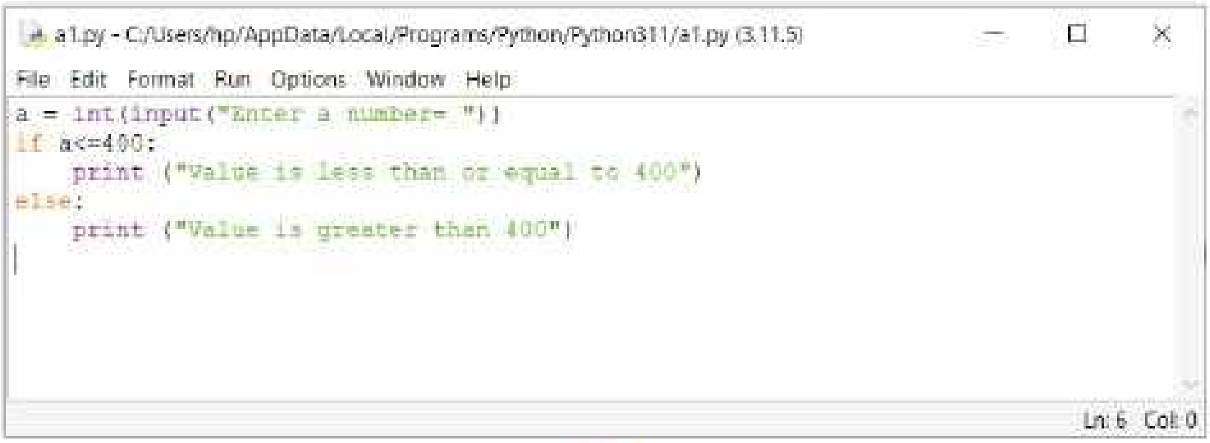
**Syntax :**

if (Test Expression) :

    Indented block

else: Indented block

**Program :** To check whether a given number is greater than or less than 400.



```
IDLE Shell 3.11.5
File  Edit  Shell  Debug  Options  Window  Help
Python 3.11.5 (tags/v3.11.5:cce6ba9, Aug 24 2023, 14:38:34) [MSC v.1936 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python311/a1.py
Enter a number= 250
Value is less than or equal to 400
>>>
====== RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python311/a1.py ======
Enter a number= 500
Value is greater than 400
>>>
```

## The if...elif..else Ladder

The if...elif...else ladder is another type of if statement. If helps us to test multiple conditions and follows a top-down approach. In this, as soon as the condition of the if evaluates to be true, the indented block associated with that 'if' is executed, and the rest of the ladder is avoided. If none of the conditions evaluates to be true, then the final else statement gets executed.

The syntax of it...elif...else ladder is shown below:

**Syntax :**

if (Test Expressions_1) :

    Indented block 1

elif (Test Expression_2) :

    Indented block 2

elif (Test Expression_3) :

    Indented block 3

else :

    Indented block

**Program :** To display the name of the month according to the number given by the user (take first 5 months).

```
a1.py - C:/Users/hp/AppData/Local/Programs/Python/Python311/a1.py (3.11.5)
File  Edit  Format  Run  Options  Window  Help
print("Enter number of the month ")
num = int(input("Enter your choice="))
if num==1:
    print("January")
elif num==2:
    print("February")
elif num==3:
    print("March")
elif num==4:
    print("April")
elif num==5:
    print("May")
else:
    print("Invalid input")
```

## Let's Implement

**A** Tick (✓) the correct answer.

1. These are the values on which the operators work.

   a. Operands ◯ b. Variables ◯ c. Numbers ◯ d. None of these ◯

2. What will be the output of the expression: x>y and x==y, if x=10, y=10?

   a. True ◯ b. False ◯ c. Error ◯ d. None of these ◯

3. What will be the output of the expression: print("50" + "70")?

   a. 50 70 ◯ b. 5070 ◯ c. 120 ◯ d. Error ◯

4. Look at the following code:

```
if(age >= 21):
        print("Adult")
else:
        print("Teenager")
```

   a. Adult ◯ b. Teenager ◯ c. Nothing will print ◯ d. Error ◯

5. ______________ refers to the normal flow of control in a program and is the simplest one.

   a. Sequence ◯ b. Conditional ◯ d. Iterative ◯ d. None of these ◯

**B** Fill in the blanks.

1. In the expression 2 + 8, ______________ is the operator.
2. Operators that work on two operands are called ______________ operations.
3. ______________ and ______________ are two types of arithmetic operators.
4. ______________ operators work on text as well as numbers enclosed within inverted commas.
5. ______________ and ______________ are two types of string operators.

**C** Write **True** or **False**.

1. Preference is given to higher precedence operators. ▭
2. You can use unary operators on more than three operands. ▭
3. a > b is a relational expression. ▭
4. There is a colon (:) after the if condition. ▭
5. Decisions in a program are used when the program has conditional choices to execute code block. ▭

## D Distinguish between the following with the help of examples.

1. Unary operator and Binary operator

| Unary operator | Binary operator |
|---|---|
| | |

2. Modulus operator and Division operator

| Modulus operator | Division operator |
|---|---|
| | |

## E Answer the following questions.

1. What are operators?
2. What are conditional statements?
3. Write the syntax for the if...elif...else statement.
4. Write the syntax of the if statement.
5. How does if...else statement works?

## Refreshment Corner

### Error Finding

**Find errors in the programs and write the correct program in the first blank box and correct output in the second blank box.**

1. Program to print sum and multiplication of two numbers entered

```
num1 = input("Enter first value= ")
num2 = input("Enter second value= ")
c = num1 + num2
d = num1 * num2
print (c)
```

2. Program to show use of string operator

```
str1 = "Hello"
str2 = "Friend"
print("Concatenation is", str1*3)
print("Replication is", str1+str2)
```

3. Program to accept a number. If number is <=400, print "Number is <=400"

```
num=input("Enter a name")
if num <= 400
    print("Number is less than 400")
```

4. Program to input a number. If number is equal to 10, print "Exact" otherwise print "Not exact"

```
num=int(input("Enter a number= ")
if num = 10
    print("Exact")
else
    print("Not exact"
```

5. Program to accept a name and age. Print what age will be after five years.

```
x=int(input("Enter a name= "))
y=int(input("Enter the age= "))
print(Hello, x)
print("After 5 years age is", y+5)
```

## Practical Session

Experiential Learning

- **Write a program in Python to:**
  1. Check whether a number is even or odd.
  2. Check whether a number is positive or negative.
  3. To convert the temperature from Fahrenheit to Celsius.
  4. To check whether a given number is two-digit number, three-digit number or four-digit number.
  5. To find out if the student is selected for the college or not based on the given conditions—

     Selected if age > 18 and marks > 65

     Not selected if age > 18 and marks < 65

     Not selected if age < 18

### Competency Based Questions

1. You were writing a program in which you wanted to execute a particular set of statements depending upon a particular test condition. Which type of statement will you use for this?
2. Sonam has created a Python program on various operators taught in the chapter. Her teacher asked her to give proper explanation of each operator in the program without displaying it on the output screen. How can this be possible? Which feature of Python can she use to do so?

## RACKING YOUR BRAIN

Life Skills and Values

One of your friends is not able to understand the concept of divide and modulus operator. Then what will you do?

### Just Have a Discussion

Communication

Discuss the use of different operators in the class in detail.

### Teacher's Note

- Discuss all the topics covered in this chapter with the students.
- Tell the students, how to use indentation to define the body of the conditional statement.
- Discuss the concept of conditional statements in Python with the students.

# TRENDING TECHNOLOGY AUGMENTED REALITY AND VIRTUAL REALITY

## Outlines

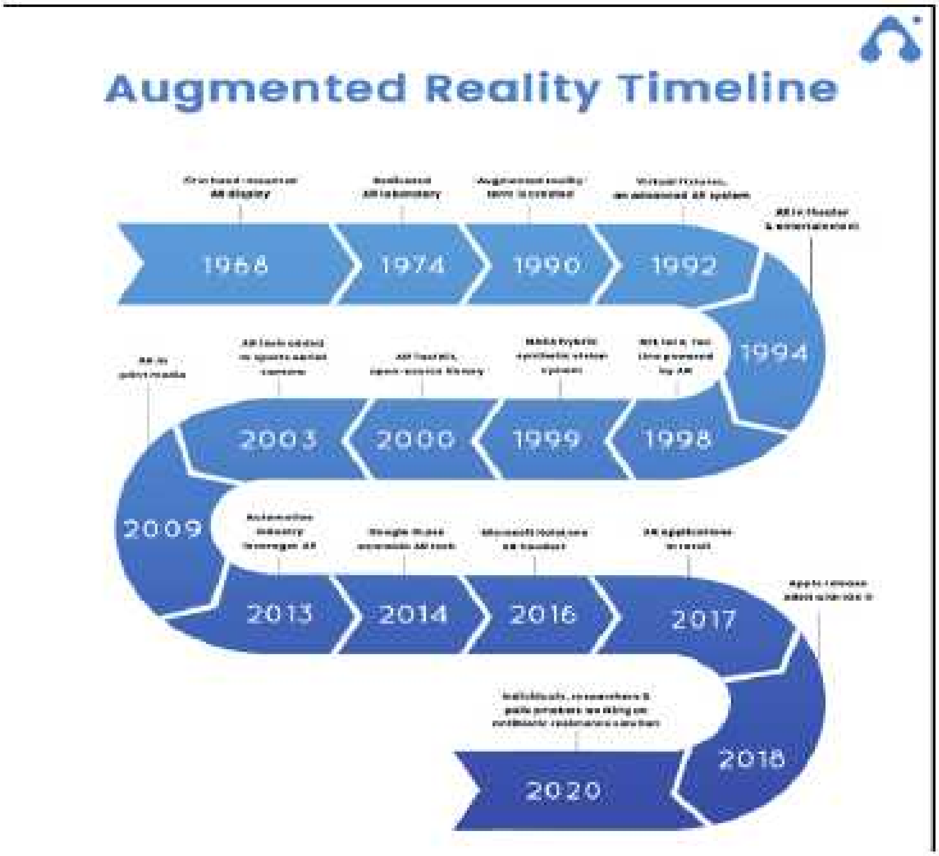
- Introduction to Augmented Reality
- Difference between Augmented Reality and Virtual Reality
- How AR and VR Technologies Work?
- Different Types of VR
- Applications of AR
- Applications of VR
- Introduction to Virtual Reality
- Advantages and Disadvantages of AR and VR
- Different Types of AR
- How VR Experiences are Created: Brief Introduction
- The Future of AR and VR

## Introduction to Augmented Reality

Till a few years back, it was difficult to imagine that users could see a product in 3D in a real-life environment and in real-time through tablets or smartphones. Augmented reality allows consumers to experience the product sitting at home. In brief, augmented reality is an enhanced and interactive version of a real-world environment through computer generated output. Augmented reality changes one's perception of the real world as it gets seamlessly interwoven with the physical world. If you use an AR app in your mobile, you can see your favourite video game characters becoming a part of your surroundings. You can also play the game in real-world surroundings, along with the characters in the game, such as Pokémon Go, which is a mobile game. There are many examples of AR technology in your surroundings, such as Snapchat Lenses and Google Maps AR.

### History and Development of AR

The idea of augmented reality has a much longer history than most people are aware of. In the last few decades, augmented reality technology has reshaped the way that we consume content in the real world.




## Introduction to Virtual Reality

Instead of bringing the video game characters into your world, would you like to be a part of their world? This too is possible, with the help of a technology called Virtual Reality (VR).

For playing some of the latest video games, you may have used a headset, comprising something that looks like a pair of glasses, and covers the eyes, and has settings for sound too. When you wear these glasses, the fictional world with various characters from the game come alive. When you play the game, you feel as if you are a part of that world! This is nothing but an example of Virtual Reality, where you are immersed in a 'Virtual' world, created by simulation.

History and Development of VR

Let us see the landmarks in the development of Virtual Reality technology with AR technology :

- **1838:** Sir Charles Wheatstone created a stereoscope that used a pair of mirrors at 45 degrees JAN1 angle to the user's eyes, each reflecting a picture located off to the side.
- **1935:** American science fiction writer, Stanley Weinbaum predicted the advent of VR in 1930s, in his short story Pygmalion's Spectacles.
- **1956:** Morton Heilig created Sensorama, the first VR machine (patented in 1962). It combined multiple technologies to stimulate all the senses.

- **1965-1968:** Ivan Sutherland, a computer scientist presented his vision of the Ultimate Display. In 1968, he created the first virtual reality HMD, named the 'Sword of Damocles'.
- **1972:** General Electric Corporation built a computerised flight simulator which featured a 180-degree field of vision.
- **1979:** Mc Donnell-Douglas Corporation integrated VR into its HMD, the VITAL helmet for military use.
- **1985:** VPL Research Inc. became the first company to sell VR goggles and gloves.
- **1991:** Antonio Medina, a NASA scientist designed a VR system to drive the Mars robot rovers from Earth. This system is called Computer Simulated Teleoperator.
- **2007:** Google introduced Street View.

## Difference between Augmented Reality and Virtual Reality

Let us understand the difference between Augmented and Virtual Reality with an example. Example: In architecture, AR can be used to show the structure and virtual elements of a new building superimposed on a real-life view, while VR can be used to create a digital simulation of the inside of that building.

| Augmented Reality | Virtual Reality |
|---|---|
| • It makes changes in the real-world. | • It replaces the real-world. |
| • The user can see and feel the real-world with virtual elements. | • The user is out of the real-world and focuses only on the virtual world. |
| • Virtual elements are superimposed on real-world. | • The user is transferred to a virtual environment. |
| • It is generally used with the help of a mobile phone screen. | • It is generally used with the help of a head-mounted display. |
| • The vision is limited to the screen of the device. | • The vision is completely off the real-world. |

## Advantages and Disadvantages of AR and VR

Some of the advantages and disadvantages of AR and VR are as follows:

| | Advantages | Disadvantages |
|---|---|---|
| **AR** | • Interesting way of providing information and knowledge by | • Quite expensive |

| | | |
|---|---|---|
| | blending it with a real-world environment<br>• Mobile apps required for AR are well established.<br>• Can be used to demonstrate the relevance of a product, for example, how a piece of furniture looks in the home of a customer | • Limited field of vision<br>• Difficult to control actions in some situations<br>• Less accessible for small business houses |
| **VR** | • Very engaging for the senses<br>• Great way to demonstrate new complex products<br>• Latest technology for creating new vivid environments | • Quite expensive<br>• Requires unlimited imagination to create VR experiences<br>• Content is specific for the platform for which it is to be created<br>• Can isolate the user from the outside world, or even from the society<br>• Slow and time-consuming method for demonstration |

## How AR and VR Technologies Work?

Let us understand how these technologies work:

### How AR Works?

Augmented Reality superimposes pictures, GIFs, animations, text or even 3D images on the real-world background. For doing this, it captures the surroundings of the user using a smartphone camera, on the screen of the phone.

Phones which are compatible for AR, often have sensors to sense solid objects in their surroundings and generate images of them. These images may be in the form of icons and animated GIFS of 3D images. The AR enabled phones have a CPU (Central Processing Unit) and GPU (Graphics Processing Unit) for processing this information. They also have a GPS (Global Positioning System) to find the relative location and position of the user and the elements on the screen. The user can move these images on the real scene captured on the screen, using controls on the phone.

**Smart Tip**

A GPU is a specialised electronic circuit for rendering all images on the computer screen. These are used for manipulating computer graphics and processing images in mobile phones, personal computers and game consoles.

In case of AR headsets, a projector is also required. The data from the sensors, which is processed, is projected on the background view. The headset may contain curved mirrors, to enable the user to see the 3D images. The main function of these mirrors is to align the superimposed objects with the view of the surroundings.

### How VR Works?

When you wear a VR headset, you can actually feel as if you are in another world, since the virtual scene seems to surround you on all sides!

The VR headset is connected to a computer by a High-Definition Multimedia Interface (HDMI) cable making it the source of the view in front of your eyes. In some cases, the software for generating the view is a part of the headset. The visuals on the screen are rendered either by using a mobile phone or HDMI cable connected to a computer.

Though you see a continuous image surrounding you, it is actually two different images for two different eyepieces of the headset. Lenses are also a part of the headset. They are placed between your eyes and the screen. These lenses help to create a stereoscopic vision. A VR headset often offers a 110 degrees field of vision. This is enough for user to feel surrounded on all sides by the scene. For tricking the user to believe that the scene is real, the view generally changes 60 to 90 frames per second.

Apart from the image, there are certain other elements, such as audio, that go into creating an immersive VR experience, making users completely engrossed in the virtual environment. Sound effects, when synced with the visuals, can create very engaging effects.

## Different Types of AR

There are different types of AR, which are discussed below:

### Marker-Based AR

You may have scanned a QR code using an app specially designed for the purpose. The QR code is a two-dimensional barcode that is only recognised by this app. This is called image recognition. Once scanned, the QR code is processed, and you can go to the correct web link. This image recognition is classified as Marker-based recognition. Scanning special signs that are recognised as images, is also a type of Marker-based AR.

### Marker-Less AR

You often use Google maps to find the location of a place, in relation to your present location. If you are on the move, Google maps update your position with reference to your destination from time to time. It basically uses the actual environment as an input. This application of AR is called marker-less AR.

A GPS used in a car that shows information about nearby landmarks, may be classified as a marker-less AR system. Hence, it is also called location-based AR. To find the geographic location of the user, with the help of GPS, it uses a direction indicator (compass). The system also includes a gyroscope and an accelerometer. Depending on your location, the Facebook app in your mobile may provide the information about nearby events on that day. This is also an example of a location-based AR system.

### AR Based on Projection

Suppose, you choose to spend time, during your commute, by playing an AR based video game. The video game uses the images of your surroundings to create a hologram or a 3D effect, and projects it on the mobile screen. This is an example of a Projection-based AR system. The game involves your interaction with this image, projected on the mobile screen.

### AR Based on Superimposition

Suppose, your video game involves characters attacking you, these characters will be included in your view of your surroundings on the screen. Thus, the real-life scene is augmented.

You may have used the IKEA app, where you can make a choice of furniture you want to buy, by superimposing the images of it on the scene of your room at home. These are some applications of superimposition-based AR.

## Different Types of VR

There are different types of VR, which are discussed below:

### Non-Immersive VR

Playing a simulated game of tennis in front of your wall mounted personal computer, using a pair of headphones for sound effects, and a joystick instead of a racket, is an example of non-immersive VR. It is called so, because you are partially aware of your surroundings while playing the game. You are not completely transferred to the virtual tennis court.

### Completely or Fully Immersive VR

Playing the game of tennis using VR headset along with headphones for sound effects, and performing the strokes of tennis using the joystick, is an example of completely immersive VR. In this game, you get a feeling of being on the virtual tennis court. It detaches you from your real-world surroundings, and while hitting the tennis ball, you may end up knocking an object around you. So, be cautious!

### Semi-Immersive VR

Semi-immersive experiences provide the users with a partially virtual environment to interact with, while remaining connected to the real-world surroundings. This type of VR is mainly used for educational and training purposes and the experience is made possible with graphical computing and large projector systems. One such example of semi-immersive virtual reality is a flight simulator.

### Collaborative VR

In the above examples, you have been playing tennis against the computer. But, have you imagined playing tennis with your friend, while each one of you is in front of your respective personal computers? This is possible, if both of you are connected to the virtual tennis court at the same time. This is an example of collaborative VR.

## How VR Experiences are Created: Brief Introduction

You can create a VR experience by using any of the following VR technologies:

### Head Tracking

This is a VR technology used to change the image in front of you when you, wear a VR headset. It is called 'Six degrees of Freedom', and divides your head's dimensions along three axes, X, Y and Z. The scene that you see changes when you move your head upwards, downwards, side to side or when you tilt your head. The head movements are measured along the three axes, and the scene changes accordingly.

### Motion Tracking

In this type of VR, you can see your hands and track your own movements in the virtual world. This gives you the feeling of being present at the virtual scene, thus making the user experience more believable.

This type of VR may use sensors, so if you pull the trigger of any object using some function on your joystick, your senses give you an experience of pulling a real trigger.

### Eye Tracking

Eye tracking is a sensor technology which uses infrared light to enable a computer or other device to know where a person is looking. Infrared sensors fitted in headsets monitor the movements of the eyes to make the video game characters move. The senses of the virtual characters are sharper than yours, since they know exactly in which direction you are looking. This is more accurate than the movement of the head. So, you cannot deceive your virtual opponent in the game. This would be the ultimate gaming experience!

## Applications of AR

You can experience AR in various areas, some of which are discussed below:

### Military Training

The Heads-up Display (HUD) is used to train fighter pilots. In this AR technology, a transparent scene is superimposed on the pilot's view from the plane. The data display on the transparent screen includes altitude, speed, direction, latitude and longitude, GPS location of the enemy and other important information. It is called Heads-up Display because the pilot can look straight, instead of looking down at a screen.

### Entertainment and Gaming

Video games that use AR have been around for a long time now. A well-known example is Pokémon Go. These games work by superimposing virtual characters in your surroundings, with whom you can play the game, even when on a move! This is how AR games augment the world around you, and make it more interesting, while allowing you to remain aware of your real surroundings.

### Education

AR can make the process of learning very captivating. For example, you can learn about an ancient civilisation by taking a tour of a city using AR. You can learn how people of that civilisation traded using the barter system and you may even exchange a few goods with the virtual traders, to gain some ancient artefacts, of course, virtually!

Currently, in some institutes, AR is being used to train medical students to perform difficult surgeries, by providing them a view of the intricate organs in the human body. AR is also being used in the study of astrophysics.

### Robotics and Engineering

With the advent of Artificial Intelligence, as robots become more and more intelligent, the technology of AR can be used to expose these robots to different environments, so that they can learn to react on these environments. AR can be used to challenge the robots with virtual problems in the real world and train them to solve these problems.

## Applications of VR

You can experience VR in various areas, some of them are discussed below:

### Education

Like AR, VR too can be used to make the studies more interesting. Instead of going on an educational tour to a nearby wildlife sanctuary, imagine going on a virtual safari to Kenya, or visiting the Amazon River basin to study its biodiversity while sitting in your classroom! While studying the circulatory system, imagine travelling inside the blood vessels, to take a look at different types of blood cells, and understand the path taken by the blood from the heart to the lungs, back to the heart and to the tip of the toes. This could be made possible by VR!

### Travel and Tourism

When you approach a travel agency to make bookings for your forthcoming vacation, it usually gets difficult for you to choose a tour package. Well, in this case, the travel agent can take you on a virtual tour of the most exciting tourist places included in different tours, to help you make a decision. The agent may even introduce you to new travel destinations, to broaden your choice. Thus, VR can be used to promote travel and tourism.

### Healthcare Facilities

VR can prove to be of great help to medical practitioners while performing complex procedures and treatments. For example, Virtual Reality Therapy (VRT) can be used to diagnose and treat psychological conditions that cause difficulties for patients. Professionals often treat patients with phobias or related disorders by gradually exposing them to the cause of the issue. This technology uses simulated interactive and immersive environments as a tool to give the patient a simulated experience.

### Space Projects

Many spacecrafts launch special space vehicles on the surface of the moon. These space vehicles, called rovers, are able to get images of the surface of the moon and allow the scientists to view the surface in real time. A virtual reality interface is used to allow one operator to change the focus between various narrow targets of interest within the wide set of fused data available from the sensors for exploration.

### Military Training

Before invading the enemy in unknown territory, it is important to know the geographical details of the place and the strategic position of the enemy. This information can be obtained by using GPS, and a virtual scene of the territory that can be simulated using VR to train the military personnel before the invasion.

## The Future of AR and VR

When cinema and television were introduced for the first time, they created quite a stir, since these were a new medium of communication and information. Similarly, AR and VR can be seen as new media, not only to take you to the fictional world that does not exist, as in gaming, but also to explore the inaccessible parts of the world that exists in reality. You have seen how AR and VR have a large range of uses beyond entertainment. But, before they can be used on a large scale for more relevant and constructive purposes, these media need to become more financially and technically feasible.

Nowadays, new and emerging technologies are continuing to come to market and making new environments accessible to the masses. Such examples are Mixed Reality (MR) and Extended Reality (XR). Mixed reality is a blend of both AR and VR technologies. Whereas, extended reality can be defined as an umbrella, which brings all the three immersive technologies (AR, VR, MR) together under one term.

As media, they can engage our visual and auditory senses. However, in the future we might create AR and VR gadgets that will also engage our sense of smell and touch, thus making them even more lifelike. But, excessive use of these media can be detrimental to our brain, since they misguide our nervous system. It may lead to a stage where humans won't be able to differentiate between the virtual and real sensations. There is going to be a dynamic change in people's lives in the future, as these technologies have a lot to offer.

## COMPUTER ETIQUETTES

Always cover a computer when it is not in use.

## Let's Implement

**A Tick (✓) the correct answer.**

1. In Augmented Reality, which of the following elements is superimposed on the real-world background?
   a. Virtual ◯ b. Real ◯ c. Both a and b ◯ d. None of these ◯
2. Which is the most popularly used equipment in VR?
   a. Headset ◯ b. Microphone ◯ c. Stereoscope ◯ d. None of these ◯
3. Which of the following is a type of virtual reality?
   a. Marker-less ◯ b. Collaborative ◯ c. Marker-based ◯ d. None of these ◯
4. Which of the following can be used to diagnose and treat psychological conditions?
   a. AR ◯ b. Video Games ◯ c. VR ◯ d. None of these ◯
5. Which of the following is defined as an umbrella that brings all the three immersive technologies together under one team?
   a. VR ◯ b. Extended Reality ◯
   c. Mixed Reality ◯ d. None of these ◯

**B Fill in the blanks.**

1. The term ______________ reality was coined by Caudell.
2. The comparatively affordable version of AR introduced by Microsoft in 2015 is called ______________.
3. A heads-up display is used to train ______________.

4. The _____________ reality is quite engaging, but can isolate the user from the outside world.
5. You can play a game of tennis against your friend on a virtual tennis court using _____________ VR.

**C** Write **True** or **False**.

1. Augmented reality can be used to create a digital simulation. [ ]
2. AR and VR technologies are used in gaming only. [ ]
3. The AR enabled phones have a CPU and GPU for processing the information. [ ]
4. Motion tracking monitors a user's head position and orientation. [ ]
5. Overuse of AR and VR can affect the ability of a human being to distinguish between virtual and real sensations. [ ]

**D** Answer the following questions.

1. Which technologies are used to create VR experience? Explain briefly.
2. What are the different types of Augmented Reality?
3. Name any five applications of VR.
4. Differentiate between AR and VR.
5. How does VR work? Explain in brief.

## Refreshment Corner

- Write the applications of AR and VR.

| AR | VR |
|---|---|
| 1. ____________________ | 1. ____________________ |
| 2. ____________________ | 2. ____________________ |
| 3. ____________________ | 3. ____________________ |

## RACKING YOUR BRAIN

Life Skills and Values

Imagine, you have been on a virtual tour of the ancient empire of Rome. Conduct a brainstorming session on what great monuments you saw there, the historical figures you met, and any major historical events that you witnessed.

## Just Have a Discussion

Communication

Discuss the difference between Augmented Reality and Virtual Reality in the class.

### Teacher's Note

The teacher should demonstrate to the students the different types and applications of AR and VR.

# QUICK REVIEW 2

**Tick (✓) the correct option.**

1. Which of the following is a Container tag?

   a. <HR> ◯ b. <BR> ◯

   c. <HEAD> ◯ d. All of these ◯

2. Gmail is a ______________ app.

   a. native ◯ b. web ◯ c. hybrid ◯ d. None of these ◯

3. A ______________ is a legal right granted by law to the creator for his original work.

   a. Copyright ◯ b. Patent ◯

   c. Trademark ◯ d. None of these ◯

4. What will be the output of the expression: print("50"+ "70")?

   a. 50 70 ◯ b. 5070 ◯ c. 120 ◯ d. Error ◯

5. Which of the following is defined as an umbrella that brings all the three immersive technologies together under one team?

   a. VR ◯ b. Extended Reality ◯

   c. Mixed Reality ◯ d. None of these ◯

6. It is not an educational app.

   a. Duolingo ◯ b. Lumosity ◯ c. Photomath ◯ d. Temple Run ◯

7. ______________ is the process of transforming data into an unreadable code.

   a. Encoding ◯ b. Encryption ◯ c. Decryption ◯ d. None of these ◯

8. ______________ refers to the normal flow of control in a program and is the simplest one.

   a. Sequence ◯ b. Conditional ◯

   d. Iterative ◯ d. None of these ◯

9. Which of the following is a Text Editor?

   a. PowerPoint ◯ b. Internet Explorer ◯

   c. Notepad ◯ d. MS-QBasic ◯

10. In Augmented Reality, which of the following elements are superimposed on the real-world background?

    a. Virtual ◯ b. Real ◯ c. Both a and b ◯ d. None of these ◯

# TEST PAPER 1

## A Fill in the blanks.

1. The base of the binary number system is ______________ .
2. ______________ is a powerful tool for consolidating, summarising and presenting data.
3. Databases in MS Access 2016 are composed of ______________ main objects.
4. The ______________ Selection tool works as a freehand selection tool.
5. _____________ is the row containing values under each field for one entry.

## B Write True or False.

1. The numbers used in hexadecimal number system are 0 to 15. [ ]
2. You can sort data only in ascending order in a worksheet. [ ]
3. Value is the data item which is the smallest unit in a database. [ ]
4. Krita offers various panels that can either be grouped, stacked or docked. [ ]
5. Filtering means sorting data. [ ]

## C Name the following.

1. Two components of Krita ______________ ______________
2. Two Brush tools ______________ ______________
3. Two Drawing tools ______________ ______________
4. Two Selection Tools ______________ ______________

## D Answer the following questions.

1. Explain number system. Name different types of number system.
2. State the use of PivotTable.
3. Explain RDBMS.
4. What is the use of Similar Color Selection tool?
5. Write a short note on octal number system.

# TEST PAPER 2

## A Fill in the blanks.

1. ______________ is a text editor.
2. The ______________ software allows free app development.
3. ______________ is an act of harming harassing or targeting a person by another person using Internet, in a deliberate manner.
4. ______________ operators work on text as well as numbers enclosed within inverted commas.
5. You can play a game of tennis against your friend on a virtual tennis court using ______________ VR.

## B Write **True** or **False**.

1. HTML is used for website creation.
2. App Development refers to the process of developing apps that can run on desired mobile platforms.
3. A patent grants the inventor the sole right to make, use and sell that invention, for an unlimited period.
4. a > b is a relational expression.
5. AR and VR technologies are used in gaming only.

## C Distinguish between the following with the help of examples.

1. Unary operator and Binary operator

| Unary operator | Binary operator |
| --- | --- |
| | |

## D Answer the following questions.

1. Define HTML.
2. Write a short note on:

   a. Facebook b. Instagram c. WhatsApp d. Twitter
3. Write a short note on:

   a. Spamming b. Cybercrime c. Digital Footprint d. Hacking
4. Write the syntax for the if...elif...else statement.
5. What are the different types of Augmented Reality?